De La Salle University

SEMPHY01 — Semiconductor Physics

Semiconductor Physics

Models, Computation, and Validation

Eugene Joseph M. Ragasa

Development manuscript for `dlsu_AY20262026T1`.

The notes connect physical models, analytical reductions, numerical implementations, and validation evidence.

November 12, 2026

Preface

A semiconductor course can become a list of formulas remarkably quickly. A band gap appears, two effective densities of states are quoted, a Fermi level is placed somewhere in the gap, and a concentration falls out of an exponential. The calculation may be correct, yet the connection between the crystal, the model, and the reported number remains hidden.

These notes take the opposite route. Each important quantity is attached to a model and each model is attached to evidence. The derivation identifies the origin of the band energies, the range replaced by a parabola, the information preserved by an effective mass, the multiplicity of the available states, the partition function governing occupation, and the charge constraint determining the chemical potential. Computation provides an inspectable representation of the physical and mathematical argument.

The recurring modeling chain is

$$H(\boldsymbol{k}) \longrightarrow E_n(\boldsymbol{k}) \longrightarrow \text{state multiplicity} \longrightarrow g(E) \longrightarrow Z, \Xi \longrightarrow f(E, \mu, T) \longrightarrow n, p \longrightarrow Q(\mu) = 0.$$

Every arrow carries assumptions, units, conventions, and a range of validity. Semiconductor physics requires selecting reductions that remain accurate in the regime under study.

Each treatment states the physical model and its domain, gives an inspectable analytic or numerical representation, and supplies validation evidence tied to the defining physical relations.

The initial chapters connect the particle in a box to Bloch states and finite HOMO–LUMO spectra to tight-binding bands. They then establish spectral multiplicity, density of states, partition functions, occupation, chemical potential, and equilibrium. The intrinsic and extrinsic semiconductor chapters then apply that foundation before later treatments of transport, recombination, interfaces, and junctions. Each treatment uses the simplest explicit model that preserves the physics required by the problem.

The principal textbook conventions follow standard treatments of solid-state and semiconductor physics, including Refs. [3, 5, 10] and the device-oriented perspectives of Refs. [7, 8]. Specific material claims require material-specific primary or curated sources with declared provenance.

Contents

Preface **i**

I Foundations and Equilibrium **1**

1 From a Particle in a Box to Bloch States **2**

1.1 The one-dimensional infinite box 2

1.2 Spectral representation of the box Hamiltonian 3

1.3 Periodic boundary conditions and plane waves 4

1.4 Translation as a symmetry 5

1.5 Bloch's theorem 6

1.6 A finite crystal selects discrete Bloch wave vectors 7

1.7 Three-dimensional form 7

1.8 Chapter summary 7

2 From HOMO–LUMO Levels to Tight-Binding Bands **8**

2.1 Localized orbitals as a basis 8

2.2 Two sites: bonding and antibonding orbitals 9

2.3 HOMO and LUMO in a finite system 10

2.4 A finite chain of localized orbitals 10

2.5 The periodic chain and the Bloch basis 11

2.6 Several orbitals per cell 12

2.7 A two-orbital cell and a band gap 12

2.8 From HOMO and LUMO to valence and conduction bands 13

2.9 Model content 14

2.10 Chapter summary 14

3 Spectral Representation and the Density of States **15**

3.1 Spectral representation and discrete multiplicity 15
3.2 Bloch-band information and state density 17
3.3 Allowed wave vectors in a finite periodic crystal 18
3.4 The general density of states of a band 18
3.5 Constant-energy surfaces and band velocity 19
3.6 Exact example: a one-dimensional tight-binding band 19
3.7 Parabolic bands in one, two, and three dimensions 21
3.7.1 One dimension 21
3.7.2 Two dimensions 21
3.7.3 Three dimensions 22
3.8 Anisotropic valleys and the density-of-states mass 22
3.9 Several bands at the same energy 23
3.10 Calculating a density of states from sampled bands 23
3.11 Normalization, units, and convergence 24
3.12 Computational laboratory: bands, state counting, and DOS 25
3.13 Chapter summary 25

4 Statistical Mechanics of Carrier Populations **26**

4.1 From equipartition to a probability distribution 26
4.2 Microstates, macrostates, and constraints 27
4.3 From reservoir multiplicity to the partition function 27
4.3.1 Recovering equipartition from the partition function 28
4.3.2 Two-state canonical example 29
4.4 One electron distributed over a band 29
4.5 Many electrons and the chemical potential 30
4.6 Mean occupation of one spin-resolved band state 31
4.7 Many independent fermionic states 33
4.8 From a state sum to carrier concentrations 33
4.9 The Boltzmann limit derived from the partition function 34
4.10 The chemical potential is selected by a constraint 35
4.11 Localized states and internal degeneracy 36
4.12 A disciplined route through a carrier calculation 36
4.13 Chapter summary 36

5 Determining the Chemical Potential from the Available States 37

5.1 Thermodynamic inputs beyond the band structure . . . 37
5.2 The finite discrete-state problem . . . 38
5.2.1 Zero-temperature limit . . . 38
5.2.2 A symmetric two-level example . . . 38
5.3 The continuum limit for energy bands . . . 39
5.4 Charge neutrality as the semiconductor constraint . . . 39
5.5 Intrinsic material as the first closure problem . . . 40
5.6 Energy references must be shared . . . 40
5.7 Choosing the statistical and band model . . . 41
5.7.1 Nondegenerate or Fermi–Dirac statistics . . . 41
5.7.2 Parabolic or full bands . . . 41
5.7.3 Complete or incomplete dopant ionization . . . 41
5.7.4 Fixed bands or charge-dependent bands . . . 41
5.8 Spatially varying systems . . . 41
5.9 A reliable mathematical solution procedure . . . 42
5.10 Physical closure . . . 42
5.11 Chapter summary . . . 42

6 The Intrinsic Semiconductor 43

6.1 What intrinsic means . . . 43
6.2 Why a solid has bands rather than isolated levels . . . 44
6.2.1 Why neither an empty nor a full band conducts . . . 44
6.2.2 Why an empty valence state is called a hole . . . 45
6.3 From a periodic solid to band edges . . . 45
6.3.1 Quadratic expansion at a conduction minimum . . . 46
6.3.2 Valence maxima and positive hole masses . . . 46
6.4 Applying state counting to intrinsic band edges . . . 46
6.4.1 The conduction-band ellipsoid . . . 47
6.4.2 The valence-band density of states . . . 49
6.5 Combining available states with statistical occupation . . . 49
6.5.1 One-state partition-function recap . . . 49
6.5.2 Spectral carrier densities . . . 50
6.6 Exact parabolic-band carrier integrals . . . 50

6.7 The nondegenerate limit . . . 52
6.8 Deriving the intrinsic chemical potential . . . 53
6.9 Deriving the intrinsic carrier concentration . . . 54
6.10 Temperature dependence . . . 54
6.11 Worked silicon estimate at $3.00 \times 10^2\,\mathrm{K}$ . . . 55
6.12 When the full Fermi integrals are required . . . 56
6.13 Assumptions exposed . . . 56
6.14 Verification checklist . . . 57
6.15 Chapter summary . . . 57

7 The Extrinsic Semiconductor 58
7.1 What doping changes—and what it does not . . . 58
7.2 Substitutional donors and acceptors . . . 59
7.3 Why shallow dopants resemble hydrogen . . . 60
7.4 Localized-state statistics and degeneracy . . . 61
7.4.1 Donor partition argument . . . 61
7.4.2 Acceptor partition argument . . . 61
7.5 The master charge-neutrality problem . . . 62
7.6 Nondegenerate carrier relations . . . 63
7.7 Complete ionization and compensation . . . 63
7.7.1 Strong n-type limit . . . 64
7.7.2 Strong p-type limit . . . 65
7.7.3 Fully compensated material . . . 65
7.8 Fermi level in the complete-ionization regime . . . 65
7.9 Incomplete ionization and freeze-out . . . 66
7.10 Three temperature regimes . . . 67
7.11 Worked n-type silicon estimate at $3.00 \times 10^2\,\mathrm{K}$ . . . 67
7.12 Degenerate doping and impurity-band limits . . . 68
7.13 Why local neutrality fails for nonuniform doping . . . 69
7.14 Computational laboratory: solving neutrality . . . 69
7.15 Verification checklist . . . 70
7.16 Chapter summary . . . 71

II Transport and Nonequilibrium 72

8 Brownian Motion, Diffusion, and the Heat Equation 73

8.1 The random-walk limit . . . 73
8.2 Brownian motion and drift–diffusion dynamics . . . 74
8.3 From stochastic paths to an ensemble equation . . . 76
8.4 The master equation and Fick's law . . . 76
8.5 The heat equation . . . 77
8.6 Matrix representation and exponential evolution . . . 77
8.7 Range of validity . . . 78
8.8 Chapter summary . . . 79

9 From Langevin Dynamics to the Maxwell–Boltzmann Gas 80

9.1 Underdamped Langevin dynamics . . . 80
9.2 Velocity memory and the Brownian limit . . . 81
9.3 Chapter summary . . . 81

10 The Drude and Drude–Sommerfeld Electron Gases 82

10.1 The classical Drude response . . . 82
10.2 The Sommerfeld correction . . . 83
10.3 Range of validity . . . 84
10.4 Chapter summary . . . 84

11 Effective Mass and the Semiconductor Carrier Model 85

11.1 Crystal bands and effective mass . . . 85
11.2 The semiconductor carrier model . . . 86
11.3 Range of validity . . . 87
11.4 Chapter summary . . . 87

12 Mobility, Resistivity, and the Hall Effect 88

12.1 Notation and units . . . 89
12.2 Scattering-limited mobility . . . 89
12.3 Conductivity and resistivity . . . 90
12.4 Hall response . . . 90
12.4.1 Lorentz-force balance and sign convention . . . 90

12.4.2 Hall voltage and extracted quantities . . . 91
12.4.3 Hall scattering factor . . . 92
12.4.4 Electron–hole conduction . . . 92
12.5 Range of validity . . . 93
12.6 Chapter summary . . . 93

13 Generation, Recombination, and Carrier Continuity 94
13.1 Notation and units . . . 95
13.2 Excess carriers and lifetime . . . 95
13.3 Direct and recombination-center-assisted processes . . . 97
13.3.1 Band-to-band recombination and detailed balance . . . 97
13.3.2 The two-step recombination-center mechanism . . . 97
13.3.3 Detailed balance and center degeneracy . . . 98
13.3.4 Nonequilibrium occupancy and the SRH rate . . . 99
13.3.5 Depletion and high-injection limits . . . 100
13.4 Quasi-Fermi levels . . . 100
13.5 Continuity and diffusion length . . . 101
13.6 Drift, diffusion, and recombination of a packet . . . 101
13.7 Finite-difference continuity and dynamic center kinetics . . . 103
13.7.1 Nonlinear steady SRH boundary-value problem . . . 103
13.7.2 Time-dependent center occupation . . . 104
13.8 Range of validity . . . 104
13.9 Chapter summary . . . 105

14 Barrier Transport and Space-Charge-Limited Current 106
14.1 Notation and units . . . 106
14.2 Thermionic emission . . . 107
14.3 Quantum tunneling through a barrier . . . 107
14.4 Space charge from injected carriers . . . 107
14.5 Finite-difference barrier and space-charge calculations . . . 108
14.5.1 Open-boundary Schrödinger equation . . . 109
14.5.2 Poisson equation for prescribed charge . . . 110
14.5.3 Current-constrained nonlinear space charge . . . 110
14.6 Range of validity . . . 111

14.7 Chapter summary 111

15 High-Field Carrier Transport **112**

15.1 Notation and units 112

15.2 Velocity saturation 112

15.3 Transferred-electron effect 113

15.4 Impact ionization 113

15.5 Transport-regime hierarchy 114

15.6 Chapter summary 114

References **115**

Part I

Foundations and Equilibrium

Chapter 1

From a Particle in a Box to Bloch States

A quantum particle confined to a finite region has a discrete spectrum. A particle in a periodic crystal also has discrete states when the crystal is finite, but lattice translation organizes those states by a Bloch wave vector. This chapter develops that connection from boundary conditions, translation symmetry, and the spectral representation of the one-electron Hamiltonian [3, 6, 10].

1.1 The one-dimensional infinite box

Consider one electron on the interval $0 < x < L$ with infinitely high walls. The Hamiltonian inside the box is

$$\hat{H} = -\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2}, \tag{1.1}$$

and its domain imposes

$$\psi(0) = \psi(L) = 0. \tag{1.2}$$

The stationary Schrödinger equation is

$$-\frac{\hbar^2}{2m}\frac{\mathrm{d}^2\psi}{\mathrm{d}x^2} = E\psi. \tag{1.3}$$

For $E > 0$, define

$$k = \frac{\sqrt{2mE}}{\hbar}. \tag{1.4}$$

The general interior solution is

$$\psi(x) = A\sin(kx) + B\cos(kx). \tag{1.5}$$

The boundary condition at $x = 0$ gives $B = 0$. The condition at $x = L$ gives

$$\sin(kL) = 0, \tag{1.6}$$

so

$$k_n = \frac{n\pi}{L}, \qquad n = 1, 2, 3, \ldots. \tag{1.7}$$

Normalization produces

$$\boxed{\phi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right),} \tag{1.8}$$

and the energies are

$$\boxed{E_n = \frac{\hbar^2 \pi^2 n^2}{2mL^2}.} \tag{1.9}$$

The discrete spectrum follows from the boundary conditions. Each allowed standing wave must contain an integer number of half wavelengths inside the box:

$$L = n\frac{\lambda_n}{2}. \tag{1.10}$$

The finite length selects a discrete set of wavelengths and therefore a discrete set of energies.

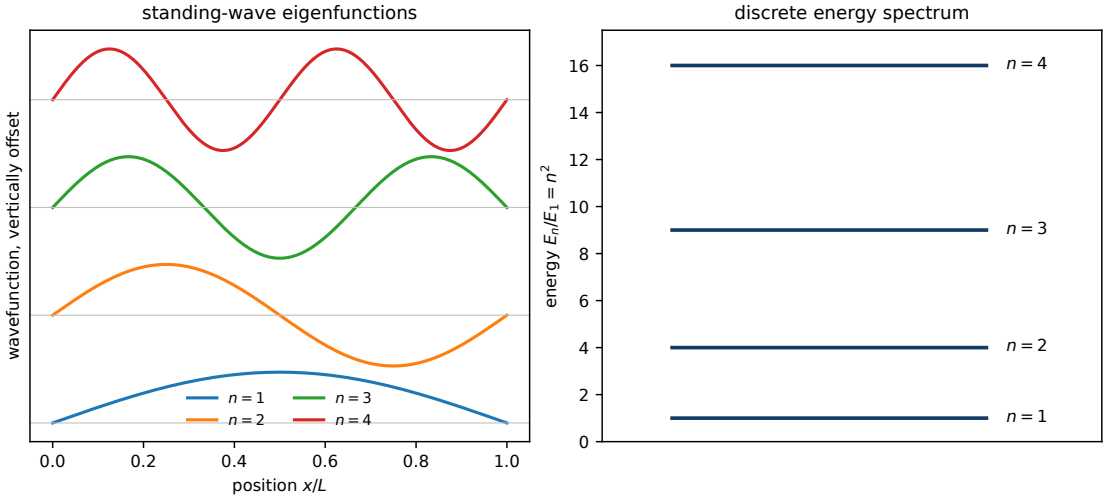


Figure 1.1: The first four particle-in-a-box eigenfunctions and their energies. The number of interior nodes increases with n, while the spectrum grows as n^2.[1]

1.2 Spectral representation of the box Hamiltonian

The eigenfunctions form an orthonormal basis:

$$\langle m|n\rangle = \delta_{mn}, \tag{1.11}$$

and resolve the identity on the box Hilbert space:

$$\hat{I} = \sum_{n=1}^{\infty} |n\rangle \langle n|. \tag{1.12}$$

The Hamiltonian has the spectral representation

$$\boxed{\hat{H} = \sum_{n=1}^{\infty} E_n |n\rangle \langle n|.} \tag{1.13}$$

[1] *Student reproduction notebook:* figures/01_particle-box-to-bloch/fig_01_01.ipynb.

An arbitrary state expands as

$$|\psi\rangle = \sum_{n=1}^{\infty} c_n |n\rangle , \qquad c_n = \langle n|\psi\rangle . \tag{1.14}$$

The probability of measuring E_n is

$$\mathbb{P}(E_n) = |c_n|^2. \tag{1.15}$$

This finite-region example establishes the main spectral objects: eigenvalues, eigenstates, projectors, and multiplicities.

1.3 Periodic boundary conditions and plane waves

Replace the hard walls by the periodic condition

$$\psi(x + L) = \psi(x). \tag{1.16}$$

The free-particle eigenfunctions are plane waves

$$\phi_k(x) = \frac{1}{\sqrt{L}} e^{ikx}. \tag{1.17}$$

Periodicity requires

$$e^{ikL} = 1, \tag{1.18}$$

which gives

$$\boxed{k_m = \frac{2\pi m}{L}, \qquad m \in \mathbb{Z}.} \tag{1.19}$$

The energies are

$$E_m = \frac{\hbar^2 k_m^2}{2m}. \tag{1.20}$$

The spacing of neighboring wave vectors is

$$\Delta k = \frac{2\pi}{L}. \tag{1.21}$$

As L grows, the discrete wave vectors become dense while the state density $L/(2\pi)$ per unit k remains finite.

The hard-wall box uses standing waves because reflection fixes nodes at two boundaries. The periodic box uses traveling waves because translation through L returns the same state. Both spectra are discrete for finite L; their boundary conditions organize the eigenstates differently.

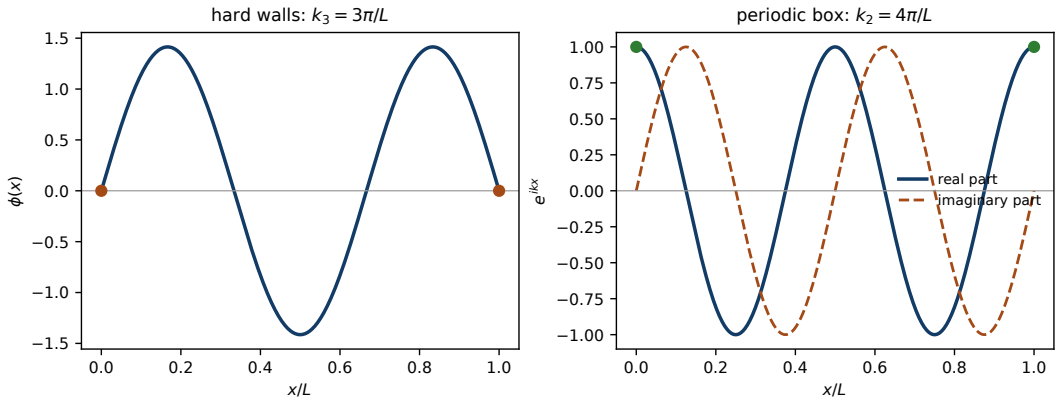


Figure 1.2: Hard-wall and periodic boundary conditions select different eigenfunctions. The hard-wall box supports standing waves with $k_n = n\pi/L$; the periodic box supports plane waves with $k_m = 2\pi m/L$.[2]

1.4 Translation as a symmetry

The translation operator through a displacement a acts as

$$(\hat{T}_a\psi)(x) = \psi(x + a). \tag{1.22}$$

Successive translations satisfy

$$\hat{T}_a\hat{T}_b = \hat{T}_{a+b}. \tag{1.23}$$

For a periodic potential,

$$V(x + a) = V(x), \tag{1.24}$$

the Hamiltonian

$$\hat{H} = -\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2} + V(x) \tag{1.25}$$

commutes with lattice translation:

$$[\hat{H}, \hat{T}_a] = 0. \tag{1.26}$$

Energy eigenstates can therefore be chosen as translation eigenstates:

$$\hat{T}_a\psi_k = \lambda_k\psi_k. \tag{1.27}$$

Translation preserves normalization, so $|\lambda_k| = 1$. Write

$$\lambda_k = e^{ika}. \tag{1.28}$$

The eigenvalue equation becomes

$$\boxed{\psi_k(x + a) = e^{ika}\psi_k(x).} \tag{1.29}$$

[2] *Student reproduction notebook:* figures/01_particle-box-to-bloch/fig_01_02.ipynb.

1.5 Bloch's theorem

Define

$$u_k(x) = e^{-ikx}\psi_k(x). \tag{1.30}$$

Using equation (1.29),

$$\begin{aligned} u_k(x+a) &= e^{-ik(x+a)}\psi_k(x+a) \\ &= e^{-ikx}e^{-ika}e^{ika}\psi_k(x) \\ &= u_k(x). \end{aligned} \tag{1.31}$$

Thus u_k has the lattice period and

$$\boxed{\psi_{nk}(x) = e^{ikx}u_{nk}(x), \qquad u_{nk}(x+a) = u_{nk}(x).} \tag{1.32}$$

The band index n distinguishes different eigenstates at the same k.

Substitution into the Schrödinger equation gives the cell-periodic eigenproblem

$$\boxed{\hat{H}(k)u_{nk} = E_n(k)u_{nk},} \tag{1.33}$$

with

$$\hat{H}(k) = e^{-ikx}\hat{H}e^{ikx} = \frac{(\hat{p}+\hbar k)^2}{2m} + V(x). \tag{1.34}$$

For each k, diagonalization of $\hat{H}(k)$ gives the band energies $E_n(k)$.

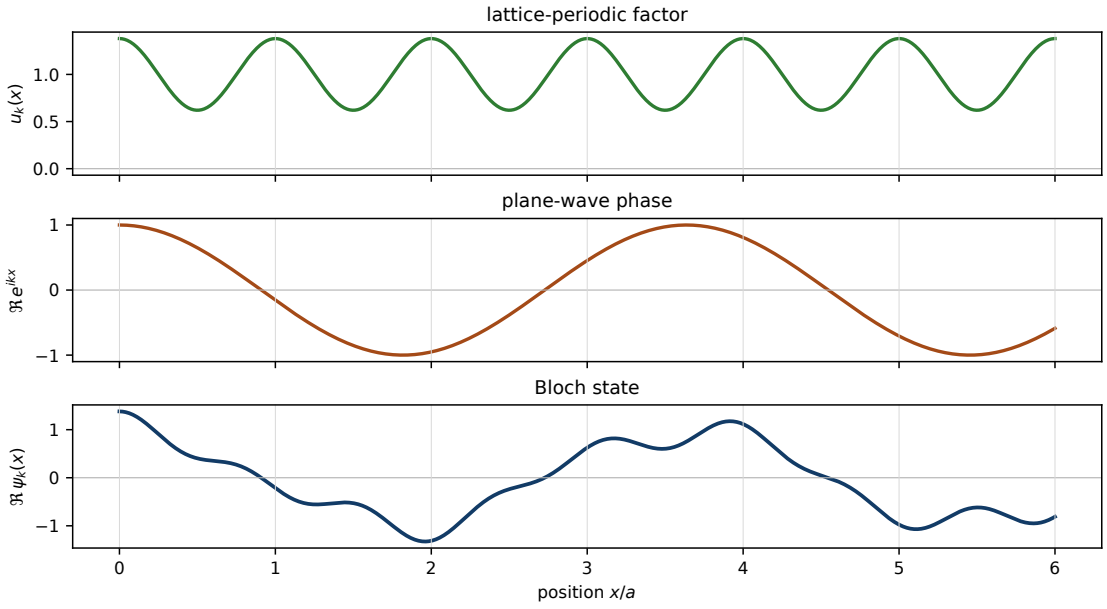


Figure 1.3: A Bloch state is the product of a plane-wave phase and a lattice-periodic function. Translation through one cell changes the full state by the phase e^{ika} while reproducing the periodic factor.[3]

[3] *Student reproduction notebook:* figures/01_particle-box-to-bloch/fig_01_03.ipynb.

1.6 A finite crystal selects discrete Bloch wave vectors

Let the crystal contain N cells, so $L = Na$, and impose periodicity across the whole crystal:

$$\psi_k(x + Na) = \psi_k(x). \tag{1.35}$$

Repeated application of Bloch's condition gives

$$\psi_k(x + Na) = e^{ikNa}\psi_k(x). \tag{1.36}$$

Therefore

$$e^{ikNa} = 1, \tag{1.37}$$

and

$$k_m = \frac{2\pi m}{Na}. \tag{1.38}$$

There are N distinct k values in one Brillouin zone for each band. This matches the N primitive cells in the finite crystal.

A reciprocal-lattice vector in one dimension is

$$G = \frac{2\pi\ell}{a}, \qquad \ell \in \mathbb{Z}. \tag{1.39}$$

The translation phase is unchanged by $k \to k + G$:

$$e^{i(k+G)a} = e^{ika}e^{i2\pi\ell} = e^{ika}. \tag{1.40}$$

Distinct crystal momenta can therefore be represented in the first Brillouin zone,

$$-\frac{\pi}{a} \le k < \frac{\pi}{a}. \tag{1.41}$$

1.7 Three-dimensional form

For lattice vectors $\boldsymbol{R}$, Bloch's theorem becomes

$$\psi_{n\boldsymbol{k}}(\boldsymbol{r} + \boldsymbol{R}) = e^{i\boldsymbol{k}\cdot\boldsymbol{R}}\psi_{n\boldsymbol{k}}(\boldsymbol{r}), \tag{1.42}$$

or

$$\psi_{n\boldsymbol{k}}(\boldsymbol{r}) = e^{i\boldsymbol{k}\cdot\boldsymbol{r}}u_{n\boldsymbol{k}}(\boldsymbol{r}), \qquad u_{n\boldsymbol{k}}(\boldsymbol{r} + \boldsymbol{R}) = u_{n\boldsymbol{k}}(\boldsymbol{r}). \tag{1.43}$$

Reciprocal vectors $\boldsymbol{G}$ satisfy

$$e^{i\boldsymbol{G}\cdot\boldsymbol{R}} = 1, \tag{1.44}$$

so $\boldsymbol{k}$ and $\boldsymbol{k} + \boldsymbol{G}$ represent the same lattice-translation character. The first Brillouin zone contains one distinct $\boldsymbol{k}$ value per primitive cell for each band in a finite periodic crystal.

1.8 Chapter summary

Hard-wall confinement converts the free-particle continuum into discrete standing-wave energies. Periodic boundary conditions produce discrete plane waves and a uniform reciprocal-space spacing. A lattice-periodic Hamiltonian commutes with translation, so its one-electron eigenstates carry translation phases $e^{i\boldsymbol{k}\cdot\boldsymbol{R}}$. Bloch's theorem separates each state into a plane-wave factor and a lattice-periodic factor. A finite crystal supplies one allowed $\boldsymbol{k}$ value per primitive cell in each band. The next chapter shows how coupling localized orbitals produces those Bloch bands and connects finite HOMO–LUMO spectra to bulk valence and conduction bands.

Chapter 2

From HOMO–LUMO Levels to Tight-Binding Bands

A finite molecule has discrete molecular orbitals. A periodic solid has Bloch bands. Tight binding connects these descriptions by coupling localized atomic or molecular orbitals on neighboring sites. One localized level on each of N sites produces N one-electron eigenvalues; as N grows, those closely spaced levels form a band. The same spectral construction identifies bonding and antibonding orbitals in a molecule, HOMO and LUMO levels in a finite system, and valence and conduction bands in a crystal [3, 6, 10].

2.1 Localized orbitals as a basis

Let $|j\rangle$ denote a normalized localized orbital centered on site j. The simplest orthonormal tight-binding basis satisfies

$$\langle i|j\rangle = \delta_{ij}. \tag{2.1}$$

The on-site energy is

$$\epsilon_j = \langle j|\, \hat{H}\, |j\rangle\,, \tag{2.2}$$

and the coupling between distinct sites is

$$H_{ij} = \langle i|\, \hat{H}\, |j\rangle\,. \tag{2.3}$$

A common sign convention writes the nearest-neighbor matrix element as

$$\langle j|\, \hat{H}\, |j+1\rangle = -t, \qquad t > 0. \tag{2.4}$$

The sign convention places the symmetric bonding combination below the antisymmetric combination.

The tight-binding Hamiltonian is a one-electron Hamiltonian expressed in a localized basis. Its eigenvectors are molecular orbitals for a finite cluster and Bloch orbitals for a periodic crystal.

2.2 Two sites: bonding and antibonding orbitals

For two equivalent sites A and B with on-site energy ϵ, the Hamiltonian is

$$H = \begin{pmatrix} \epsilon & -t \\ -t & \epsilon \end{pmatrix}. \tag{2.5}$$

The characteristic equation is

$$\det(H - EI) = (\epsilon - E)^2 - t^2 = 0. \tag{2.6}$$

The eigenvalues are

$$E_{\mathrm{b}} = \epsilon - t, \qquad E_{\mathrm{a}} = \epsilon + t. \tag{2.7}$$

The corresponding normalized eigenvectors are

$$|\mathrm{b}\rangle = \frac{|A\rangle + |B\rangle}{\sqrt{2}}, \tag{2.8}$$

$$|\mathrm{a}\rangle = \frac{|A\rangle - |B\rangle}{\sqrt{2}}. \tag{2.9}$$

The symmetric state has enhanced amplitude between the sites and is the bonding orbital. The antisymmetric state has a node between the sites and is the antibonding orbital. Coupling splits one localized energy ϵ into two molecular-orbital energies separated by

$$\Delta E = 2t. \tag{2.10}$$

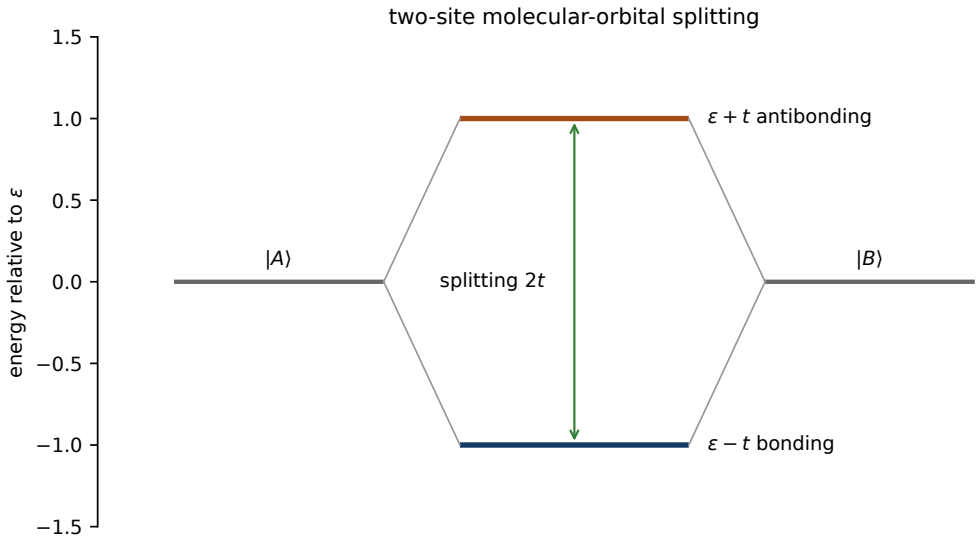


Figure 2.1: Two equivalent localized orbitals at energy ϵ form a lower bonding orbital and an upper antibonding orbital. The hopping magnitude t sets the splitting $2t$.[1]

[1] *Student reproduction notebook:* figures/02_homo-lumo-tight-binding/fig_02_01.ipynb.

2.3 HOMO and LUMO in a finite system

A finite molecule or cluster has discrete one-electron orbitals

$$E_1 \le E_2 \le \cdots \le E_M. \tag{2.11}$$

An independent-electron ground-state filling with a prescribed electron number selects the occupied orbitals. The highest occupied molecular orbital is the HOMO, with energy

$$E_{\mathrm{H}} = E_{N_{\mathrm{occ}}}, \tag{2.12}$$

and the lowest unoccupied molecular orbital is the LUMO,

$$E_{\mathrm{L}} = E_{N_{\mathrm{occ}}+1}. \tag{2.13}$$

The one-electron HOMO–LUMO gap is

$$\boxed{\Delta_{\mathrm{HL}} = E_{\mathrm{L}} - E_{\mathrm{H}}.} \tag{2.14}$$

HOMO and LUMO are occupation labels applied to a finite one-electron spectrum. The spectrum comes from the Hamiltonian; the electron inventory determines which levels receive those labels.

The one-electron HOMO–LUMO difference is a model energy spacing. Optical and charged excitation energies can include relaxation, electron–hole attraction, and many-body corrections. Their evaluation requires the corresponding excitation model.

2.4 A finite chain of localized orbitals

Consider N equivalent sites with open ends. The nearest-neighbor Hamiltonian is

$$\hat{H} = \epsilon \sum_{j=1}^{N} |j\rangle \langle j| - t \sum_{j=1}^{N-1} \left(|j\rangle \langle j+1| + |j+1\rangle \langle j| \right). \tag{2.15}$$

Write an eigenstate as

$$|\psi_r\rangle = \sum_{j=1}^{N} c_j^{(r)} |j\rangle . \tag{2.16}$$

The interior coefficients satisfy

$$-t c_{j-1}^{(r)} + \epsilon c_j^{(r)} - t c_{j+1}^{(r)} = E_r c_j^{(r)}. \tag{2.17}$$

The open boundaries are represented by

$$c_0^{(r)} = c_{N+1}^{(r)} = 0. \tag{2.18}$$

The normalized solutions are standing waves on the site index:

$$c_j^{(r)} = \sqrt{\frac{2}{N+1}} \sin\left(\frac{r\pi j}{N+1} \right), \qquad r = 1, \ldots, N, \tag{2.19}$$

with energies

$$\boxed{E_r = \epsilon - 2t\cos\left(\frac{r\pi}{N+1}\right).} \tag{2.20}$$

One localized orbital per site has produced exactly N molecular orbitals. All energies lie in

$$\epsilon - 2t < E_r < \epsilon + 2t. \tag{2.21}$$

The width approaches

$$W = 4t \tag{2.22}$$

as N increases.

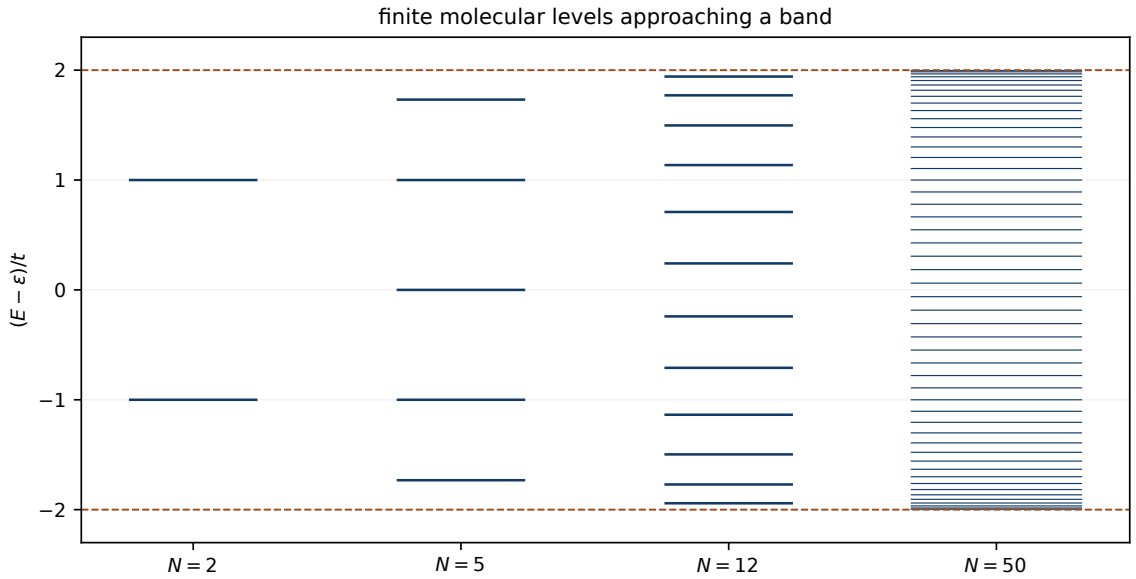


Figure 2.2: One localized level on each site produces N molecular-orbital energies. Increasing N fills the finite interval $\epsilon - 2t < E < \epsilon + 2t$ with progressively closer levels.[2]

2.5 The periodic chain and the Bloch basis

Join the final site to the first and label sites modulo N. The Hamiltonian is

$$\hat{H} = \epsilon \sum_{j=0}^{N-1} |j\rangle\langle j| - t \sum_{j=0}^{N-1} \left(|j\rangle\langle j+1| + |j+1\rangle\langle j|\right). \tag{2.23}$$

The normalized Bloch basis is

$$\boxed{|k\rangle = \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} e^{ikja} |j\rangle.} \tag{2.24}$$

The allowed wave vectors satisfy

$$k = \frac{2\pi m}{Na}. \tag{2.25}$$

[2] *Student reproduction notebook:* figures/02_homo-lumo-tight-binding/fig_02_02.ipynb.

Acting with the Hamiltonian gives

$$\begin{aligned}\hat{H}\,|k\rangle &= \epsilon\,|k\rangle - te^{ika}\,|k\rangle - te^{-ika}\,|k\rangle \\ &= \left[\epsilon - 2t\cos(ka)\right]|k\rangle\,.\end{aligned} \tag{2.26}$$

Therefore

$$\boxed{E(k) = \epsilon - 2t\cos(ka).} \tag{2.27}$$

The Bloch transformation is the discrete Fourier transformation from localized site orbitals to translation eigenstates. It diagonalizes every translation-invariant one-orbital nearest-neighbor Hamiltonian.

For finite N, equation (2.27) is sampled at N discrete k values. In the thermodynamic limit those values become dense in the first Brillouin zone and the molecular-orbital levels become a continuous band.

2.6 Several orbitals per cell

Let each primitive cell contain q localized orbitals labeled by α. A Bloch basis is formed for each orbital:

$$|\alpha, \boldsymbol{k}\rangle = \frac{1}{\sqrt{N}} \sum_{\boldsymbol{R}} e^{i\boldsymbol{k}\cdot\boldsymbol{R}}\,|\alpha, \boldsymbol{R}\rangle\,. \tag{2.28}$$

The Hamiltonian becomes a $q \times q$ matrix at each $\boldsymbol{k}$:

$$H_{\alpha\beta}(\boldsymbol{k}) = \sum_{\boldsymbol{R}} e^{i\boldsymbol{k}\cdot\boldsymbol{R}}\,\langle\alpha, \boldsymbol{0}|\,\hat{H}\,|\beta, \boldsymbol{R}\rangle\,. \tag{2.29}$$

Diagonalization produces q bands,

$$H(\boldsymbol{k})u_{n\boldsymbol{k}} = E_n(\boldsymbol{k})u_{n\boldsymbol{k}}, \qquad n = 1, \ldots, q. \tag{2.30}$$

One localized orbital per primitive cell produces one band; q localized orbitals per cell produce q bands before additional spin multiplicity is counted.

2.7 A two-orbital cell and a band gap

A chain with two orbitals A and B per cell provides the simplest connection between molecular bonding–antibonding splitting and separated solid-state bands. Let t_1 couple A and B inside one cell and t_2 couple neighboring cells. For equal on-site energy ϵ, the Bloch Hamiltonian is

$$H(k) = \begin{pmatrix} \epsilon & -(t_1 + t_2 e^{-ika}) \\ -(t_1 + t_2 e^{ika}) & \epsilon \end{pmatrix}. \tag{2.31}$$

The characteristic equation gives

$$(E - \epsilon)^2 = |t_1 + t_2 e^{ika}|^2 = t_1^2 + t_2^2 + 2t_1 t_2 \cos(ka). \tag{2.32}$$

The two bands are

$$\boxed{E_\pm(k) = \epsilon \pm \sqrt{t_1^2 + t_2^2 + 2t_1 t_2 \cos(ka)}.} \tag{2.33}$$

At the Brillouin-zone boundary $k = \pi/a$,

$$E_{\pm}\left(\frac{\pi}{a}\right) = \epsilon \pm |t_1 - t_2|, \tag{2.34}$$

so the band gap is

$$\boxed{E_g = 2|t_1 - t_2|.} \tag{2.35}$$

The finite two-site bonding and antibonding levels have become lower and upper Bloch bands. Alternating couplings separate those bands by a gap.

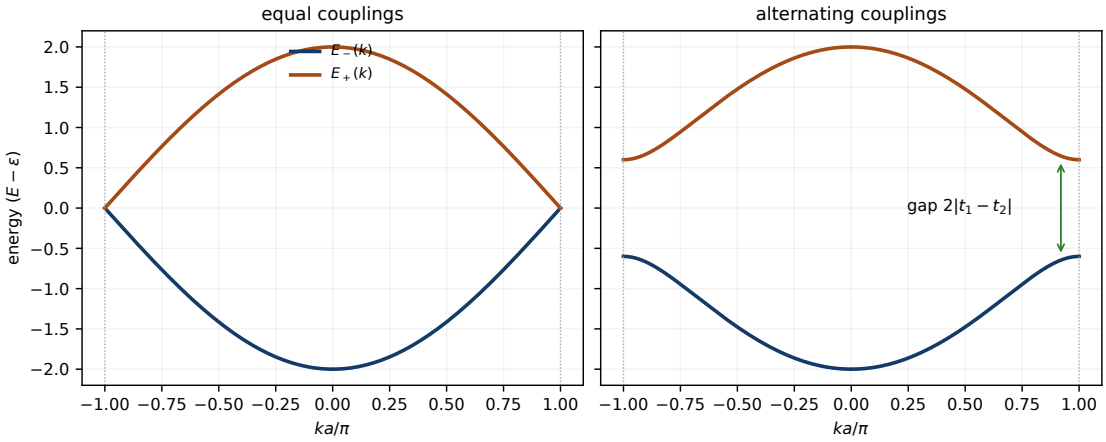


Figure 2.3: Two orbitals per cell produce two Bloch bands. Equal couplings close the gap at the zone boundary; unequal intracell and intercell couplings open the gap $2|t_1 - t_2|$.[3]

2.8 From HOMO and LUMO to valence and conduction bands

For a finite chain, the electron inventory divides the discrete molecular orbitals into occupied and empty sets. Their boundary is the HOMO–LUMO gap. As the number of cells grows, each family of molecular levels becomes a Bloch band. A completely occupied lower band and an empty upper band define a valence band and conduction band separated by a band gap.

A single spin-degenerate orbital per cell produces one band containing two one-electron states per cell. Its electronic character depends on filling. A partially filled band has available states at the occupation boundary. A completely filled band requires another band to define the lowest empty states and a possible gap. The number of orbitals per cell, the hopping matrix, and the electron inventory jointly determine whether the independent-electron spectrum is metallic or insulating.

The bulk analogues of HOMO and LUMO are the valence-band maximum and conduction-band minimum:

$$E_V = \max_{n \in \text{valence}, \boldsymbol{k}} E_n(\boldsymbol{k}), \tag{2.36}$$

$$E_C = \min_{n \in \text{conduction}, \boldsymbol{k}} E_n(\boldsymbol{k}). \tag{2.37}$$

[3] *Student reproduction notebook:* `figures/02_homo-lumo-tight-binding/fig_02_03.ipynb`.

The fundamental independent-particle gap is

$$\boxed{E_g = E_C - E_V.} \tag{2.38}$$

The extrema can occur at the same or different wave vectors, producing direct or indirect gaps.

2.9 Model content

Tight binding preserves the localized-orbital basis, on-site energies, selected hopping amplitudes, lattice symmetry, and Bloch phase relations. Additional orbitals and longer-range hopping extend the spectral range and shape of the bands. A nonorthogonal localized basis leads to the generalized eigenproblem

$$H(\boldsymbol{k})c_{n\boldsymbol{k}} = E_n(\boldsymbol{k})S(\boldsymbol{k})c_{n\boldsymbol{k}}, \tag{2.39}$$

where $S(\boldsymbol{k})$ is the overlap matrix. Spin–orbit coupling, magnetism, and interaction corrections enlarge the Hamiltonian according to the physical system.

The tight-binding parameters define a reduced one-electron model. Their connection to measured or higher-level calculated bands requires an explicit parameter source and a stated fitting range.

2.10 Chapter summary

Coupling two localized orbitals produces bonding and antibonding molecular orbitals. Coupling N sites produces N molecular levels within a finite energy interval. Periodic translation diagonalizes the Hamiltonian in a Bloch basis and converts those levels into the band $E(k) = \epsilon - 2t\cos(ka)$. Several orbitals per cell produce several bands. A two-orbital cell shows explicitly how molecular bonding–antibonding splitting becomes a valence–conduction band separation. HOMO and LUMO label the occupied boundary of a finite spectrum; valence- and conduction-band edges provide the bulk counterpart. The next chapter uses the spectral representation of these Bloch bands to derive their density of states.

Chapter 3

Spectral Representation and the Density of States

A one-electron Hamiltonian determines its available states through its spectral representation. The density of states is the spectral multiplicity per energy and per physical size. For a periodic crystal, Bloch's theorem resolves the spectrum into bands $E_n(\boldsymbol{k})$, and reciprocal-space integration converts those bands into an energy density. Statistical mechanics enters in the next chapter to determine occupation.

The spectral and state-counting development follows Kittel and the semiconductor treatments of Sze and Lee, Sze and Ng, and Rudan [3, 6–8]. The analytic representation also defines the quantities evaluated numerically later in the chapter.

3.1 Spectral representation and discrete multiplicity

Let the finite one-electron Hamiltonian $\hat{H}$ have distinct eigenvalues E_j. An orthonormal basis for the eigenspace at E_j is

$$\{|j,a\rangle : a = 1, \ldots, M_j\}, \tag{3.1}$$

where M_j is the multiplicity of that eigenvalue. The projector onto the complete eigenspace is

$$\hat{P}_j = \sum_{a=1}^{M_j} |j,a\rangle \langle j,a| . \tag{3.2}$$

Its trace equals the multiplicity:

$$M_j = \dim \operatorname{Ran} \hat{P}_j = \operatorname{Tr} \hat{P}_j. \tag{3.3}$$

The spectral representations of the Hamiltonian and identity are

$$\boxed{\hat{H} = \sum_j E_j \hat{P}_j, \qquad \hat{I} = \sum_j \hat{P}_j.} \tag{3.4}$$

A function of the Hamiltonian acts on each eigenspace through its eigenvalue:

$$F(\hat{H}) = \sum_j F(E_j) \hat{P}_j. \tag{3.5}$$

For the spectral delta distribution,

$$\delta(E - \hat{H}) = \sum_j \delta(E - E_j)\hat{P}_j. \tag{3.6}$$

Taking the trace gives the discrete density of states:

$$\boxed{G(E) = \operatorname{Tr} \delta(E - \hat{H}) = \sum_j M_j \delta(E - E_j).} \tag{3.7}$$

This trace is basis independent. It counts the full eigenspace at each energy and is determined by the spectrum of $\hat{H}$. Integrating across an interval counts the states inside it:

$$\int_{E_a}^{E_b} G(E)\, \mathrm{d}E = \#\{\alpha : E_a \le E_\alpha \le E_b\}, \tag{3.8}$$

with the usual endpoint convention understood.

A finite energy measurement has a resolution ΔE. If energy bin b contains M_b states, the state density in that bin is

$$g(E_b) \simeq \frac{M_b}{\mathcal{V}_d\, \Delta E}, \tag{3.9}$$

where $\mathcal{V}_d$ is the system's length, area, or volume in d dimensions. As the physical system becomes large, neighboring levels become closely spaced. The discrete multiplicities within narrow energy intervals then approach a smooth density:

$$\boxed{g(E) = \lim_{\Delta E \to 0} \frac{M(E, E + \Delta E)}{\mathcal{V}_d\, \Delta E}.} \tag{3.10}$$

The limit is understood together with the thermodynamic limit: the physical system grows so that every small but finite energy interval contains enough closely spaced states to define a density.

The cumulative or integrated density of states is the trace of the spectral projector below E:

$$\mathcal{N}(E) = \frac{1}{\mathcal{V}_d} \operatorname{Tr} \Theta(E - \hat{H}) = \frac{1}{\mathcal{V}_d} \sum_j M_j \Theta(E - E_j). \tag{3.11}$$

It is a staircase for a finite system. The height of each step is the multiplicity of the corresponding energy, divided by physical size. In the continuum limit the steps become dense and

$$g(E) = \frac{\mathrm{d}\mathcal{N}}{\mathrm{d}E}. \tag{3.12}$$

This discrete-to-continuous construction is the meaning of "density" in density of states.

Definition 3.1: Density of states

For a system of physical size $\mathcal{V}_d$ in d spatial dimensions, the density of states $g(E)$ is defined so that

$$g(E)\, \mathrm{d}E \tag{3.13}$$

is the number of one-particle states per $\mathcal{V}_d$ with energies in $[E, E + \mathrm{d}E]$. Thus $g(E)$ is the continuum form of discrete level multiplicity and describes state availability.

3.2 Bloch-band information and state density

For a periodic effective one-electron Hamiltonian, Bloch's theorem gives

$$\psi_{n\boldsymbol{k}}(\boldsymbol{r}) = e^{i\boldsymbol{k}\cdot\boldsymbol{r}} u_{n\boldsymbol{k}}(\boldsymbol{r}), \qquad u_{n\boldsymbol{k}}(\boldsymbol{r}+\boldsymbol{R}) = u_{n\boldsymbol{k}}(\boldsymbol{r}), \tag{3.14}$$

and the Bloch eigenvalue problem gives

$$H(\boldsymbol{k}) u_{n\boldsymbol{k}} = E_n(\boldsymbol{k}) u_{n\boldsymbol{k}}. \tag{3.15}$$

For a finite periodic crystal, the spectral representation is

$$\hat{H} = \sum_{n,\boldsymbol{k}} E_n(\boldsymbol{k}) \left|n, \boldsymbol{k}\right\rangle \left\langle n, \boldsymbol{k}\right|, \tag{3.16}$$

with an additional sum or degeneracy factor for spin. Its spectral trace is

$$G(E) = g_s \sum_{n,\boldsymbol{k}} \delta[E - E_n(\boldsymbol{k})]. \tag{3.17}$$

The general object used to calculate the DOS is the resulting Bloch band $E_n(\boldsymbol{k})$. Plane-wave, tight-binding, $\boldsymbol{k}\cdot\boldsymbol{p}$, and first-principles Hamiltonians are different models for calculating that Bloch band. The nearest-neighbor tight-binding band introduced later is an exactly solvable Bloch-band example.

A band dispersion is a function

$$E_n : \boldsymbol{k} \longmapsto E_n(\boldsymbol{k}), \tag{3.18}$$

where n labels the band and $\boldsymbol{k}$ lies in the first Brillouin zone. A graph of $E_n(\boldsymbol{k})$ shows how energy changes with Bloch wave vector. The density of states supplies the corresponding state count on the energy axis. Two portions of a band can span the same energy interval while contributing very different numbers of states.

The distinction becomes visible in one dimension. Suppose neighboring allowed wave vectors are separated by Δk. If a band is steep, successive states differ substantially in energy:

$$\Delta E \simeq \left|\frac{\mathrm{d}E}{\mathrm{d}k}\right| \Delta k. \tag{3.19}$$

The number of states per energy interval is consequently proportional to

$$\frac{\Delta k}{\Delta E} \simeq \frac{1}{|\,\mathrm{d}E/\,\mathrm{d}k|}. \tag{3.20}$$

A flat band has many nearby wave vectors within a small energy interval and therefore a large density of states. A steep band has a smaller density of states.

For bulk material, $\mathcal{V}_3$ is a volume and $g(E)$ commonly has units $1/(\mathrm{eV\,cm^3})$. In a two-dimensional sheet it is quoted per area, and in a one-dimensional channel it is quoted per length. A density of states "per primitive cell" converts to a volumetric density through the primitive-cell volume before combination with concentrations per cubic centimetre.

3.3 Allowed wave vectors in a finite periodic crystal

Begin with a one-dimensional crystal of length $L = N_a a$, where a is the lattice period and N_a is the number of primitive cells. Impose periodic boundary conditions on a Bloch factor:

$$e^{ik(x+L)} = e^{ikx}. \tag{3.21}$$

Cancelling e^{ikx} gives

$$e^{ikL} = 1, \tag{3.22}$$

which requires

$$k_m = \frac{2\pi m}{L}, \qquad m \in \mathbb{Z}. \tag{3.23}$$

Neighboring values are separated by

$$\Delta k = \frac{2\pi}{L}. \tag{3.24}$$

Consequently, the number of allowed k values per interval $\mathrm{d}k$ is

$$\mathrm{d}N_k = \frac{L}{2\pi}\,\mathrm{d}k. \tag{3.25}$$

Including a spin degeneracy g_s multiplies this count by g_s.

For a periodic rectangular crystal with $V = L_x L_y L_z$, the allowed vectors are

$$k_i = \frac{2\pi m_i}{L_i}, \qquad i \in \{x, y, z\}, \tag{3.26}$$

and one allowed $\boldsymbol{k}$ point occupies reciprocal-space volume

$$\Delta^3 k = \Delta k_x \Delta k_y \Delta k_z = \frac{(2\pi)^3}{V}. \tag{3.27}$$

Thus

$$\mathrm{d}N_k = g_s \frac{V}{(2\pi)^3}\,\mathrm{d}^3 k. \tag{3.28}$$

The artificial finite crystal makes the allowed states discrete. In the thermodynamic limit, $L_i \to \infty$, the spacings Δk_i approach zero and the sum over allowed vectors becomes an integral. The state-counting factor $V/(2\pi)^3$ remains.

Model check

The first Brillouin zone contains one allowed $\boldsymbol{k}$ point per primitive cell for each band, before spin is counted. Therefore one nondegenerate band integrated over the complete Brillouin zone contains one state per primitive cell; with ordinary spin degeneracy it contains two. This provides a global normalization check for every calculated density of states.

3.4 The general density of states of a band

The Dirac delta distribution provides a concise way to select the wave vectors whose energy equals E. For bands $E_n(\boldsymbol{k})$, the density of states per unit volume is

$$\boxed{g(E) = \frac{g_s}{(2\pi)^3} \sum_n \int_{\mathrm{BZ}} \delta[E - E_n(\boldsymbol{k})]\,\mathrm{d}^3 k.} \tag{3.29}$$

The integral is over the first Brillouin zone. The band sum is included because different bands can contribute states at the same energy.

To verify the meaning of this expression, integrate it over an energy interval $[E_1, E_2]$:

$$\int_{E_1}^{E_2} g(E)\,\mathrm{d}E = \frac{g_s}{(2\pi)^3}\sum_n \int_{\mathrm{BZ}} \mathrm{d}^3k \int_{E_1}^{E_2} \mathrm{d}E\,\delta[E - E_n(\boldsymbol{k})]\,. \tag{3.30}$$

The inner integral is one if $E_n(\boldsymbol{k})$ lies inside the interval and zero otherwise. Equation (3.30) therefore counts exactly the reciprocal-space states in that energy window.

3.5 Constant-energy surfaces and band velocity

For a regular energy value, the set

$$S_n(E) = \{\boldsymbol{k} : E_n(\boldsymbol{k}) = E\} \tag{3.31}$$

is a surface in reciprocal space. The multidimensional delta-function identity converts the Brillouin-zone integral into a surface integral:

$$\boxed{g(E) = \frac{g_s}{(2\pi)^3}\sum_n \int_{S_n(E)} \frac{\mathrm{d}S}{|\nabla_{\boldsymbol{k}} E_n(\boldsymbol{k})|}.} \tag{3.32}$$

This equation combines the area $\mathrm{d}S$ of the constant-energy surface with the inverse rate $1/|\nabla_{\boldsymbol{k}} E_n|$ at which energy changes normal to that surface. The semiclassical group velocity is

$$\boldsymbol{v}_n(\boldsymbol{k}) = \frac{1}{\hbar}\nabla_{\boldsymbol{k}} E_n(\boldsymbol{k}). \tag{3.33}$$

Consequently, slowly varying portions of a band contribute strongly to the density of states. At a critical point where $\nabla_{\boldsymbol{k}} E_n = 0$, the surface formula signals a nonanalytic feature called a van Hove singularity. Its detailed form depends on dimensionality and on whether the critical point is a minimum, maximum, or saddle point.

3.6 Exact example: a one-dimensional tight-binding band

Consider one orbital per primitive cell with nearest-neighbor hopping $t > 0$:

$$E(k) = E_0 - 2t\cos(ka), \qquad -\frac{\pi}{a} \le k < \frac{\pi}{a}. \tag{3.34}$$

The allowed energies occupy the finite interval

$$E_0 - 2t \le E \le E_0 + 2t. \tag{3.35}$$

Differentiate the dispersion:

$$\frac{\mathrm{d}E}{\mathrm{d}k} = 2ta\sin(ka). \tag{3.36}$$

For an energy strictly inside the band, there are two wave-vector roots with equal derivative magnitude. The one-dimensional delta-function identity

$$\delta[E - E(k)] = \sum_{k_i : E(k_i) = E} \frac{\delta(k - k_i)}{|\,\mathrm{d}E/\,\mathrm{d}k|_{k_i}} \tag{3.37}$$

gives, per primitive cell,

$$g(E) = \frac{g_s}{2\pi} \sum_{k_i} \frac{a}{|\,\mathrm{d}E/\,\mathrm{d}k|_{k_i}}. \tag{3.38}$$

From

$$\cos(ka) = \frac{E_0 - E}{2t}, \tag{3.39}$$

it follows that

$$|\sin(ka)| = \sqrt{1 - \left(\frac{E - E_0}{2t}\right)^2}. \tag{3.40}$$

Substitution of the two roots gives

$$\boxed{g(E) = \frac{g_s}{\pi\sqrt{4t^2 - (E - E_0)^2}}, \qquad |E - E_0| < 2t.} \tag{3.41}$$

The density of states diverges at both band edges because the band slope vanishes there. The divergence is integrable: the total number of states is finite,

$$\int_{E_0 - 2t}^{E_0 + 2t} g(E)\ \mathrm{d}E = g_s. \tag{3.42}$$

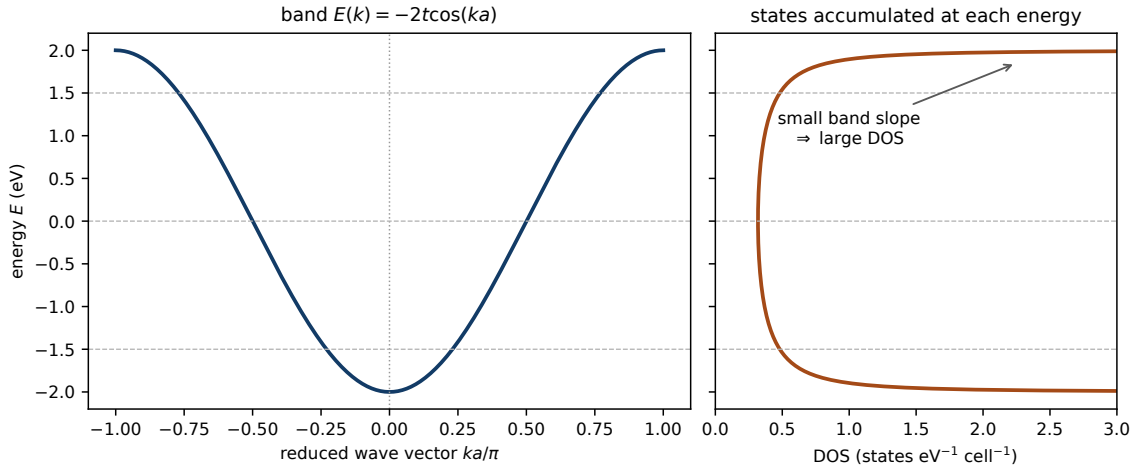


Figure 3.1: The same one-dimensional tight-binding band shown as a dispersion and as a density of states. Near a band edge, many neighboring k states occupy a narrow energy interval because $|\,\mathrm{d}E/\,\mathrm{d}k|$ is small.[1]

[1] *Student reproduction notebook:* figures/03_spectral-density-of-states/fig_03_01.ipynb.

3.7 Parabolic bands in one, two, and three dimensions

Near a nondegenerate isotropic minimum, approximate the band by

$$E(\boldsymbol{k}) = E_0 + \frac{\hbar^2 k^2}{2m^*}. \tag{3.43}$$

Define the excess energy

$$\varepsilon = E - E_0 \geq 0. \tag{3.44}$$

The constant-energy radius is

$$k(\varepsilon) = \frac{\sqrt{2m^*\varepsilon}}{\hbar}. \tag{3.45}$$

In d dimensions, the volume of a radius-k ball is

$$\Omega_d(k) = \frac{\pi^{d/2}}{\Gamma(d/2+1)} k^d. \tag{3.46}$$

The number of states per physical d-dimensional measure below energy $E_0 + \varepsilon$ is therefore

$$\frac{\mathcal{N}_d(\varepsilon)}{\mathcal{V}_d} = \frac{g}{(2\pi)^d} \frac{\pi^{d/2}}{\Gamma(d/2+1)} \left(\frac{2m^*\varepsilon}{\hbar^2}\right)^{d/2}, \tag{3.47}$$

where g contains the declared spin, valley, and band degeneracies. Differentiating with respect to energy gives

$$g_d(E) \propto (E - E_0)^{d/2-1} \Theta(E - E_0). \tag{3.48}$$

Dimensionality determines the band-edge power law.

3.7.1 One dimension

For $d = 1$,

$$\boxed{g_{1\mathrm{D}}(E) = \frac{g}{\pi\hbar} \sqrt{\frac{m^*}{2(E - E_0)}} \Theta(E - E_0).} \tag{3.49}$$

The inverse-square-root edge divergence is the parabolic version of the one-dimensional van Hove singularity.

3.7.2 Two dimensions

For $d = 2$,

$$\boxed{g_{2\mathrm{D}}(E) = \frac{g m^*}{2\pi\hbar^2} \Theta(E - E_0).} \tag{3.50}$$

Each ideal two-dimensional parabolic subband contributes a constant step. Quantum wells consequently exhibit a staircase-like density of states when several confinement subbands are included.

3.7.3 Three dimensions

For $d = 3$,

$$\boxed{g_{3\mathrm{D}}(E) = \frac{g}{4\pi^2}\left(\frac{2m^*}{\hbar^2}\right)^{3/2}\sqrt{E - E_0}\,\Theta(E - E_0).} \tag{3.51}$$

For spin degeneracy $g_s = 2$ and no additional valley degeneracy, this becomes

$$g_{3\mathrm{D}}(E) = \frac{1}{2\pi^2}\left(\frac{2m^*}{\hbar^2}\right)^{3/2}\sqrt{E - E_0}\,\Theta(E - E_0). \tag{3.52}$$

This is the square-root form used for bulk conduction and valence bands.

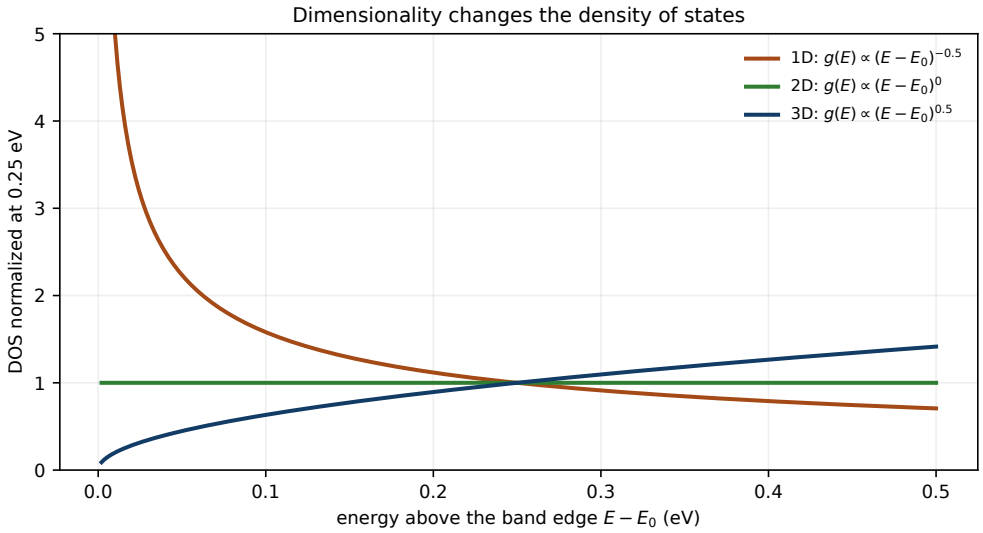


Figure 3.2: The parabolic density of states depends qualitatively on spatial dimensionality. Normalization at a common energy places the three shapes on one dimensionless scale.[2]

3.8 Anisotropic valleys and the density-of-states mass

Near an anisotropic three-dimensional minimum, choose principal axes so that

$$E(\boldsymbol{q}) = E_0 + \frac{\hbar^2}{2}\left(\frac{q_1^2}{m_1} + \frac{q_2^2}{m_2} + \frac{q_3^2}{m_3}\right). \tag{3.53}$$

The constant-energy surface is an ellipsoid with semiaxes

$$q_i^{\max} = \frac{\sqrt{2m_i\varepsilon}}{\hbar}. \tag{3.54}$$

Its reciprocal-space volume is

$$\begin{aligned}\Omega_k(\varepsilon) &= \frac{4\pi}{3}q_1^{\max}q_2^{\max}q_3^{\max} \\ &= \frac{4\pi}{3}\left(\frac{2\varepsilon}{\hbar^2}\right)^{3/2}\sqrt{m_1m_2m_3}.\end{aligned} \tag{3.55}$$

[2] *Student reproduction notebook:* figures/03_spectral-density-of-states/fig_03_02.ipynb.

Define

$$\boxed{m_d = (m_1 m_2 m_3)^{1/3}.} \tag{3.56}$$

Then $m_d^{3/2} = \sqrt{m_1 m_2 m_3}$, so the anisotropic state count has the same form as an isotropic calculation with mass m_d.

This replacement preserves the reciprocal-space volume enclosed by a constant- energy surface. Direction-dependent velocity and conductivity retain the full effective-mass tensor. The density-of-states mass governs state counting; the transport mass governs directional response.

If there are g_v equivalent valleys, their contributions add:

$$g_c(E) = \frac{g_s g_v}{4\pi^2} \left(\frac{2m_d}{\hbar^2}\right)^{3/2} \sqrt{E - E_C}\, \Theta(E - E_C). \tag{3.57}$$

A mass convention that incorporates g_v already represents the complete valley count.

3.9 Several bands at the same energy

The total density of states is additive:

$$g(E) = \sum_n g_n(E). \tag{3.58}$$

For heavy- and light-hole parabolic bands sharing a valence-band maximum,

$$g_v(E) = \frac{g_s}{4\pi^2} \left(\frac{2}{\hbar^2}\right)^{3/2} \left(m_{hh}^{3/2} + m_{lh}^{3/2}\right) \sqrt{E_V - E}. \tag{3.59}$$

It is convenient to define a combined mass $m_{v,d}$ by

$$m_{v,d}^{3/2} = m_{hh}^{3/2} + m_{lh}^{3/2}, \tag{3.60}$$

or

$$m_{v,d} = \left(m_{hh}^{3/2} + m_{lh}^{3/2}\right)^{2/3}. \tag{3.61}$$

This combined mass is a compact representation of the additive state count of two distinct physical bands.

3.10 Calculating a density of states from sampled bands

An analytic expression is rarely available for an entire realistic band structure. Suppose band energies have been calculated on wave vectors $\boldsymbol{k}_j$ with normalized reciprocal-space weights w_j:

$$\sum_j w_j = 1. \tag{3.62}$$

The discrete approximation to the DOS per primitive cell is

$$g(E) \simeq \sum_n g_n \sum_j w_j \delta[E - E_n(\boldsymbol{k}_j)]\,, \tag{3.63}$$

where g_n is the degeneracy assigned to band n.

A finite sample of wave vectors produces delta peaks. To display a continuous curve, replace each delta distribution by a normalized kernel. For a Gaussian,

$$\delta_\sigma(E - E_j) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(E - E_j)^2}{2\sigma^2}\right]. \tag{3.64}$$

Because

$$\int_{-\infty}^{\infty} \delta_\sigma(E - E_j)\, \mathrm{d}E = 1, \tag{3.65}$$

the integrated state count is preserved when the energy window contains the broadened tails.

The width σ defines the numerical energy resolution. A separate scattering model is required to interpret a width as a physical lifetime. Large σ values smooth spectral features, while small values resolve the discreteness of the sampled $\boldsymbol{k}$ mesh. Convergence requires coordinated refinement of reciprocal-space sampling and energy resolution.

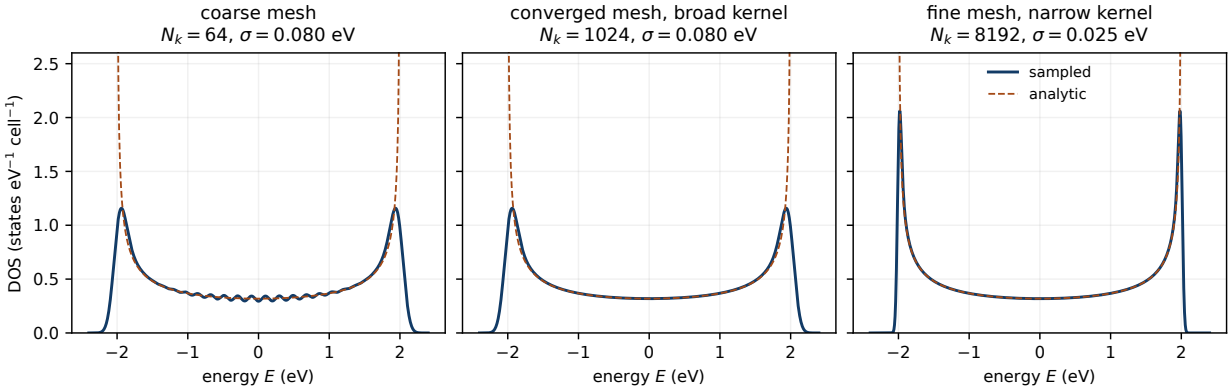


Figure 3.3: Convergence of a DOS calculated from sampled tight-binding energies. A coarse reciprocal-space mesh produces visible sampling structure; excessive energy smoothing rounds the band-edge singularity; a fine mesh with a narrower normalized kernel approaches the analytic result.[3]

An alternative is to partition energy into bins. If bin b has width ΔE_b, then

$$g(E_b) \simeq \frac{1}{\Delta E_b} \sum_{n,j\,:\,E_n(\boldsymbol{k}_j)\in b} g_n w_j. \tag{3.66}$$

Division by bin width converts state counts into a density of states. The tetrahedron method in three dimensions instead uses local interpolation of the bands to perform the Brillouin-zone integral.

3.11 Normalization, units, and convergence

Every DOS calculation should satisfy a state-count sum rule. For a set of complete bands represented per primitive cell,

$$\boxed{\int_{-\infty}^{\infty} g(E)\, \mathrm{d}E = \sum_n g_n.} \tag{3.67}$$

[3] *Student reproduction notebook:* figures/03_spectral-density-of-states/fig_03_03.ipynb.

For a DOS per volume, the right side is divided by the primitive-cell volume. A partial band or partial Brillouin-zone domain carries the corresponding fractional state count.

A second check compares the integrated DOS

$$\mathcal{N}(E) = \int_{-\infty}^{E} g(E')\, \mathrm{d}E' \tag{3.68}$$

with an independent count of states below E. Unlike the DOS, which can be singular or noisy, $\mathcal{N}(E)$ is monotonic and often reveals normalization errors clearly.

A third check varies the physical and numerical resolution. The wave-vector mesh is refined together with the energy-bin or smoothing width, and the energy interval is expanded until the normalization integral is stable. The final integral uses explicit energy and spatial units and counts every degeneracy once.

3.12 Computational laboratory: bands, state counting, and DOS

The three figure notebooks in

`figures/03_spectral-density-of-states/`

are organized by physical relation. The first constructs a sampled one-dimensional band and verifies its integrated state count, the second compares the dimensional band-edge power laws, and the third examines reciprocal-space and energy resolution. Each notebook presents its direct NumPy calculation before calling the reusable implementation that writes the published figure.

The reusable numerical expressions are collected in `figures/src/semphy/density_of_states.py`, with figure recipes in the same package and tests in `figures/tests/semphy/`. Every numerical result is compared with an analytic limit or a state-count sum rule.

3.13 Chapter summary

The density of states is the spectral trace $\mathcal{V}_d^{-1} \operatorname{Tr} \delta(E - \hat{H})$. The trace counts the multiplicity of each eigenspace. Bloch's theorem resolves a periodic Hamiltonian into bands, and periodic boundary conditions establish the reciprocal-space state density. The delta-function expression converts the band map $E_n(\boldsymbol{k})$ into an energy density, while its constant-energy-surface form explains why slowly varying bands and critical points produce large or singular contributions. Parabolic bands give inverse-square-root, constant, and square-root behavior in one, two, and three dimensions, respectively. Anisotropy enters the three-dimensional state count through the geometric mean of the principal masses. For sampled bands, reciprocal-space weights, energy resolution, degeneracy, normalization, and convergence must all be declared. The next chapter applies statistical mechanics to determine the occupation of these available states.

Chapter 4

Statistical Mechanics of Carrier Populations

The preceding chapter determined the one-particle states available at each energy. A carrier calculation also requires the equilibrium occupation probability of every state. The partition function and the macroscopic constraints determine that probability.

The purpose of this chapter is to derive the sequence

$$\text{microstates} \longrightarrow \text{partition function} \longrightarrow \text{occupation} \longrightarrow n, p \longrightarrow \text{charge neutrality} \longrightarrow \mu. \tag{4.1}$$

This provides the statistical foundation for the intrinsic and extrinsic semiconductors treated in the following chapters [3, 6–8].

4.1 From equipartition to a probability distribution

A familiar starting point in thermal physics is the equipartition theorem. In the classical regime, every independent quadratic term in the energy has mean value

$$\left\langle ax^2 \right\rangle = \frac{1}{2} k_{\mathrm{B}} T. \tag{4.2}$$

A free particle has three quadratic translational contributions,

$$E_{\mathrm{kin}} = \frac{p_x^2}{2m} + \frac{p_y^2}{2m} + \frac{p_z^2}{2m}, \tag{4.3}$$

so

$$\langle E_{\mathrm{kin}} \rangle = \frac{3}{2} k_{\mathrm{B}} T. \tag{4.4}$$

This gives an intuitive meaning to temperature: thermal energy is distributed among accessible degrees of freedom.

Equipartition applies to classical continuous quadratic degrees of freedom. A band gap is a finite quantum excitation energy, and a degree of freedom becomes thermally frozen when its energy spacing exceeds $k_{\mathrm{B}} T$. Semiconductor carrier populations therefore require state probabilities for one electron and mean state occupations for many electrons.

The partition function assigns equilibrium probabilities to microscopic states with different energies. Its classical quadratic limit reproduces equipartition.

4.2 Microstates, macrostates, and constraints

A *microstate* specifies the complete microscopic state of the system. For one electron in a crystal, a microstate is one particular band state, which may be labeled by band index, wave vector, and spin:

$$\alpha = (n, \boldsymbol{k}, s), \qquad E_\alpha = E_{ns}(\boldsymbol{k}). \tag{4.5}$$

The band calculation supplies this list of one-electron energies. A canonical partition function describes one electron on that spectrum; a many-electron occupation law is introduced separately.

A *macrostate* specifies macroscopic constraints such as temperature, volume, and total charge. Statistical mechanics assigns probabilities to the microstates compatible with those constraints.

The permitted exchanges determine the equilibrium distribution. An isolated system has fixed energy and particle number. A canonical system exchanges energy with a reservoir at fixed particle number. A grand-canonical system exchanges energy and particles and is specified by T, V, and chemical potential μ.

A one-electron band and a many-electron population are different levels of description. The canonical partition function describes one electron thermally distributed over the one-electron states. Fermi–Dirac statistics describes many electrons distributed over that same list of states. The grand-canonical description then assigns a mean occupation to each one-electron state, while the chemical potential is adjusted so that the total expectation obeys the specimen's fixed charge.

4.3 From reservoir multiplicity to the partition function

Consider a small system in thermal contact with a much larger reservoir. The combined system is isolated with fixed energy E_{tot}. If the small system occupies microstate s with energy E_s, the reservoir has energy

$$E_R = E_{\mathrm{tot}} - E_s. \tag{4.6}$$

Let $\Omega_R(E_R)$ be the number of reservoir microstates available at that energy. Every combined microstate is equally probable, so the probability of small-system state s is proportional to the reservoir multiplicity:

$$\mathbb{P}_s \propto \Omega_R(E_{\mathrm{tot}} - E_s). \tag{4.7}$$

Because the reservoir is large and $E_s \ll E_{\mathrm{tot}}$, expand the logarithm of its multiplicity:

$$\begin{aligned} \ln \Omega_R(E_{\mathrm{tot}} - E_s) \simeq\ & \ln \Omega_R(E_{\mathrm{tot}}) \\ & - E_s \left.\frac{\partial \ln \Omega_R}{\partial E_R}\right|_{E_{\mathrm{tot}}}. \end{aligned} \tag{4.8}$$

Use Boltzmann's entropy relation

$$S_R = k_{\mathrm{B}} \ln \Omega_R \tag{4.9}$$

and the thermodynamic definition

$$\frac{1}{T} = \left(\frac{\partial S_R}{\partial E_R}\right)_{V,N}. \tag{4.10}$$

Then

$$\frac{\partial \ln \Omega_R}{\partial E_R} = \frac{1}{k_{\mathrm{B}} T} \equiv \beta. \tag{4.11}$$

Exponentiating equation (4.8) gives

$$\Omega_R(E_{\text{tot}} - E_s) \propto e^{-\beta E_s}. \tag{4.12}$$

The Boltzmann factor therefore follows from the multiplicity of the reservoir.

For a system with fixed particle number whose microstates have energies E_s, normalization gives the canonical probability

$$\mathbb{P}_s = \frac{e^{-\beta E_s}}{Z}, \qquad \beta = \frac{1}{k_{\text{B}}T}, \tag{4.13}$$

where

$$\boxed{Z(T, V, N) = \sum_s e^{-\beta E_s}} \tag{4.14}$$

is the canonical partition function. The normalization follows immediately:

$$\sum_s \mathbb{P}_s = \frac{1}{Z} \sum_s e^{-\beta E_s} = 1. \tag{4.15}$$

For any observable A with value A_s in microstate s, the equilibrium expectation is

$$\langle A \rangle = \sum_s A_s \mathbb{P}_s. \tag{4.16}$$

In particular,

$$\begin{aligned} \langle E \rangle &= \frac{1}{Z} \sum_s E_s e^{-\beta E_s} \\ &= -\frac{\partial \ln Z}{\partial \beta}. \end{aligned} \tag{4.17}$$

The Helmholtz free energy is

$$F = -k_{\text{B}}T \ln Z. \tag{4.18}$$

The partition function normalizes the state probabilities, and its derivatives generate equilibrium thermodynamic quantities.

4.3.1 Recovering equipartition from the partition function

For one classical quadratic coordinate with energy

$$E(x) = ax^2, \qquad a > 0, \tag{4.19}$$

the continuous canonical partition integral is

$$\begin{aligned} Z_x &= \int_{-\infty}^{\infty} e^{-\beta a x^2} \, \mathrm{d}x \\ &= \sqrt{\frac{\pi}{\beta a}}. \end{aligned} \tag{4.20}$$

Its mean energy is

$$\begin{aligned} \langle ax^2 \rangle &= -\frac{\partial \ln Z_x}{\partial \beta} \\ &= -\frac{\partial}{\partial \beta} \left[\frac{1}{2} \ln \pi - \frac{1}{2} \ln \beta - \frac{1}{2} \ln a \right] \\ &= \frac{1}{2\beta} = \frac{1}{2} k_{\text{B}} T. \end{aligned} \tag{4.21}$$

Thus the partition function reproduces equipartition when the energy is classical, continuous, and quadratic. Unlike equipartition, the partition function also applies to discrete and quantum spectra, which is why it is the appropriate starting point for electronic occupations.

4.3.2 Two-state canonical example

Let a fixed particle occupy either a ground state of energy 0 or an excited state of energy Δ. Then

$$Z = 1 + e^{-\beta\Delta}. \tag{4.22}$$

The excited-state probability is

$$\mathbb{P}_{\text{exc}} = \frac{e^{-\beta\Delta}}{1 + e^{-\beta\Delta}}. \tag{4.23}$$

At low temperature, $\beta\Delta \gg 1$ and the excitation is rare. At high temperature, the two states approach equal probability. This elementary example already contains the central logic: enumerate microstates, assign Boltzmann weights, sum them to form Z, and normalize.

4.4 One electron distributed over a band

Let E_α be the one-electron energies obtained from a band calculation. For exactly one electron in thermal equilibrium, the canonical partition function is

$$Z_1 = \sum_\alpha e^{-\beta E_\alpha}. \tag{4.24}$$

If the distinct energies E_j have multiplicities M_j, the same sum is

$$Z_1 = \sum_j M_j e^{-\beta E_j}. \tag{4.25}$$

The probability that the electron occupies one particular state α is

$$\mathbb{P}_\alpha = \frac{e^{-\beta E_\alpha}}{Z_1}, \tag{4.26}$$

and the probability that its energy is E_j is

$$\mathbb{P}(E_j) = \frac{M_j e^{-\beta E_j}}{Z_1}. \tag{4.27}$$

The factor M_j arises because any one of M_j distinct states gives the same energy. This one-electron calculation uses the canonical Boltzmann weights directly.

In a macroscopic band, the sum becomes

$$Z_1 = \mathcal{V}_d \int g(E) e^{-\beta E}\, \mathrm{d}E, \tag{4.28}$$

where $g(E)$ is the one-electron density of states per physical size. This equation connects the discrete multiplicities of the preceding chapter to the canonical statistics of one electron.

4.5 Many electrons and the chemical potential

If a microstate can contain different particle numbers N_s, the reservoir must account for both energy and particles. The statistical weight becomes

$$e^{-\beta(E_s-\mu N_s)}. \tag{4.29}$$

The grand partition function is

$$\boxed{\Xi(T,V,\mu)=\sum_s e^{-\beta(E_s-\mu N_s)}.} \tag{4.30}$$

The probability of microstate s is

$$\mathbb{P}_s=\frac{e^{-\beta(E_s-\mu N_s)}}{\Xi}. \tag{4.31}$$

The factor $-\mu N_s$ controls the statistical cost of changing particle number. Adding a particle with energy E changes the exponent by

$$-\beta(E-\mu). \tag{4.32}$$

A state far above μ is expensive to occupy, while a state far below μ is strongly favored.

The grand potential is

$$\Omega=-k_{\mathrm{B}}T\ln\Xi, \tag{4.33}$$

and the expected particle number follows from

$$\boxed{\langle N\rangle=\frac{1}{\beta}\frac{\partial\ln\Xi}{\partial\mu}.} \tag{4.34}$$

To verify the sign and factor, differentiate directly:

$$\begin{aligned}\frac{\partial\ln\Xi}{\partial\mu}&=\frac{1}{\Xi}\sum_s\beta N_s e^{-\beta(E_s-\mu N_s)}\\&=\beta\langle N\rangle.\end{aligned} \tag{4.35}$$

The fundamental thermodynamic relation for one particle species is

$$\mathrm{d}U=T\,\mathrm{d}S-P\,\mathrm{d}V+\mu\,\mathrm{d}N. \tag{4.36}$$

Therefore

$$\mu=\left(\frac{\partial U}{\partial N}\right)_{S,V}. \tag{4.37}$$

For the Helmholtz free energy $F=U-TS$,

$$\mathrm{d}F=-S\,\mathrm{d}T-P\,\mathrm{d}V+\mu\,\mathrm{d}N, \tag{4.38}$$

so the form used at fixed temperature is

$$\boxed{\mu=\left(\frac{\partial F}{\partial N}\right)_{T,V}.} \tag{4.39}$$

Definition 4.1: Chemical potential

The chemical potential μ is the change in the appropriate thermodynamic energy when the particle number changes while the natural external variables are held fixed. It is the variable conjugate to particle number. In the grand-canonical weight, $E - \mu$ is the statistical cost of placing an electron in a one-electron state of energy E.

The derivative is understood in the thermodynamic limit, where changing N by one is represented by a continuous derivative. In a semiconductor, charge neutrality or a prescribed total electron number fixes the value of μ.

Definition 4.2: Fermi level

At thermal equilibrium, the Fermi level is the semiconductor-physics name for the electronic chemical potential $\mu(T)$. For a nondegenerate spin-resolved state governed by the Fermi–Dirac occupation,

$$f(E = \mu, T) = \frac{1}{2}. \tag{4.40}$$

The Fermi level can lie in a band gap, where no one-electron state exists.

Definition 4.3: Fermi energy

The Fermi energy is the zero-temperature electronic chemical potential:

$$\boxed{E_F = \mu(T = 0).} \tag{4.41}$$

For a noninteracting metal with a partially filled band, it is the boundary between occupied and empty one-electron states at zero temperature.

In a gapped system, the zero-temperature particle number can remain unchanged while μ moves within an interval containing no states. The chemical- potential definition remains primary; the phrase “highest occupied energy” is a special metallic interpretation of E_F. Some semiconductor texts use *Fermi energy* and *Fermi level* interchangeably at finite temperature. This manuscript reserves E_F for the zero-temperature quantity and uses $\mu(T)$ for the finite-temperature chemical potential.

4.6 Mean occupation of one spin-resolved band state

A semiconductor contains many electrons even when its bands are calculated from an effective one-electron Hamiltonian. The independent-quasiparticle model treats the resulting band states as available one-electron orbitals and constructs many-electron configurations by assigning occupations to them. The effective band model incorporates its selected interaction effects into the one-electron energies. The independent-quasiparticle treatment then assigns occupations while omitting residual correlations.

Model boundary

The canonical distribution gives the state probabilities of one electron. Fermi–Dirac statistics gives the mean occupations when many independent fermionic quasiparticles fill the same one-electron spectrum. The one-electron energies and the many-electron occupation law are separate assumptions.

For one complete spin-resolved state of energy E, fermionic antisymmetry permits two occupation possibilities:

Occupation n	Energy	Particle number
0	0	0
1	E	1

Their grand-canonical weights are

$$w_0 = 1, \qquad w_1 = e^{-\beta(E-\mu)}. \tag{4.42}$$

The single-state grand partition function is

$$\Xi_E = 1 + e^{-\beta(E-\mu)}. \tag{4.43}$$

The expected occupation is

$$\begin{aligned} \langle n_E \rangle &= 0\frac{w_0}{\Xi_E} + 1\frac{w_1}{\Xi_E} \\ &= \frac{e^{-\beta(E-\mu)}}{1 + e^{-\beta(E-\mu)}} \\ &= \frac{1}{1 + e^{\beta(E-\mu)}}. \end{aligned} \tag{4.44}$$

Thus

$$\boxed{f(E,\mu,T) = \frac{1}{1 + \exp[(E-\mu)/(k_{\mathrm{B}}T)]}.} \tag{4.45}$$

The plus sign in the denominator follows from the two mutually exclusive occupations $n = 0$ and $n = 1$. This restriction belongs to the many-electron filling of the one-electron spectrum. The band $E_n(\boldsymbol{k})$ and its density of states are calculated at the one-electron level.

The probability that the state is empty is

$$1 - f(E,\mu,T) = \frac{1}{1 + \exp[(\mu-E)/(k_{\mathrm{B}}T)]}. \tag{4.46}$$

This vacancy probability will become the hole occupation in a valence band.

Three limits provide immediate checks:

$$E = \mu: \quad f = \frac{1}{2}, \tag{4.47}$$

$$E - \mu \gg k_{\mathrm{B}}T: \quad f \to 0, \tag{4.48}$$

$$\mu - E \gg k_{\mathrm{B}}T: \quad f \to 1. \tag{4.49}$$

The statement $f = 1/2$ at $E = \mu$ applies to one nondegenerate band state. A localized donor or acceptor with unequal charge-state degeneracies acquires an additional statistical factor, as shown later.

4.7 Many independent fermionic states

Let the one-particle energies be E_i. A many-electron microstate is specified by occupations $n_i \in \{0, 1\}$, with

$$E_s = \sum_i n_i E_i, \qquad N_s = \sum_i n_i. \tag{4.50}$$

The grand partition function is

$$\begin{aligned} \Xi &= \sum_{\{n_i\}} \exp\left[-\beta \sum_i n_i (E_i - \mu)\right] \\ &= \sum_{\{n_i\}} \prod_i e^{-\beta n_i (E_i - \mu)}. \end{aligned} \tag{4.51}$$

Because each n_i can be summed independently,

$$\begin{aligned} \Xi &= \prod_i \sum_{n_i=0}^{1} e^{-\beta n_i (E_i - \mu)} \\ &= \boxed{\prod_i \left[1 + e^{-\beta(E_i - \mu)}\right]}. \end{aligned} \tag{4.52}$$

Taking the logarithm turns the product into a sum:

$$\ln \Xi = \sum_i \ln\left[1 + e^{-\beta(E_i - \mu)}\right]. \tag{4.53}$$

Using equation (4.34),

$$\begin{aligned} \langle N \rangle &= \frac{1}{\beta} \frac{\partial}{\partial \mu} \ln \Xi \\ &= \sum_i \frac{1}{1 + e^{\beta(E_i - \mu)}} \\ &= \sum_i f(E_i, \mu, T). \end{aligned} \tag{4.54}$$

The total population is therefore a sum of single-state occupation probabilities.

4.8 From a state sum to carrier concentrations

The density of states converts the discrete state sum into an energy integral:

$$\sum_i A(E_i) \longrightarrow \mathcal{V}_d \int g(E) A(E)\, \mathrm{d}E. \tag{4.55}$$

For conduction-band electrons in a bulk semiconductor,

$$\boxed{n(\mu, T) = \int_{E_C}^{\infty} g_c(E) f(E, \mu, T)\, \mathrm{d}E.} \tag{4.56}$$

This equation has two independent ingredients:

$$\underbrace{g_c(E)}_{\text{available states}} \times \underbrace{f(E,\mu,T)}_{\text{occupied fraction}} . \tag{4.57}$$

Their product is the occupied-state density per energy.

A hole is a missing electron in an otherwise occupied valence band. The probability that a valence state is missing its electron is $1 - f$, so

$$\boxed{p(\mu,T) = \int_{-\infty}^{E_V} g_v(E)[1 - f(E,\mu,T)]\,\mathrm{d}E.} \tag{4.58}$$

The hole expression counts vacancies in electronic valence states.

Model check

If $g(E)$ is in states per electronvolt per cubic centimetre and $\mathrm{d}E$ is in electronvolts, then the integral gives carriers per cubic centimetre. If the DOS is instead per joule per cubic metre, both the energy and spatial units must change together. The occupation probability is dimensionless.

4.9 The Boltzmann limit derived from the partition function

For a state several thermal energies above the chemical potential, define

$$y = \frac{E - \mu}{k_{\mathrm{B}}T} \gg 1. \tag{4.59}$$

Then

$$\begin{aligned} f(E) &= \frac{e^{-y}}{1 + e^{-y}} \\ &\simeq e^{-y} = \exp\left[-\frac{E - \mu}{k_{\mathrm{B}}T}\right]. \end{aligned} \tag{4.60}$$

This is the low-occupation limit of the fermionic partition function. The relative overestimate made by using e^{-y} is

$$\frac{e^{-y} - f}{f} = e^{-y}. \tag{4.61}$$

At $y = 3$ it is about five percent; at $y = 5$ it is below one percent.

Using a three-dimensional parabolic conduction-band DOS,

$$g_c(E) = A_c\sqrt{E - E_C}, \tag{4.62}$$

and the Boltzmann occupation gives

$$\begin{aligned} n &\simeq \int_{E_C}^{\infty} A_c\sqrt{E - E_C}\, e^{-(E-\mu)/(k_{\mathrm{B}}T)}\,\mathrm{d}E \\ &= e^{-(E_C-\mu)/(k_{\mathrm{B}}T)} A_c \int_0^{\infty} \sqrt{\varepsilon}\, e^{-\varepsilon/(k_{\mathrm{B}}T)}\,\mathrm{d}\varepsilon. \end{aligned} \tag{4.63}$$

Let $x = \varepsilon/(k_{\mathrm{B}}T)$. Then

$$\begin{aligned}\int_0^\infty \sqrt{\varepsilon} e^{-\varepsilon/(k_{\mathrm{B}}T)}\,\mathrm{d}\varepsilon &= (k_{\mathrm{B}}T)^{3/2}\int_0^\infty x^{1/2}e^{-x}\,\mathrm{d}x \\ &= (k_{\mathrm{B}}T)^{3/2}\Gamma\left(\frac{3}{2}\right) \\ &= (k_{\mathrm{B}}T)^{3/2}\frac{\sqrt{\pi}}{2}.\end{aligned} \tag{4.64}$$

Collecting the state-counting factors defines the effective conduction density of states N_C and yields

$$\boxed{n = N_C \exp\left[-\frac{E_C - \mu}{k_{\mathrm{B}}T}\right].} \tag{4.65}$$

The exponential comes from the partition function; N_C comes from integrating the band density of states. Confusing these two origins makes the formula seem arbitrary.

The same reasoning for valence-band vacancies gives

$$\boxed{p = N_V \exp\left[-\frac{\mu - E_V}{k_{\mathrm{B}}T}\right].} \tag{4.66}$$

4.10 The chemical potential is selected by a constraint

Equations (4.56) and (4.58) give populations for a trial μ. A particle-number or charge constraint closes these population equations and determines μ.

For an intrinsic neutral semiconductor,

$$n(\mu, T) = p(\mu, T). \tag{4.67}$$

For a uniform semiconductor with ionized donors and acceptors,

$$n(\mu, T) + N_A^-(\mu, T) = p(\mu, T) + N_D^+(\mu, T). \tag{4.68}$$

The equilibrium chemical potential is the value that satisfies the appropriate constraint. Thus a "Fermi-level calculation" is fundamentally a constrained statistical-mechanics calculation.

Under nondegenerate conditions, multiplication of equations (4.65) and (4.66) gives

$$np = N_C N_V e^{-(E_C - E_V)/(k_{\mathrm{B}}T)}. \tag{4.69}$$

Defining

$$n_i^2 = N_C N_V e^{-E_g/(k_{\mathrm{B}}T)} \tag{4.70}$$

produces

$$np = n_i^2. \tag{4.71}$$

The mass-action law is therefore a consequence of equilibrium, parabolic-band state counting, and the Boltzmann limit. Degenerate populations require the full Fermi–Dirac carrier integrals.

4.11 Localized states and internal degeneracy

A localized dopant has a finite set of charge configurations. The partition sum includes the internal degeneracy of each configuration.

Suppose a donor has an ionized configuration D^+ with reference weight one and a neutral configuration D^0 with degeneracy ratio g_D. Their weights are

$$w(D^+) = 1, \qquad w(D^0) = g_D e^{(\mu - E_D)/(k_\mathrm{B}T)}. \tag{4.72}$$

Normalization gives

$$\mathbb{P}(D^+) = \frac{1}{1 + g_D e^{(\mu - E_D)/(k_\mathrm{B}T)}}. \tag{4.73}$$

Therefore

$$N_D^+ = N_D \mathbb{P}(D^+). \tag{4.74}$$

The degeneracy factor originates in the partition sum over microscopic configurations. Acceptor statistics follow from the same procedure with their own charge states and degeneracies. The extrinsic-semiconductor chapter develops both cases fully.

4.12 A disciplined route through a carrier calculation

A transparent equilibrium calculation begins with the bands $E_n(\boldsymbol{k})$ and their degeneracies. Their spectral multiplicities determine the densities of available states. Partition sums for band and localized states determine occupation and vacancy probabilities. Energy integration then gives n, p, and the charged dopant populations. Total particle number or charge neutrality fixes μ. Normalization, limiting behavior, and the residual of the closing constraint verify the result.

4.13 Chapter summary

The canonical partition function normalizes Boltzmann weights for fixed particle number. For one electron, it assigns probabilities directly to the one-electron band states. Particle exchange in a many-electron subsystem introduces the grand partition function and the chemical potential. Summing the empty and occupied configurations of each spin-resolved state produces the Fermi–Dirac mean occupation. For independent states, the grand partition function factorizes and the expected particle number is the sum of those occupations. The density of states converts that sum into the carrier integrals for conduction electrons and valence-band holes. The Boltzmann carrier formulas follow from the low-occupation limit and the integrated parabolic density of states. Finally, charge neutrality selects the chemical potential. This chain is the statistical-mechanical foundation of all equilibrium carrier-concentration formulas used in the following chapters.

Chapter 5

Determining the Chemical Potential from the Available States

An energy band specifies the available one-particle states and their energies. The partition function gives their occupations for a trial chemical potential. Particle-number conservation or charge neutrality selects the physical value of μ.

This chapter develops the general constrained problem before specializing to intrinsic and extrinsic semiconductors. The central equation is either a fixed particle-number condition,

$$N(\mu, T) = N_{\text{target}}, \tag{5.1}$$

or a fixed-charge condition,

$$Q(\mu, T) = Q_{\text{target}}. \tag{5.2}$$

The chemical potential is the value that satisfies the declared constraint.

5.1 Thermodynamic inputs beyond the band structure

Consider two samples made from the same semiconductor. If their crystal structures and temperatures are equal, they can have essentially the same host bands $E_n(\boldsymbol{k})$ and density of states $g(E)$. If one sample is donor doped and the other acceptor doped, their equilibrium chemical potentials are different. Their total-charge constraints and dopant charge states select the different values of μ.

Even in an undoped material, temperature matters. The same fixed electron number is distributed more broadly in energy as T increases, and $\mu(T)$ can move to preserve that number. A chemical-potential calculation therefore specifies the temperature, the conserved particle number or charge, the included states and degeneracies, and a common energy reference for bands, localized levels, and μ.

Model boundary

Band energies determine the possible occupations. Temperature determines how sharply those occupations change with energy. Particle number or charge neutrality determines which chemical potential is realized.

5.2 The finite discrete-state problem

Begin with discrete one-particle energies E_i and multiplicities g_i. At thermal equilibrium, the expected electron number is

$$\boxed{N(\mu, T) = \sum_i g_i \frac{1}{1 + \exp[(E_i - \mu)/(k_{\mathrm{B}}T)]}.} \tag{5.3}$$

The multiplicity g_i counts how many distinct one-particle states have energy E_i; the Fermi–Dirac factor gives the occupied fraction of those states. The equation

$$N(\mu, T) = N_{\mathrm{target}} \tag{5.4}$$

then determines μ.

The left side is monotonic. Differentiating one occupation gives

$$\begin{aligned} \frac{\partial f_i}{\partial \mu} &= \beta \frac{e^{\beta(E_i-\mu)}}{[1 + e^{\beta(E_i-\mu)}]^2} \\ &= \beta f_i(1 - f_i) > 0. \end{aligned} \tag{5.5}$$

Therefore

$$\frac{\partial N}{\partial \mu} = \beta \sum_i g_i f_i(1 - f_i) > 0 \tag{5.6}$$

whenever at least one state is partially occupied. Raising μ always raises the expected particle number. This monotonicity normally makes the solution unique.

5.2.1 Zero-temperature limit

As $T \to 0$,

$$f(E, \mu, 0) = \begin{cases} 1, & E < \mu, \\ 0, & E > \mu. \end{cases} \tag{5.7}$$

States are filled from lowest energy upward. For a partially filled level, μ coincides with that level in the limiting description. If an integer number of complete levels is filled and a gap follows, any μ inside that gap produces the same zero-temperature occupations. The finite-temperature limit or a thermodynamic convention selects a particular limiting value.

A semiconductor chemical potential can lie inside the band gap between the allowed one-particle states. It remains the parameter controlling the relative statistical cost of adding or removing an electron.

5.2.2 A symmetric two-level example

Let a spinless particle occupy one of two levels,

$$E_- = E_0 - \frac{\Delta}{2}, \qquad E_+ = E_0 + \frac{\Delta}{2}, \tag{5.8}$$

and require one expected particle:

$$f(E_-, \mu, T) + f(E_+, \mu, T) = 1. \tag{5.9}$$

Using

$$f(E_0 - \epsilon, E_0, T) = 1 - f(E_0 + \epsilon, E_0, T), \tag{5.10}$$

the constraint gives

$$\boxed{\mu = E_0} \tag{5.11}$$

for every T. Spectral and particle–hole symmetry place the chemical potential at the arithmetic mean in this example.

5.3 The continuum limit for energy bands

For a macroscopic solid, the discrete sum becomes a density-of-states integral. If all relevant electronic bands are included,

$$\boxed{\frac{N(\mu, T)}{V} = \sum_n \int g_n(E) f(E, \mu, T)\ \mathrm{d}E.} \tag{5.12}$$

The prescribed total electron density determines μ through

$$\sum_n \int g_n(E) f(E, \mu, T)\ \mathrm{d}E = \frac{N_{\text{target}}}{V}. \tag{5.13}$$

This expression is exact for the declared independent-particle band model, but it can be inconvenient for a semiconductor. The total includes the enormous population of occupied valence states and perhaps deeper core states, while the quantities of interest are often the much smaller deviations from a neutral reference. Subtracting two large numbers to find a small carrier imbalance is also numerically fragile.

The electron–hole description reorganizes the same counting around a neutral reference. Conduction electrons are occupied states above E_C:

$$n(\mu, T) = \int_{E_C}^{\infty} g_c(E) f(E, \mu, T)\ \mathrm{d}E. \tag{5.14}$$

Valence holes are missing electrons below E_V:

$$p(\mu, T) = \int_{-\infty}^{E_V} g_v(E)[1 - f(E, \mu, T)]\ \mathrm{d}E. \tag{5.15}$$

The fixed positive charge of the host lattice is already incorporated in the neutral reference. The remaining constraint can then be written directly in terms of the mobile carrier and dopant deviations.

5.4 Charge neutrality as the semiconductor constraint

For a uniform semiconductor containing singly charged donors and acceptors,

$$\rho(\mu, T) = q\left[p + N_D^+ - n - N_A^-\right]. \tag{5.16}$$

Bulk neutrality requires

$$\boxed{n(\mu,T) + N_A^-(\mu,T) = p(\mu,T) + N_D^+(\mu,T).} \tag{5.17}$$

This one equation combines four statistical populations that depend on the same chemical potential.

Define the neutrality residual

$$R(\mu) = n(\mu,T) + N_A^-(\mu,T) - p(\mu,T) - N_D^+(\mu,T). \tag{5.18}$$

The equilibrium solution satisfies

$$R(\mu) = 0. \tag{5.19}$$

Increasing μ raises the conduction-electron population and the negatively charged acceptor population. It lowers the valence-hole population and the positively charged donor population. Every change raises $R(\mu)$ under the independent-state model. Consequently the root is monotonic and normally unique.

5.5 Intrinsic material as the first closure problem

An ideal intrinsic semiconductor contains no charged dopants:

$$N_D^+ = N_A^- = 0. \tag{5.20}$$

Neutrality reduces to

$$n(\mu_\mathrm{i},T) = p(\mu_\mathrm{i},T). \tag{5.21}$$

The intrinsic chemical potential μ_i is the value at which the statistically expected conduction-electron and valence-hole populations are equal. Equal conduction and valence state counts place it at midgap; unequal state counts displace it toward the band with fewer thermally accessible states.

Under nondegenerate parabolic-band conditions,

$$n = N_C e^{-(E_C-\mu)/(k_\mathrm{B}T)}, \tag{5.22}$$

$$p = N_V e^{-(\mu-E_V)/(k_\mathrm{B}T)}. \tag{5.23}$$

Equating them gives

$$\mu_\mathrm{i} = \frac{E_C + E_V}{2} + \frac{k_\mathrm{B}T}{2}\ln\left(\frac{N_V}{N_C}\right). \tag{5.24}$$

The following intrinsic-semiconductor chapter derives and interprets each term.

5.6 Energy references must be shared

Equilibrium occupations depend on energy differences such as $E-\mu$, $E_C-\mu$, and $E_D-\mu$. If every electronic energy is shifted by one constant C,

$$E' = E + C, \qquad \mu' = \mu + C, \tag{5.25}$$

then

$$E' - \mu' = E - \mu \tag{5.26}$$

and all populations remain unchanged.

Conduction bands, valence bands, localized levels, and μ require one aligned energy reference. Explicit offsets connect conventions based on the vacuum level, valence-band maximum, average electrostatic potential, or a numerical energy zero.

5.7 Choosing the statistical and band model

The physical regime determines the band and occupation models used in the closing equation.

5.7.1 Nondegenerate or Fermi–Dirac statistics

If μ lies several $k_{\mathrm{B}}T$ from a band edge, the Boltzmann approximation can be checked against the full Fermi integral. If μ approaches or enters a band, the full Fermi–Dirac occupation is required. Using a Boltzmann formula to calculate a chemical potential and then finding that the result lies inside the band is a failed self-consistency check.

5.7.2 Parabolic or full bands

Near a band edge, effective masses and analytic densities of states may be adequate. If thermal occupation samples nonparabolic regions, multiple remote valleys, or narrow bands, the carrier integrals must use a more complete $g(E)$ calculated from the bands.

5.7.3 Complete or incomplete dopant ionization

The replacements $N_D^+ = N_D$ and $N_A^- = N_A$ are regime-dependent approximations. At low temperature or for deeper levels, localized-state partition functions must be included in the same neutrality equation.

5.7.4 Fixed bands or charge-dependent bands

At high carrier or impurity concentration, screening, exchange, band-gap narrowing, and impurity-band formation can make the bands themselves depend on the solution. The problem is then self-consistent at two levels: occupations determine charge, and charge modifies the available states.

5.8 Spatially varying systems

A single bulk chemical potential can describe thermal equilibrium, but the local band edges can vary with electrostatic potential:

$$E_C(\boldsymbol{r}) = E_C^{(0)} - q\phi(\boldsymbol{r}), \qquad E_V(\boldsymbol{r}) = E_V^{(0)} - q\phi(\boldsymbol{r}). \tag{5.27}$$

The local carrier populations then depend on $\mu - E_C(\boldsymbol{r})$ and $E_V(\boldsymbol{r}) - \mu$. Their charge density enters Poisson's equation,

$$-\nabla \cdot [\varepsilon(\boldsymbol{r})\nabla\phi(\boldsymbol{r})] = \rho[\phi(\boldsymbol{r}), \mu, T]. \tag{5.28}$$

The equilibrium calculation must solve the occupation and electrostatic problems together. Imposing local neutrality would remove the space charge that creates depletion regions and built-in fields.

Outside equilibrium, electron and hole quasi-Fermi levels describe carrier populations driven by generation, recombination, injection, or external fields. A single chemical potential describes thermal equilibrium.

5.9 A reliable mathematical solution procedure

For a uniform equilibrium problem, all populations are expressed as functions of one trial μ and combined into the residual $R(\mu)$. The calculation uses one energy reference, a specified temperature and dimensionality, a normalized DOS with declared degeneracies, and a complete account of the conserved particle number or charge. The selected occupation laws state their physical regime. Opposite residual signs at two trial chemical potentials form a bracket containing the root.

The final value is substituted into the original number or charge equation. Energy-shift invariance checks the alignment convention, while refinement of the band integrals checks numerical convergence. The returned μ must lie inside the physical regime of every approximation used to construct the populations.

The bracket is physically interpretable. Far below all relevant electronic states, the system lacks electrons and is positively charged relative to the target. Far above them, the states fill and the system becomes negatively charged. The equilibrium chemical potential lies between these limiting charge states.

5.10 Physical closure

The one-electron spectrum and its spectral multiplicities define the normalized density of states. The partition function defines the many-electron occupation of those states. Valence vacancies define holes relative to the neutral ionic background, while localized dopant partition sums define additional charge states. A specified temperature and a conserved particle number or charge then produce one closing equation for μ on the common energy scale.

5.11 Chapter summary

The chemical potential follows from a band spectrum together with thermodynamic information. Bands and their multiplicities determine the available states; the partition function determines their expected occupation for a trial μ; and particle-number conservation or charge neutrality selects the physical value. In semiconductors, electron–hole notation recasts the total-electron problem relative to a neutral filled-valence reference and avoids subtracting enormous nearly equal charges. Exact bands, parabolic reductions, Fermi integrals, Boltzmann limits, and dopant-ionization approximations are alternative models for the same constrained calculation. The following chapters apply this framework first to intrinsic material and then to extrinsic material.

Chapter 6

The Intrinsic Semiconductor

The device-oriented development in Sze and Lee begins with energy bands, density of states, Fermi–Dirac occupation, and the equality of thermally generated electron and hole populations [7, Sec. 1.5]. This chapter follows that sequence but exposes the solid-state derivation underneath its effective densities of states. The full Fermi-integral formulation and degeneracy limits are compared with Sze and Ng [8, Sec. 1.4], while the tensor effective-mass and state-counting steps follow the more explicit periodic-crystal treatment developed by Rudan [6, Secs. 17.6.3 and 18.2–18.3].

6.1 What intrinsic means

Definition 6.1: Intrinsic semiconductor

An intrinsic semiconductor is a chemically pure, spatially uniform semiconductor whose equilibrium mobile-carrier populations are generated by the material's own electronic excitations rather than by intentionally added donors or acceptors. In the ideal bulk model,

$$N_D^+ = N_A^- = 0, \qquad n = p = n_i. \tag{6.1}$$

Intrinsic does not mean that the conduction band is always empty. At $T = 0$ in the independent-particle ground-state picture, the valence band is full and the conduction band is empty. At finite temperature, some electrons occupy conduction-band states. Each such excitation leaves one unoccupied state in the valence band, which is described as a hole. Pair creation therefore preserves

$$N_{\text{conduction electrons}} = N_{\text{valence holes}}, \tag{6.2}$$

and spatial uniformity converts the equality of particle numbers into $n = p$. The result is a counting statement and remains true even when the conduction and valence densities of states have different shapes.

Real nominally undoped samples can contain residual impurities, native defects, surface charge, or nonuniform electrostatic potentials. The intrinsic model is the reference obtained when these effects are neglected. It is a model limit, not a guarantee supplied by the word "pure" on a material label.

6.2 Why a solid has bands rather than isolated levels

An isolated atom has discrete one-electron energy levels. When many atoms are brought together to form a crystal, their orbitals overlap. The electrostatic potential experienced by an electron is no longer the potential of one isolated atom, and the Pauli exclusion principle prevents all electrons from occupying one identical state. Each atomic level therefore splits into many closely spaced crystal levels. For a macroscopic number of unit cells, the spacing is so small that the allowed levels are described as energy bands.

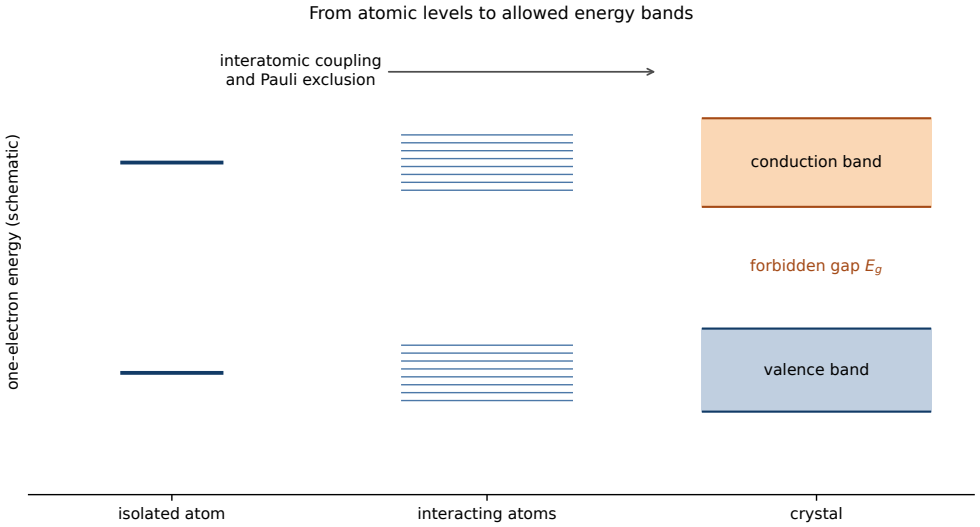


Figure 6.1: Schematic evolution from discrete atomic levels to crystal bands. The drawing is a model of level splitting, not a calculated band structure. The number, width, and ordering of real bands must come from the crystal Hamiltonian.[1]

In a covalent semiconductor such as silicon, the occupied bonding combinations form the upper valence bands and the unoccupied antibonding combinations form the lower conduction bands. The energy interval between them contains no allowed bulk eigenstates and is the forbidden gap. This bonding picture builds intuition, but quantitative carrier statistics require the dispersions $E_n(\boldsymbol{k})$ rather than only two horizontal band edges.

6.2.1 Why neither an empty nor a full band conducts

The semiclassical group velocity of a Bloch state is

$$\boldsymbol{v}_n(\boldsymbol{k}) = \frac{1}{\hbar}\nabla_{\boldsymbol{k}} E_n(\boldsymbol{k}). \tag{6.3}$$

In a nonmagnetic crystal with time-reversal symmetry,

$$E_n(\boldsymbol{k}) = E_n(-\boldsymbol{k}), \qquad \boldsymbol{v}_n(-\boldsymbol{k}) = -\boldsymbol{v}_n(\boldsymbol{k}). \tag{6.4}$$

An empty band carries no electronic current because it contains no electrons. A full band also carries no net current: every occupied state at $\boldsymbol{k}$ has an occupied partner at $-\boldsymbol{k}$ with the opposite velocity.

[1] *Student reproduction notebook:* `figures/06_intrinsic-semiconductor/fig_06_01.ipynb`.

Written as a state sum,

$$\boldsymbol{j}_n = -\frac{q}{V} \sum_{\boldsymbol{k}\,\text{occupied}} \boldsymbol{v}_n(\boldsymbol{k}) = \boldsymbol{0} \qquad \text{for a full band.} \tag{6.5}$$

Here $q > 0$ is the magnitude of the electron charge. Conductivity requires a band to be only partially occupied so that an applied field can create an imbalance between opposite velocities.

6.2.2 Why an empty valence state is called a hole

Suppose a valence band is full except for a small collection of empty states. Because the velocity sum over the full band vanishes,

$$\begin{aligned} \sum_{\text{occupied}} \boldsymbol{v}_v &= \sum_{\text{full band}} \boldsymbol{v}_v - \sum_{\text{empty}} \boldsymbol{v}_v \\ &= -\sum_{\text{empty}} \boldsymbol{v}_v. \end{aligned} \tag{6.6}$$

The electronic current of the almost-full band is consequently

$$\boldsymbol{j}_v = -\frac{q}{V} \sum_{\text{occupied}} \boldsymbol{v}_v = +\frac{q}{V} \sum_{\text{empty}} \boldsymbol{v}_v. \tag{6.7}$$

Thus the missing electronic states can be represented by positively charged quasiparticles: holes. The hole description is not the claim that a positively charged fundamental particle has appeared inside the atom. It is an exact and convenient rewriting of the dynamics of a nearly full band.

When thermal energy promotes one valence electron into the conduction band, the result is simultaneously one conduction electron and one valence hole. This solid-state picture is the origin of intrinsic pair generation and, in a uniform ideal crystal, the equality $n = p$.

6.3 From a periodic solid to band edges

For a perfect crystal, lattice translation symmetry allows one-electron states to be labeled by a band index n and crystal wave vector $\boldsymbol{k}$. Solving the Bloch Hamiltonian gives the band functions

$$H(\boldsymbol{k})\,|u_{n\boldsymbol{k}}\rangle = E_n(\boldsymbol{k})\,|u_{n\boldsymbol{k}}\rangle\,. \tag{6.8}$$

The highest occupied band at $T = 0$ is called the valence band, and the next available band is called the conduction band. Their relevant extrema are

$$E_{\mathrm{C}} = \min_{\boldsymbol{k}} E_c(\boldsymbol{k}), \tag{6.9}$$

$$E_{\mathrm{V}} = \max_{\boldsymbol{k}} E_v(\boldsymbol{k}), \tag{6.10}$$

$$E_{\mathrm{g}} = E_{\mathrm{C}} - E_{\mathrm{V}}. \tag{6.11}$$

If the extrema occur at the same crystal wave vector, the fundamental gap is direct; otherwise it is indirect. This distinction is essential for optical and recombination processes. For equilibrium carrier counting, the immediate requirements are the edge energies, the number of equivalent extrema, and the local shape of each band.

6.3.1 Quadratic expansion at a conduction minimum

Let $\boldsymbol{k}_c^{(a)}$ denote conduction minimum a, and define $\boldsymbol{q} = \boldsymbol{k} - \boldsymbol{k}_c^{(a)}$. Taylor expansion gives

$$\begin{aligned} E_c(\boldsymbol{k}_c^{(a)} + \boldsymbol{q}) = E_{\mathrm{C}} &+ \nabla_{\boldsymbol{k}} E_c|_{\boldsymbol{k}_c^{(a)}} \cdot \boldsymbol{q} \\ &+ \frac{1}{2} \boldsymbol{q}^{\mathsf{T}} \, \nabla_{\boldsymbol{k}} \nabla_{\boldsymbol{k}} E_c|_{\boldsymbol{k}_c^{(a)}} \, \boldsymbol{q} + \mathcal{O}(q^3). \end{aligned} \tag{6.12}$$

The gradient vanishes at the extremum. Define the inverse effective-mass tensor by

$$\left(\boldsymbol{M}_{c,a}^{*-1}\right)_{ij} = \frac{1}{\hbar^2} \left. \frac{\partial^2 E_c}{\partial k_i \partial k_j} \right|_{\boldsymbol{k}_c^{(a)}}. \tag{6.13}$$

Then the local dispersion is

$$E_c(\boldsymbol{k}_c^{(a)} + \boldsymbol{q}) \simeq E_{\mathrm{C}} + \frac{\hbar^2}{2} \boldsymbol{q}^{\mathsf{T}} \boldsymbol{M}_{c,a}^{*-1} \boldsymbol{q}. \tag{6.14}$$

At a genuine minimum, the Hessian and the mass tensor are positive definite. In their principal-axis frame,

$$E_c - E_{\mathrm{C}} = \frac{\hbar^2}{2} \left(\frac{q_1^2}{m_{c1}} + \frac{q_2^2}{m_{c2}} + \frac{q_3^2}{m_{c3}} \right). \tag{6.15}$$

6.3.2 Valence maxima and positive hole masses

At a valence maximum the electronic Hessian is negative definite. It is useful to measure the missing-electron energy downward from E_{V} and define a positive hole-mass tensor:

$$E_{\mathrm{V}} - E_v(\boldsymbol{k}_v^{(b)} + \boldsymbol{q}) \simeq \frac{\hbar^2}{2} \boldsymbol{q}^{\mathsf{T}} \boldsymbol{M}_{v,b}^{*-1} \boldsymbol{q}. \tag{6.16}$$

The sign change does not alter the electronic band. It changes the description from electrons near a maximum with negative curvature to holes with positive excitation energy and positive effective mass.

Model boundary

The effective-mass approximation is local. It preserves the band-edge energy and quadratic curvature, but it discards nonparabolicity, warping, remote valleys, and finite bandwidth. A carrier integral that samples far from the edge can invalidate the approximation even when the algebra is exact.

6.4 Applying state counting to intrinsic band edges

Chapter 3 derived the density of states from discrete level multiplicity and the reciprocal-space count of a general band. We now apply that construction specifically to the ellipsoidal conduction and valence band edges of an intrinsic bulk semiconductor. The repeated intermediate steps make every mass and degeneracy factor visible in this first material application.

Begin in one dimension with a cell of length L_x. Periodicity requires a plane-wave factor to reproduce itself after translation by the cell length:

$$e^{ik_x(x+L_x)} = e^{ik_x x}. \tag{6.17}$$

After cancelling $e^{ik_x x}$,

$$e^{ik_x L_x} = 1 \quad \Longrightarrow \quad k_x L_x = 2\pi n_x, \qquad n_x \in \mathbb{Z}. \tag{6.18}$$

Thus

$$k_{x,n} = \frac{2\pi n_x}{L_x}, \qquad \Delta k_x = k_{x,n} - k_{x,n-1} = \frac{2\pi}{L_x}. \tag{6.19}$$

Now place the crystal in a periodic rectangular cell of volume $V = L_x L_y L_z$. Applying the same argument independently along each axis gives

$$k_i = \frac{2\pi n_i}{L_i}, \qquad n_i \in \mathbb{Z}, \qquad i \in \{x, y, z\}. \tag{6.20}$$

Each allowed $\boldsymbol{k}$ point therefore occupies reciprocal-space volume

$$\Delta^3 k = \frac{(2\pi)^3}{V}. \tag{6.21}$$

The reciprocal-space point density is the inverse of this cell volume:

$$\frac{1}{\Delta^3 k} = \frac{V}{(2\pi)^3}. \tag{6.22}$$

Including spin degeneracy g_s, the number of one-particle states in a small reciprocal-space volume is

$$\mathrm{d}N = g_s \frac{V}{(2\pi)^3}\, \mathrm{d}^3 k. \tag{6.23}$$

The finite cell gives a discrete lattice of wave vectors. The thermodynamic limit sends each $L_i \to \infty$, so each $\Delta k_i \to 0$ and the discrete sum becomes a continuum integral. The factor $V/(2\pi)^3$ survives this limit and is the origin of the volume normalization in the density of states.

6.4.1 The conduction-band ellipsoid

For one ellipsoidal valley, let

$$\varepsilon = E - E_{\mathrm{C}} \geq 0. \tag{6.24}$$

The surface of constant ε in equation (6.15) has semiaxes

$$q_i^{\max} = \frac{\sqrt{2m_{ci}\varepsilon}}{\hbar}. \tag{6.25}$$

Its enclosed reciprocal-space volume is

$$\begin{aligned} \Omega_k(\varepsilon) &= \frac{4\pi}{3} \prod_{i=1}^{3} q_i^{\max} \\ &= \frac{4\pi}{3} \left(\frac{2\varepsilon}{\hbar^2}\right)^{3/2} \sqrt{m_{c1} m_{c2} m_{c3}}. \end{aligned} \tag{6.26}$$

Define the per-valley density-of-states mass

$$m_{c,d} = (m_{c1} m_{c2} m_{c3})^{1/3}. \tag{6.27}$$

Then $\sqrt{m_{c1} m_{c2} m_{c3}} = m_{c,d}^{3/2}$. For $g_{v,c}$ equivalent valleys, the number of states per volume with energy between E_C and $E_C + \varepsilon$ is

$$\frac{\mathcal{N}_c(\varepsilon)}{V} = \frac{g_s g_{v,c}}{6\pi^2} \left(\frac{2m_{c,d}}{\hbar^2} \right)^{3/2} \varepsilon^{3/2}. \tag{6.28}$$

The integrated count scales as $\varepsilon^{3/2}$ because it counts the volume of a three-dimensional ellipsoid. The density of states is the number added per energy interval. Since $\varepsilon = E - E_C$,

$$\begin{aligned} g_c(E) &= \frac{1}{V} \frac{\mathrm{d}\mathcal{N}_c}{\mathrm{d}E} \\ &= \frac{g_s g_{v,c}}{6\pi^2} \left(\frac{2m_{c,d}}{\hbar^2} \right)^{3/2} \frac{\mathrm{d}}{\mathrm{d}E} (E - E_C)^{3/2} \\ &= \frac{g_s g_{v,c}}{4\pi^2} \left(\frac{2m_{c,d}}{\hbar^2} \right)^{3/2} \sqrt{E - E_C}. \end{aligned} \tag{6.29}$$

Including the fact that no conduction states occur below E_C gives

$$\boxed{g_c(E) = \frac{g_s g_{v,c}}{4\pi^2} \left(\frac{2m_{c,d}}{\hbar^2} \right)^{3/2} \sqrt{E - E_C}\, \Theta(E - E_C).} \tag{6.30}$$

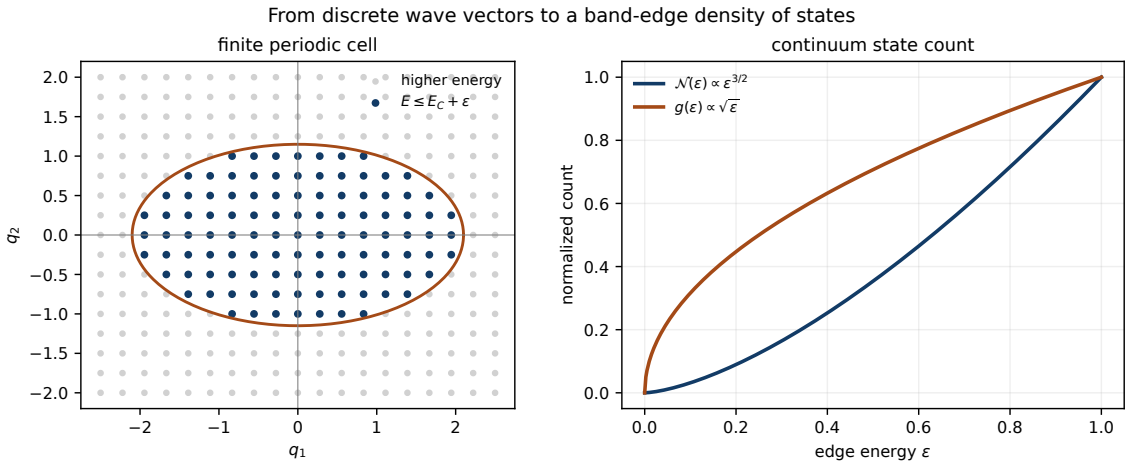


Figure 6.2: A finite periodic cell supplies discrete wave vectors. Counting points inside the constant-energy ellipsoid gives an integrated count proportional to $\varepsilon^{3/2}$; differentiation produces the familiar three-dimensional square-root density of states. Only a two-dimensional cross-section of the ellipsoid is drawn.[2]

The factor g_s counts spin, $g_{v,c}$ counts equivalent valleys, and $m_{c,d}$ accounts for the shape of each valley. A quoted effective mass that already absorbs valley degeneracy must not be multiplied by $g_{v,c}$ again.

[2] *Student reproduction notebook:* `figures/06_intrinsic-semiconductor/fig_06_02.ipynb`.

6.4.2 The valence-band density of states

The same counting argument, now using the positive hole energy $\varepsilon_h = E_{\mathrm{V}} - E$, gives

$$\boxed{g_v(E) = \frac{g_s g_{v,v}}{4\pi^2}\left(\frac{2m_{v,d}}{\hbar^2}\right)^{3/2}\sqrt{E_{\mathrm{V}} - E}\,\Theta(E_{\mathrm{V}} - E).} \tag{6.31}$$

If several inequivalent valleys or valence bands contribute, their densities of states must be summed:

$$g_c(E) = \sum_a g_{c,a}(E), \qquad g_v(E) = \sum_b g_{v,b}(E). \tag{6.32}$$

For degenerate isotropic heavy- and light-hole bands with a common maximum, this sum can be represented by

$$m_{v,d} = \left(m_{hh}^{3/2} + m_{lh}^{3/2}\right)^{2/3}, \tag{6.33}$$

provided spin and band degeneracies are handled consistently.

6.5 Combining available states with statistical occupation

The density of states answers, "How many states are available?" It contains no temperature and no chemical potential. The partition-function derivation in chapter 4 answers the independent question, "With what probability is each state occupied?" We briefly restate the single-state result before applying it to the intrinsic bands.

6.5.1 One-state partition-function recap

A one-electron state of energy E can contain either zero or one electron of a given spin label. In equilibrium with a particle and energy reservoir, its two grand-canonical weights are

$$w_0 = 1, \qquad w_1 = e^{-\beta(E-\mu)}, \qquad \beta = \frac{1}{k_{\mathrm{B}}T}. \tag{6.34}$$

The single-state grand partition function is the sum

$$\mathcal{Z}_E = w_0 + w_1 = 1 + e^{-\beta(E-\mu)}. \tag{6.35}$$

The mean occupation is the occupied value, one, times its normalized probability:

$$\begin{aligned} f(E,\mu,T) &= \frac{1\times w_1 + 0\times w_0}{\mathcal{Z}_E} \\ &= \frac{e^{-\beta(E-\mu)}}{1+e^{-\beta(E-\mu)}} \\ &= \frac{1}{1+e^{(E-\mu)/(k_{\mathrm{B}}T)}}. \end{aligned} \tag{6.36}$$

Pauli exclusion is what restricts the occupation to zero or one and produces the plus sign in the denominator.

An electronic state is empty with probability

$$1 - f(E,\mu,T) = 1 - \frac{1}{1 + e^{(E-\mu)/(k_{\mathrm{B}}T)}} = \frac{1}{1 + e^{(\mu-E)/(k_{\mathrm{B}}T)}}. \tag{6.37}$$

This is the probability used to count holes in a valence band.

6.5.2 Spectral carrier densities

The Fermi–Dirac function answers, “What is the probability that each electronic state is occupied?” The carrier spectral densities are the products of state availability and the relevant probability:

$$\backslash(E) = g_c(E) f(E,\mu,T), \tag{6.38}$$

$$\surd(E) = g_v(E)\,[1 - f(E,\mu,T)]. \tag{6.39}$$

Their areas are the carrier concentrations.

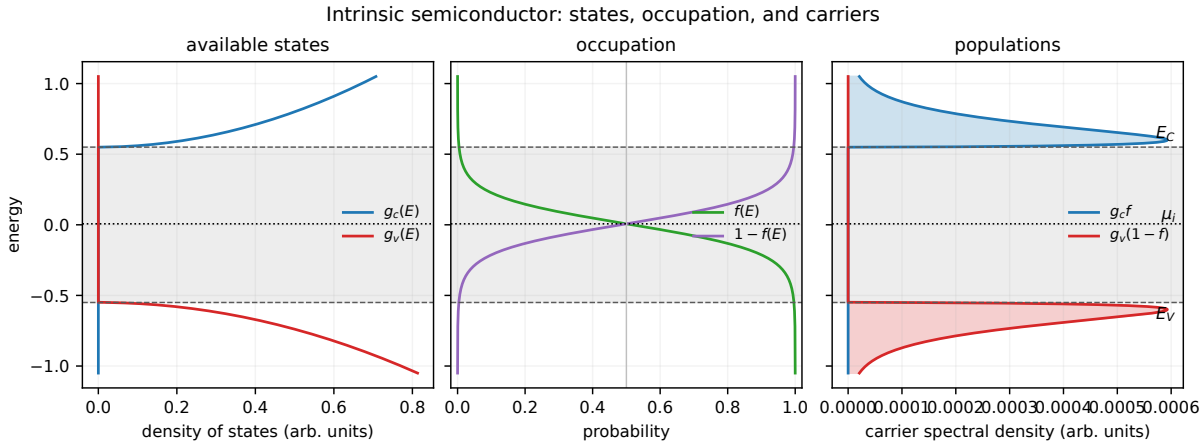


Figure 6.3: Available states, occupation probabilities, and carrier spectral densities are distinct quantities. The gray interval is the forbidden gap, where the density of states is zero. The unequal model prefactors illustrate that intrinsic neutrality requires equal areas, not mirror-symmetric curves.[3]

6.6 Exact parabolic-band carrier integrals

Start with the electron concentration

$$n = \int_{E_{\mathrm{C}}}^{\infty} \frac{g_c(E)}{1 + \exp[(E-\mu)/(k_{\mathrm{B}}T)]}\,\mathrm{d}E. \tag{6.40}$$

Introduce

$$x = \frac{E - E_{\mathrm{C}}}{k_{\mathrm{B}}T}, \qquad \eta_c = \frac{\mu - E_{\mathrm{C}}}{k_{\mathrm{B}}T}. \tag{6.41}$$

[3] *Student reproduction notebook:* figures/06_intrinsic-semiconductor/fig_06_03.ipynb.

Then $E - E_{\mathrm{C}} = x k_{\mathrm{B}} T$ and $\mathrm{d}E = k_{\mathrm{B}} T \, \mathrm{d}x$. Substitution gives

$$n = \frac{g_s g_{v,c}}{4\pi^2} \left(\frac{2 m_{c,d}}{\hbar^2} \right)^{3/2} (k_{\mathrm{B}} T)^{3/2} \times \int_0^\infty \frac{\sqrt{x}}{1 + e^{x - \eta_c}} \, \mathrm{d}x. \tag{6.42}$$

Define the normalized Fermi–Dirac integral

$$F_{1/2}(\eta) = \frac{2}{\sqrt{\pi}} \int_0^\infty \frac{\sqrt{x}}{1 + e^{x - \eta}} \, \mathrm{d}x. \tag{6.43}$$

Different books normalize Fermi integrals differently; the prefactor in equation (6.43) is part of our definition. With this convention,

$$\boxed{n = N_{\mathrm{C}} F_{1/2}(\eta_c), \qquad N_{\mathrm{C}} = g_s g_{v,c} \left(\frac{m_{c,d} k_{\mathrm{B}} T}{2\pi\hbar^2} \right)^{3/2}.} \tag{6.44}$$

For $g_s = 2$, the effective density of states can also be written

$$N_{\mathrm{C}} = 2 g_{v,c} \left(\frac{2\pi m_{c,d} k_{\mathrm{B}} T}{h^2} \right)^{3/2}. \tag{6.45}$$

Introduce the thermal de Broglie wavelength associated with the conduction DOS mass,

$$\lambda_{T,c} = \frac{h}{\sqrt{2\pi m_{c,d} k_{\mathrm{B}} T}}. \tag{6.46}$$

Then

$$N_{\mathrm{C}} = \frac{g_s g_{v,c}}{\lambda_{T,c}^3}. \tag{6.47}$$

This form gives a useful interpretation: the effective density of states is the degeneracy divided by a characteristic thermal wave-packet volume. It also makes the $T^{3/2}$ dependence immediate because $\lambda_{T,c} \propto T^{-1/2}$.

For holes, introduce

$$x = \frac{E_{\mathrm{V}} - E}{k_{\mathrm{B}} T}, \qquad \eta_v = \frac{E_{\mathrm{V}} - \mu}{k_{\mathrm{B}} T}. \tag{6.48}$$

Using

$$1 - f(E, \mu, T) = \frac{1}{1 + \exp[(\mu - E)/(k_{\mathrm{B}} T)]}, \tag{6.49}$$

produces

$$\boxed{p = N_{\mathrm{V}} F_{1/2}(\eta_v), \qquad N_{\mathrm{V}} = g_s g_{v,v} \left(\frac{m_{v,d} k_{\mathrm{B}} T}{2\pi\hbar^2} \right)^{3/2}.} \tag{6.50}$$

These equations retain Fermi–Dirac statistics while still assuming local parabolic bands.

6.7 The nondegenerate limit

An intrinsic semiconductor with $E_{\mathrm{g}} \gg k_{\mathrm{B}}T$ normally has its chemical potential several thermal energies from either band edge. To see the approximation directly, define

$$y = \frac{E - \mu}{k_{\mathrm{B}}T} > 0. \tag{6.51}$$

The exact occupation can be rewritten as

$$f(E) = \frac{e^{-y}}{1 + e^{-y}}. \tag{6.52}$$

When $e^{-y} \ll 1$, the denominator approaches one and

$$f(E) \simeq e^{-y} = \exp\left[-\frac{E - \mu}{k_{\mathrm{B}}T}\right]. \tag{6.53}$$

The relative overestimate made by replacing f with e^{-y} is especially simple:

$$\frac{e^{-y} - f}{f} = e^{-y}. \tag{6.54}$$

At $E - \mu = 3k_{\mathrm{B}}T$, it is about $e^{-3} \simeq 5\%$; farther into the tail it falls exponentially.

The same limit can be taken inside the normalized Fermi integral. If $\eta \ll -1$,

$$\frac{1}{1 + e^{x - \eta}} \simeq e^{\eta - x}. \tag{6.55}$$

Therefore

$$\begin{aligned} F_{1/2}(\eta) &\simeq \frac{2e^{\eta}}{\sqrt{\pi}} \int_0^{\infty} x^{1/2} e^{-x}\, \mathrm{d}x \\ &= \frac{2e^{\eta}}{\sqrt{\pi}} \Gamma\left(\frac{3}{2}\right) \\ &= \frac{2e^{\eta}}{\sqrt{\pi}} \frac{\sqrt{\pi}}{2} \\ &= e^{\eta}. \end{aligned} \tag{6.56}$$

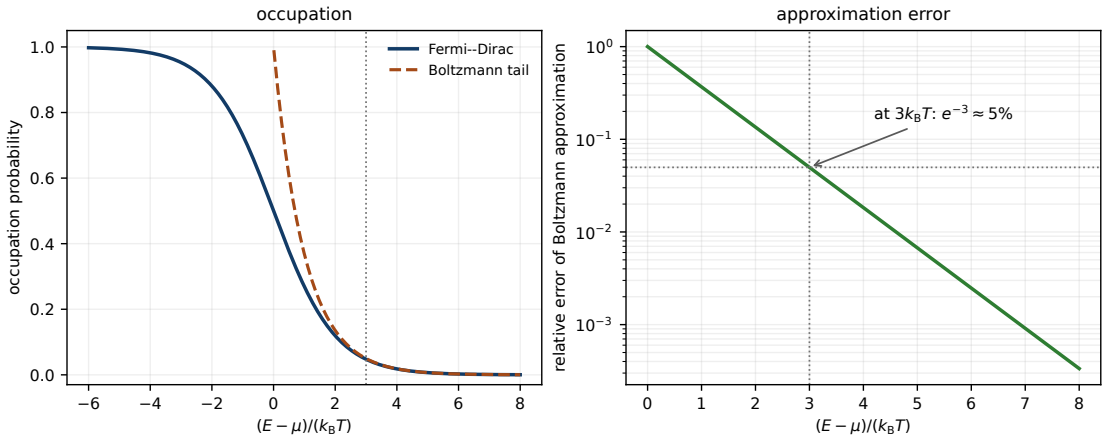


Figure 6.4: Exact Fermi–Dirac occupation and its Boltzmann-tail approximation. The right panel makes the approximation error explicit instead of treating "several $k_{\mathrm{B}}T$" as an unexamined rule.[4]

The carrier concentrations then become

$$n \simeq N_{\mathrm{C}} \exp\left[-\frac{E_{\mathrm{C}} - \mu}{k_{\mathrm{B}}T}\right], \tag{6.57}$$

$$p \simeq N_{\mathrm{V}} \exp\left[-\frac{\mu - E_{\mathrm{V}}}{k_{\mathrm{B}}T}\right]. \tag{6.58}$$

The exponential factors describe occupation; N_{C} and N_{V} describe the number of thermally accessible states near the edges. They play different roles and should not be merged conceptually.

Model check

The exact and nondegenerate calculations should be compared at selected conditions. Agreement establishes that the Boltzmann reduction is adequate for that case; it does not establish that every doped or high-temperature case will remain nondegenerate.

6.8 Deriving the intrinsic chemical potential

Set $n = p$ in equations (6.57) and (6.58):

$$N_{\mathrm{C}} e^{-(E_{\mathrm{C}} - \mu_{\mathrm{i}})/(k_{\mathrm{B}}T)} = N_{\mathrm{V}} e^{-(\mu_{\mathrm{i}} - E_{\mathrm{V}})/(k_{\mathrm{B}}T)}. \tag{6.59}$$

Taking the natural logarithm gives

$$\ln N_{\mathrm{C}} - \frac{E_{\mathrm{C}} - \mu_{\mathrm{i}}}{k_{\mathrm{B}}T} = \ln N_{\mathrm{V}} - \frac{\mu_{\mathrm{i}} - E_{\mathrm{V}}}{k_{\mathrm{B}}T}. \tag{6.60}$$

Collecting the two μ_{i} terms yields

$$2\mu_{\mathrm{i}} = E_{\mathrm{C}} + E_{\mathrm{V}} + k_{\mathrm{B}}T \ln\left(\frac{N_{\mathrm{V}}}{N_{\mathrm{C}}}\right), \tag{6.61}$$

and therefore

$$\boxed{\mu_{\mathrm{i}} = \frac{E_{\mathrm{C}} + E_{\mathrm{V}}}{2} + \frac{k_{\mathrm{B}}T}{2} \ln\left(\frac{N_{\mathrm{V}}}{N_{\mathrm{C}}}\right).} \tag{6.62}$$

The intrinsic chemical potential equals midgap only when the effective conduction and valence state counts are equal. If the spin factors cancel, substitution of the effective densities gives

$$\mu_{\mathrm{i}} - E_{\mathrm{mid}} = \frac{k_{\mathrm{B}}T}{2} \ln\left(\frac{g_{v,v}}{g_{v,c}}\right) + \frac{3k_{\mathrm{B}}T}{4} \ln\left(\frac{m_{v,d}}{m_{c,d}}\right), \tag{6.63}$$

where $E_{\mathrm{mid}} = (E_{\mathrm{C}} + E_{\mathrm{V}})/2$. More valence-band states shift μ_{i} toward the conduction band so that their smaller empty fraction still produces the same number of holes as electrons. More conduction-band states produce the opposite shift.

[4] *Student reproduction notebook:* `figures/06_intrinsic-semiconductor/fig_06_04.ipynb`.

6.9 Deriving the intrinsic carrier concentration

Multiply the nondegenerate electron and hole concentrations:

$$\begin{aligned} np &= N_{\mathrm{C}} N_{\mathrm{V}} \exp\left[-\frac{E_{\mathrm{C}} - \mu + \mu - E_{\mathrm{V}}}{k_{\mathrm{B}} T}\right] \\ &= N_{\mathrm{C}} N_{\mathrm{V}} \exp\left(-\frac{E_{\mathrm{g}}}{k_{\mathrm{B}} T}\right). \end{aligned} \tag{6.64}$$

In an intrinsic semiconductor $n = p = n_i$, hence

$$\boxed{n_i = \sqrt{N_{\mathrm{C}} N_{\mathrm{V}}} \exp\left(-\frac{E_{\mathrm{g}}}{2k_{\mathrm{B}} T}\right).} \tag{6.65}$$

The factor of two in the exponential denominator has a physical counting origin: one thermal excitation creates two mobile quasiparticles, but it costs approximately one band-gap energy to create the pair.

Equation (6.64) also gives the equilibrium mass-action law

$$np = n_i^2. \tag{6.66}$$

This simple form uses nondegenerate statistics and a common equilibrium chemical potential. In a degenerate semiconductor the exact Fermi integrals must replace the exponentials, so $np = n_i^2$ is not generally preserved in this form [8, Sec. 1.4.1].

6.10 Temperature dependence

If the DOS masses and degeneracies are treated as constants,

$$N_{\mathrm{C}} \propto T^{3/2}, \qquad N_{\mathrm{V}} \propto T^{3/2}. \tag{6.67}$$

Consequently,

$$n_i(T) \propto T^{3/2} \exp\left[-\frac{E_{\mathrm{g}}(T)}{2k_{\mathrm{B}} T}\right]. \tag{6.68}$$

The gap itself usually changes with temperature. A common interpolation is

$$E_{\mathrm{g}}(T) = E_{\mathrm{g}}(0) - \frac{\alpha T^2}{T + \beta}, \tag{6.69}$$

with material-specific parameters α and β. This compact law is useful only over the range supported by the fitted data.

Taking the logarithmic derivative of equation (6.68) gives

$$\frac{\mathrm{d} \ln n_i}{\mathrm{d}T} = \frac{3}{2T} - \frac{1}{2k_{\mathrm{B}} T} \frac{\mathrm{d}E_{\mathrm{g}}}{\mathrm{d}T} + \frac{E_{\mathrm{g}}}{2k_{\mathrm{B}} T^2}, \tag{6.70}$$

when mass temperature dependence is neglected. For an ordinary semiconductor, the last term is large and positive; a decreasing gap adds another positive contribution. This is why intrinsic concentration can change by many orders of magnitude over an experimentally modest temperature interval.

If E_g is nearly constant over a restricted range,

$$\ln\left(\frac{n_i}{T^{3/2}}\right) = \text{constant} - \frac{E_g}{2k_B}\frac{1}{T}. \tag{6.71}$$

A plot of $\ln(n_i/T^{3/2})$ against $1/T$ is then approximately linear with slope $-E_g/(2k_B)$. Curvature can reflect $E_g(T)$, temperature-dependent masses, or failure of the adopted band model.

For the illustrative silicon calculation in figure 6.5, the gap interpolation parameters follow the convention summarized by Sze and Ng [8, Fig. 1.6], and the room-temperature effective densities follow Sze and Lee [7, Sec. 1.5]. Their joint use is declared explicitly so that a later edition can replace them as one parameter set rather than silently mixing new and old values.

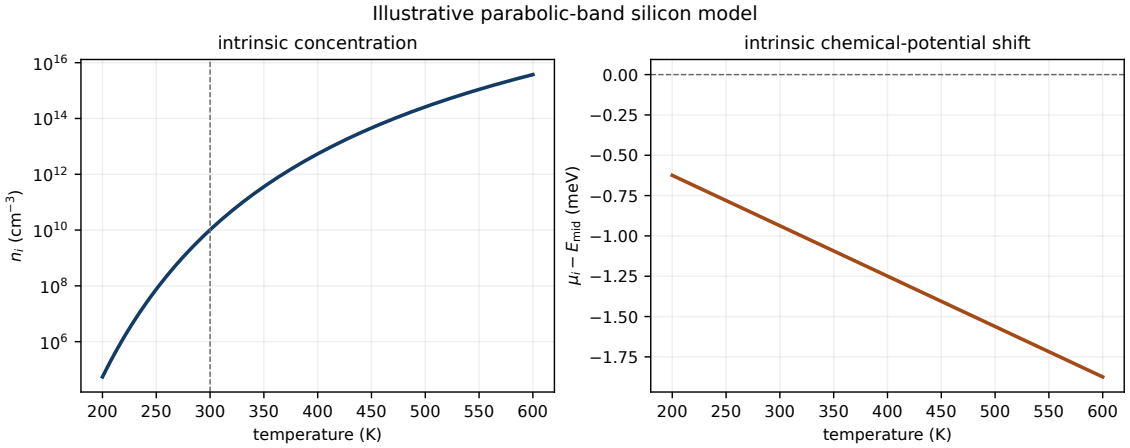


Figure 6.5: Illustrative silicon parabolic-band calculation using $E_g(0) = 1.169\,\text{eV}$, $\alpha = 4.9\times10^{-4}\,\text{eV/K}$, $\beta = 6.55\times10^{2}\,\text{K}$, $N_C(3.00\times10^{2}\,\text{K}) = 2.86\times10^{19}/\text{cm}^3$, and $N_V(3.00\times10^{2}\,\text{K}) = 2.66\times10^{19}/\text{cm}^3$. The concentration is exponentially temperature-sensitive, whereas the intrinsic-level displacement from midgap remains on the millielectronvolt scale for these parameters. The numbers illustrate one internally consistent convention rather than immutable material constants.[5]

6.11 Worked silicon estimate at $3.00\times10^{2}\,\text{K}$

Using the same Sze–Lee effective-density convention for a transparent control calculation [7, Sec. 1.5], take

$$\begin{aligned} T &= 3.00\times10^{2}\,\text{K}, & k_BT &= 2.585\times10^{-2}\,\text{eV},\\ E_g &= 1.12\,\text{eV}, & N_C &= 2.86\times10^{19}/\text{cm}^3,\\ & & N_V &= 2.66\times10^{19}/\text{cm}^3. \end{aligned} \tag{6.72}$$

The intrinsic-level displacement from midgap is

$$\begin{aligned} \mu_i - E_{mid} &= \frac{k_BT}{2}\ln\left(\frac{N_V}{N_C}\right)\\ &\simeq -9.4\times10^{-1}\,\text{meV}. \end{aligned} \tag{6.73}$$

[5] *Student reproduction notebook:* figures/06_intrinsic-semiconductor/fig_06_05.ipynb.

The negative sign means that this particular DOS convention places μ_{i} slightly below midgap. The intrinsic concentration is

$$\begin{aligned} n_i &= \sqrt{(2.86\times10^{19})(2.66\times10^{19})}\exp\left[-\frac{1.12}{2(0.02585)}\right] \\ &\simeq 1.1\times10^{10}/\mathrm{cm}^3. \end{aligned} \tag{6.74}$$

Published room-temperature values differ because the adopted gap, DOS masses, valley conventions, and temperature are not identical. Agreement should be judged only after these definitions are aligned, not by silently adjusting a constant until one preferred table is reproduced.

6.12 When the full Fermi integrals are required

The intrinsic parabolic-band problem can be solved without the Boltzmann approximation. Define

$$\eta_c = \frac{\mu_{\mathrm{i}} - E_{\mathrm{C}}}{k_{\mathrm{B}}T}, \qquad \eta_v = \frac{E_{\mathrm{V}} - \mu_{\mathrm{i}}}{k_{\mathrm{B}}T}. \tag{6.75}$$

Their sum is constrained by the gap:

$$\eta_c + \eta_v = -\frac{E_{\mathrm{g}}}{k_{\mathrm{B}}T}. \tag{6.76}$$

Neutrality requires

$$N_{\mathrm{C}}F_{1/2}(\eta_c) = N_{\mathrm{V}}F_{1/2}(\eta_v). \tag{6.77}$$

Equations (6.76) and (6.77) determine μ_{i} without assuming $F_{1/2}(\eta) \simeq e^{\eta}$. Ordinary intrinsic Si and GaAs near room temperature are strongly nondegenerate because $E_{\mathrm{g}}/k_{\mathrm{B}}T$ is large, but the exact form is the appropriate reference for narrow-gap materials, sufficiently high temperatures, or numerical validation.

6.13 Assumptions exposed

The compact intrinsic formula equation (6.65) depends on the following assumptions:

1. the material is spatially uniform and in thermal equilibrium;
2. donor, acceptor, and charged-defect contributions are absent;
3. band edges are represented by quadratic dispersions;
4. all relevant valleys and valence bands are counted consistently;
5. the chemical potential lies far enough from the bands for nondegenerate statistics;
6. the reported gap and DOS parameters refer to the same material phase and temperature convention; and
7. many-body effects such as band-gap narrowing are negligible.

Removing one assumption does not merely change a prefactor. It can change the neutrality equation, the required statistics, or even the meaning of the band edges.

6.14 Verification checklist

A trustworthy intrinsic-semiconductor calculation should demonstrate that

1. the DOS integrates to units of concentration;
2. $n(\mu, T)$ increases and $p(\mu, T)$ decreases monotonically with μ;
3. the solved intrinsic state satisfies $n - p = 0$ to a declared tolerance;
4. a rigid shift of E_{C}, E_{V}, and μ leaves n and p unchanged;
5. the full Fermi-integral result approaches the Boltzmann result in the nondegenerate limit;
6. constant-mass effective densities scale as $T^{3/2}$; and
7. the silicon control case is compared with a source using compatible gap and DOS conventions.

6.15 Chapter summary

The intrinsic concentration is the endpoint of a solid-state derivation. Band curvature defines effective-mass tensors; periodic-boundary state counting turns the resulting ellipsoids into square-root densities of states; Fermi–Dirac statistics converts available states into occupied conduction states and empty valence states; and intrinsic neutrality selects the chemical potential. Only then does the familiar expression

$$n_i = \sqrt{N_{\mathrm{C}} N_{\mathrm{V}}}\, e^{-E_{\mathrm{g}}/(2k_{\mathrm{B}}T)}$$

appear. Its compactness should not hide the chain of assumptions that makes it valid.

Chapter 7

The Extrinsic Semiconductor

Sze and Lee introduce extrinsic semiconductors through substitutional donors and acceptors, impurity levels, complete ionization, compensation, and the three temperature regimes of freeze-out, extrinsic behavior, and intrinsic behavior [7, Sec. 1.6]. This chapter retains that device-oriented sequence while making the statistical closure explicit. The localized-state degeneracy factors and exact charge-neutrality problem are compared with Sze and Ng [8, Secs. 1.4.2–1.4.3], and Rudan's treatment is used to connect the dopant states to the periodic crystal, full Fermi integrals, nonuniform electrostatics, compensation, and band-gap narrowing [6, Secs. 18.4–18.6].

7.1 What doping changes—and what it does not

Definition 7.1: Extrinsic semiconductor

An extrinsic semiconductor is a semiconductor whose equilibrium carrier populations are intentionally modified by electrically active impurities or defects. Donors tend to provide conduction-band electrons; acceptors tend to provide valence-band holes. The actual populations are determined by the band structure, dopant charge states, temperature, chemical potential, and charge neutrality.

Doping does not simply append a prescribed number of mobile particles. A concentration N_D of donor atoms is not automatically the same as a concentration $n = N_D$ of conduction electrons. The equality requires at least complete donor ionization, negligible compensation, and a regime in which intrinsic carriers do not dominate. Similarly, $p = N_A$ is a limiting result for acceptors, not the definition of an acceptor.

Dilute substitutional impurities also do not erase the host bands. To first approximation, the periodic host still supplies its conduction and valence bands, while each dilute impurity supplies a localized state in or near the gap. At high impurity concentrations this separation fails: localized states overlap, impurity bands form, the band edges change, and many-body corrections become important.

7.2 Substitutional donors and acceptors

In group-IV silicon, a substitutional group-V atom such as phosphorus has five valence electrons. Four participate in covalent bonds with neighboring silicon atoms; the fifth is weakly bound. Removing that electron to the conduction band leaves a fixed positive donor ion:

$$D^0 \longrightarrow D^+ + e_{\mathrm{C}}^-. \tag{7.1}$$

A group-III atom such as boron has only three valence electrons. It can accept an electron from the valence band, leaving one mobile hole and one fixed negative acceptor ion:

$$A^0 \longrightarrow A^- + h_{\mathrm{V}}^+. \tag{7.2}$$

Common group-V donors in silicon include phosphorus, arsenic, and antimony; common group-III acceptors include boron, aluminum, gallium, and indium [7, Sec. 1.6.1]. The mobile carriers belong to extended band states. The ionized dopant charges remain attached to lattice sites and contribute to electrostatics, not to band transport as mobile particles.

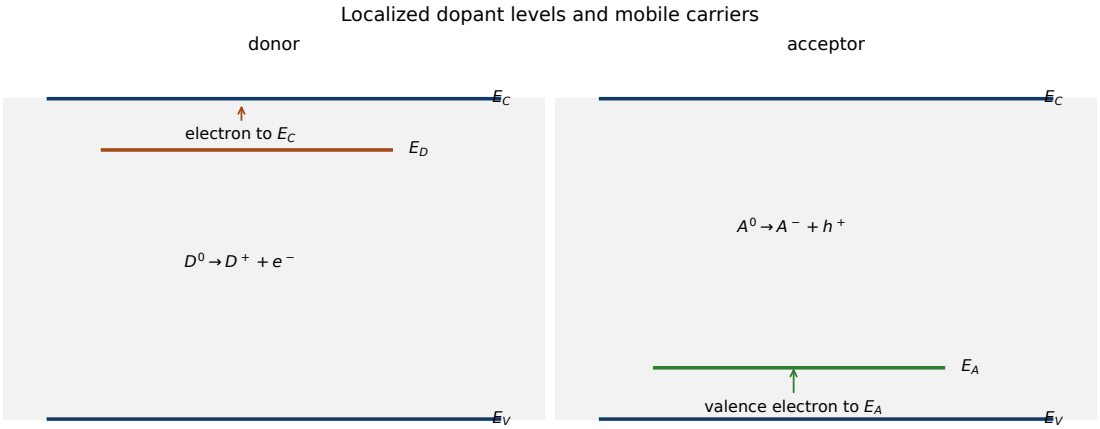


Figure 7.1: Schematic donor and acceptor ionization. A shallow donor level lies below E_{C} by the donor binding energy; a shallow acceptor level lies above E_{V} by the acceptor binding energy. The drawing is not to scale.[1]

Table 7.1: Charge-state bookkeeping for singly ionizable dopants.

Species	State	Localized electronic occupancy	Charge relative to host
Donor	D^0	donor electron bound	neutral
Donor	D^+	donor state empty	$+q$
Acceptor	A^0	acceptor state empty	neutral
Acceptor	A^-	acceptor state filled by an electron	$-q$

The notation prevents a common sign mistake: an ionized donor is positive because it has lost an electron, while an ionized acceptor is negative because it has gained one.

[1] *Student reproduction notebook:* `figures/07_extrinsic-semiconductor/fig_07_01.ipynb`.

7.3 Why shallow dopants resemble hydrogen

A shallow donor electron is weakly bound and extends over many unit cells. In the simplest effective-mass model, the rapidly varying lattice-scale part has already been absorbed into the band effective mass, and the remaining envelope obeys a screened hydrogenic equation:

$$\left[-\frac{\hbar^2}{2m_c^*}\nabla^2 - \frac{q^2}{4\pi\varepsilon_s r}\right] F(\boldsymbol{r}) = -E_{B,D}F(\boldsymbol{r}). \tag{7.3}$$

Relative to vacuum hydrogen, two replacements have occurred:

$$m_0 \longrightarrow m_c^*, \qquad \varepsilon_0 \longrightarrow \varepsilon_s. \tag{7.4}$$

The ground-state binding energy is therefore

$$\begin{aligned} E_{B,D} &= \frac{m_c^* q^4}{2(4\pi\varepsilon_s)^2\hbar^2} \\ &= \left(\frac{m_c^*}{m_0}\right)\left(\frac{\varepsilon_0}{\varepsilon_s}\right)^2 1.36\times 10^{1}\,\mathrm{eV}. \end{aligned} \tag{7.5}$$

The donor energy level is

$$E_D = E_\mathrm{C} - E_{B,D}. \tag{7.6}$$

The effective Bohr radius is

$$\begin{aligned} a_D^* &= \frac{4\pi\varepsilon_s\hbar^2}{m_c^* q^2} \\ &= \left(\frac{\varepsilon_s}{\varepsilon_0}\right)\left(\frac{m_0}{m_c^*}\right) a_0. \end{aligned} \tag{7.7}$$

A large dielectric constant weakens the Coulomb attraction, and a small effective mass raises the kinetic cost of localization. Both effects reduce the binding energy and enlarge the orbital, explaining why shallow-donor wave functions can cover many lattice sites.

The analogous acceptor model replaces the conduction electron mass by an appropriate hole mass and gives

$$E_A = E_\mathrm{V} + E_{B,A}. \tag{7.8}$$

Within this scale estimate, Sze and Lee quote donor ionization energies of about $2.5\times10^{-2}\,\mathrm{eV}$ in silicon and $7\times10^{-3}\,\mathrm{eV}$ in gallium arsenide, and an acceptor scale of about $5\times10^{-2}\,\mathrm{eV}$ in both materials [7, Sec. 1.6.1]. These are pedagogical hydrogenic estimates rather than universal species-resolved measurements. The $4.5\times10^{1}\,\mathrm{meV}$ value used later in the illustrative silicon calculation is a representative shallow-donor binding scale, deliberately showing that a computational model must declare its chosen dopant parameter.

Model boundary

The hydrogenic model predicts the correct scale for many shallow impurities, but it is not a universal impurity theory. Anisotropic and multivalley bands, valence-band mixing, central-cell corrections, chemical bonding, multiple charge states, and deep defects require more detailed models. A deep level should not be forced into a shallow-dopant formula merely because both appear inside the gap.

7.4 Localized-state statistics and degeneracy

Band states form a continuum and use the Fermi–Dirac occupation derived from the grand partition function in chapter 4. A dilute donor or acceptor level is a localized state with a small number of charge configurations. Its occupation therefore includes the degeneracies of those configurations.

This chapter adopts the Sze–Ng convention [8, Sec. 1.4.3]:

$$\boxed{N_D^+ = \frac{N_D}{1 + g_D \exp[(\mu - E_D)/(k_\mathrm{B}T)]}, \qquad g_D = 2,} \tag{7.9}$$

and

$$\boxed{N_A^- = \frac{N_A}{1 + g_A \exp[(E_A - \mu)/(k_\mathrm{B}T)]}, \qquad g_A = 4.} \tag{7.10}$$

The numerical degeneracy factor is inseparable from the convention used to place it in the formula. Rudan, for example, parameterizes the localized-state probabilities with an inverse degeneracy coefficient [6, Secs. 18.4.1–18.4.2]. Copying a bare value such as 2 or 4 between conventions without copying its defining equation gives the wrong ionized fraction.

7.4.1 Donor partition argument

A singly ionizable donor has two relevant configurations: the empty, positively charged state D^+ and the occupied, neutral state D^0. Let the statistical weight of D^+ define the reference. The occupied state has the relative grand canonical weight

$$\frac{w(D^0)}{w(D^+)} = g_D \exp\left(\frac{\mu - E_D}{k_\mathrm{B}T}\right). \tag{7.11}$$

Normalizing the two weights gives

$$\begin{aligned} \mathbb{P}(D^+) &= \frac{w(D^+)}{w(D^+) + w(D^0)} \\ &= \frac{1}{1 + g_D \exp[(\mu - E_D)/(k_\mathrm{B}T)]}, \end{aligned} \tag{7.12}$$

which yields equation (7.9) after multiplication by N_D. Raising μ makes electron occupation more favorable and therefore reduces the positive ionized fraction.

7.4.2 Acceptor partition argument

For an acceptor, the negatively charged state A^- contains the captured electron. Under the adopted convention, the ratio of neutral to ionized weights is

$$\frac{w(A^0)}{w(A^-)} = g_A \exp\left(\frac{E_A - \mu}{k_\mathrm{B}T}\right). \tag{7.13}$$

Therefore

$$\begin{aligned} \mathbb{P}(A^-) &= \frac{w(A^-)}{w(A^-) + w(A^0)} \\ &= \frac{1}{1 + g_A \exp[(E_A - \mu)/(k_\mathrm{B}T)]}, \end{aligned} \tag{7.14}$$

which gives equation (7.10). Raising μ now makes capture of an electron more favorable and increases the negative ionized fraction.

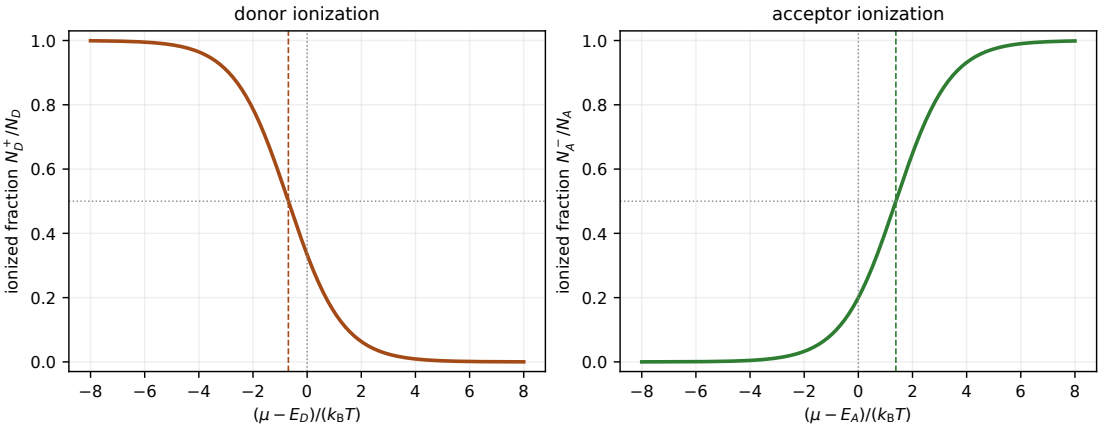


Figure 7.2: Ionized donor and acceptor fractions using the Sze–Ng degeneracy convention. The half-ionization point is displaced from $\mu = E_D$ or $\mu = E_A$ by the logarithm of the corresponding degeneracy factor.[2]

Indeed, setting either fraction to one half gives

$$\mu_{D,1/2} = E_D - k_B T \ln g_D, \tag{7.15}$$

$$\mu_{A,1/2} = E_A + k_B T \ln g_A. \tag{7.16}$$

This is why the statement "a level is half occupied when μ equals the level energy" is not automatically valid for a degenerate localized state.

7.5 The master charge-neutrality problem

For a spatially uniform semiconductor containing both donors and acceptors, the charge density is

$$\rho = q[p + N_D^+ - n - N_A^-]. \tag{7.17}$$

Thermal equilibrium in a neutral bulk region requires

$$\boxed{n + N_A^- = p + N_D^+.} \tag{7.18}$$

This is the central closure equation of the extrinsic problem. Every term is a concentration of charge with a declared sign. The equation does not say that $n = N_D$ or $p = N_A$; those approximations may emerge after solving it.

With parabolic bands and full Fermi–Dirac statistics,

$$n(\mu, T) = N_C F_{1/2}\left(\frac{\mu - E_C}{k_B T}\right), \tag{7.19}$$

$$p(\mu, T) = N_V F_{1/2}\left(\frac{E_V - \mu}{k_B T}\right). \tag{7.20}$$

[2] *Student reproduction notebook:* figures/07_extrinsic-semiconductor/fig_07_02.ipynb.

Substituting equations (7.9) and (7.10) produces one nonlinear equation for μ:

$$R(\mu) = N_{\mathrm{C}} F_{1/2}\left(\frac{\mu - E_{\mathrm{C}}}{k_{\mathrm{B}}T}\right) + \frac{N_A}{1 + g_A e^{(E_A-\mu)/(k_{\mathrm{B}}T)}} - N_{\mathrm{V}} F_{1/2}\left(\frac{E_{\mathrm{V}} - \mu}{k_{\mathrm{B}}T}\right) - \frac{N_D}{1 + g_D e^{(\mu-E_D)/(k_{\mathrm{B}}T)}} = 0. \tag{7.21}$$

As μ increases, n and N_A^- increase while p and N_D^+ decrease. Consequently $R(\mu)$ is monotonic under this model, making a bracketed scalar root solver a natural numerical method. The returned chemical potential is acceptable only if the neutrality residual is small compared with the largest charge concentration in the calculation.

7.6 Nondegenerate carrier relations

When the chemical potential lies several $k_{\mathrm{B}}T$ from both band edges,

$$n = N_{\mathrm{C}} \exp\left[-\frac{E_{\mathrm{C}} - \mu}{k_{\mathrm{B}}T}\right], \tag{7.22}$$

$$p = N_{\mathrm{V}} \exp\left[-\frac{\mu - E_{\mathrm{V}}}{k_{\mathrm{B}}T}\right]. \tag{7.23}$$

Using the intrinsic quantities derived in chapter 6, these may be rewritten as

$$n = n_i \exp\left(\frac{\mu - \mu_{\mathrm{i}}}{k_{\mathrm{B}}T}\right), \tag{7.24}$$

$$p = n_i \exp\left(-\frac{\mu - \mu_{\mathrm{i}}}{k_{\mathrm{B}}T}\right). \tag{7.25}$$

Multiplication gives the equilibrium mass-action law

$$np = n_i^2. \tag{7.26}$$

A donor raises μ, increasing n exponentially while decreasing p by the same exponential factor. An acceptor lowers μ and reverses those roles. The dopant changes the balance between carrier types; it does not violate thermal equilibrium.

7.7 Complete ionization and compensation

Suppose all shallow donors and acceptors are ionized:

$$N_D^+ \simeq N_D, \qquad N_A^- \simeq N_A. \tag{7.27}$$

Define the net doping

$$\Delta N = N_D - N_A. \tag{7.28}$$

Charge neutrality becomes

$$n - p = \Delta N. \tag{7.29}$$

Use $p = n_i^2/n$ to obtain

$$n^2 - \Delta N\, n - n_i^2 = 0. \tag{7.30}$$

The physical positive root is

$$\boxed{n = \frac{\Delta N + \sqrt{\Delta N^2 + 4n_i^2}}{2}.} \tag{7.31}$$

Similarly,

$$\boxed{p = \frac{-\Delta N + \sqrt{\Delta N^2 + 4n_i^2}}{2}.} \tag{7.32}$$

These two equations cover n-type, intrinsic, compensated, and p-type limits without changing formulas.

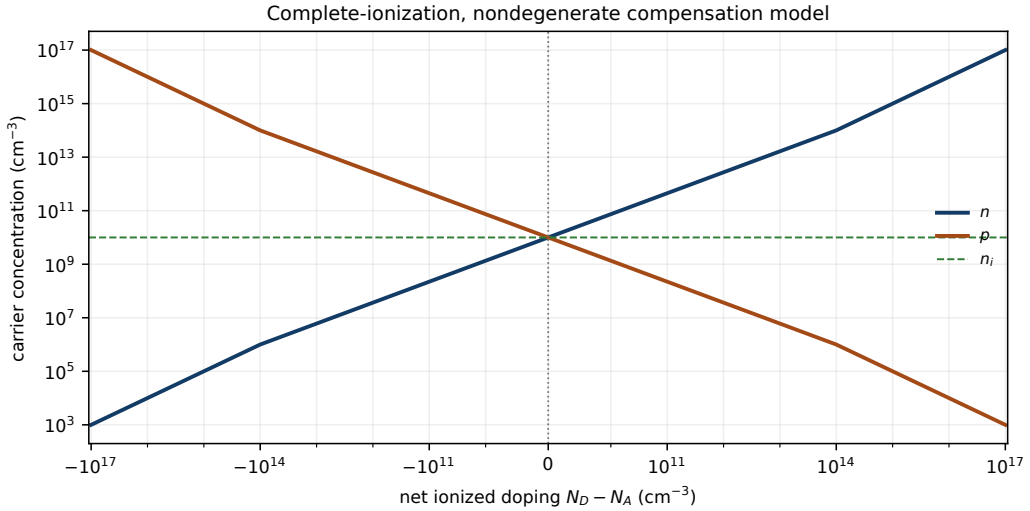


Figure 7.3: Carrier populations versus net ionized doping in the complete- ionization, nondegenerate model. The transition through compensation is continuous; the majority-carrier approximation fails when $|N_D - N_A|$ is comparable to n_i.[3]

7.7.1 Strong n-type limit

If $\Delta N \gg n_i$, expand

$$\sqrt{\Delta N^2 + 4n_i^2} = \Delta N \sqrt{1 + \frac{4n_i^2}{\Delta N^2}} \simeq \Delta N + \frac{2n_i^2}{\Delta N}. \tag{7.33}$$

Equations (7.31) and (7.32) reduce to

$$n \simeq \Delta N, \qquad p \simeq \frac{n_i^2}{\Delta N}. \tag{7.34}$$

Electrons are majority carriers and holes are minority carriers. The minority population is small but not identically zero.

[3] *Student reproduction notebook:* `figures/07_extrinsic-semiconductor/fig_07_03.ipynb`.

7.7.2 Strong p-type limit

If $-\Delta N \gg n_i$,

$$p \simeq -\Delta N = N_A - N_D, \qquad n \simeq \frac{n_i^2}{N_A - N_D}. \tag{7.35}$$

Holes are now majority carriers and electrons are minority carriers.

7.7.3 Fully compensated material

When $N_D = N_A$ under complete ionization, $\Delta N = 0$ and

$$n = p = n_i. \tag{7.36}$$

The crystal is not chemically intrinsic: it contains charged donors and acceptors. It is electrically compensated in this ideal bulk model. The impurities can still affect scattering, lifetime, disorder, and band edges even though the mobile equilibrium concentrations equal their intrinsic values.

7.8 Fermi level in the complete-ionization regime

In a strongly n-type, nondegenerate semiconductor with negligible compensation,

$$n \simeq N_D. \tag{7.37}$$

Substituting into the conduction-band carrier relation gives

$$\boxed{E_{\mathrm{C}} - \mu = k_{\mathrm{B}} T \ln\left(\frac{N_{\mathrm{C}}}{N_D}\right).} \tag{7.38}$$

Equivalently,

$$\mu - \mu_{\mathrm{i}} = k_{\mathrm{B}} T \ln\left(\frac{N_D}{n_i}\right). \tag{7.39}$$

Increasing N_D moves μ upward toward E_{C}. For p-type material,

$$\boxed{\mu - E_{\mathrm{V}} = k_{\mathrm{B}} T \ln\left(\frac{N_{\mathrm{V}}}{N_A}\right),} \tag{7.40}$$

and

$$\mu_{\mathrm{i}} - \mu = k_{\mathrm{B}} T \ln\left(\frac{N_A}{n_i}\right). \tag{7.41}$$

Increasing N_A moves μ downward toward E_{V}.

These logarithmic equations contain their own warning. If a trial doping makes μ approach a band edge within a few $k_{\mathrm{B}}T$, the Boltzmann approximation used to derive them is failing, and the full Fermi integral must replace it.

7.9 Incomplete ionization and freeze-out

At sufficiently low temperature, many dopant-bound carriers lack the thermal energy required to reach the relevant band. Consider uncompensated donor doping, neglect holes, and retain nondegenerate band statistics. Neutrality is

$$n = N_D^+. \tag{7.42}$$

The band electron concentration gives

$$e^{(\mu - E_\mathrm{C})/(k_\mathrm{B}T)} = \frac{n}{N_\mathrm{C}}. \tag{7.43}$$

Let the donor ionization energy be

$$E_{I,D} = E_\mathrm{C} - E_D > 0. \tag{7.44}$$

Then

$$\begin{aligned} e^{(\mu - E_D)/(k_\mathrm{B}T)} &= e^{(\mu - E_\mathrm{C})/(k_\mathrm{B}T)} e^{(E_\mathrm{C} - E_D)/(k_\mathrm{B}T)} \\ &= \frac{n}{N_\mathrm{C}} e^{E_{I,D}/(k_\mathrm{B}T)}. \end{aligned} \tag{7.45}$$

Substitution into the donor-ionization formula and use of $n = N_D^+$ gives

$$\frac{g_D}{N_\mathrm{C}} e^{E_{I,D}/(k_\mathrm{B}T)} n^2 + n - N_D = 0. \tag{7.46}$$

The positive solution is

$$n = \frac{N_\mathrm{C} e^{-E_{I,D}/(k_\mathrm{B}T)}}{2g_D} \left[\sqrt{1 + \frac{4 g_D N_D}{N_\mathrm{C}} e^{E_{I,D}/(k_\mathrm{B}T)}} - 1 \right]. \tag{7.47}$$

Deep in freeze-out, the quadratic term dominates the linear term in equation (7.46), producing

$$\boxed{n \simeq \sqrt{\frac{N_D N_\mathrm{C}}{g_D}} \exp\left(-\frac{E_{I,D}}{2k_\mathrm{B}T} \right).} \tag{7.48}$$

The activation energy in this ideal uncompensated asymptote is one half of the donor binding energy because the chemical potential moves as ionization proceeds. Reading a freeze-out slope as $E_{I,D}$ without deriving the relevant limit can therefore introduce a factor-of-two error.

For uncompensated acceptors, define

$$E_{I,A} = E_A - E_\mathrm{V} > 0. \tag{7.49}$$

The same calculation yields

$$p \simeq \sqrt{\frac{N_A N_\mathrm{V}}{g_A}} \exp\left(-\frac{E_{I,A}}{2k_\mathrm{B}T} \right) \tag{7.50}$$

under the same idealized assumptions. Compensation can change both the prefactor and the observed activation behavior.

7.10 Three temperature regimes

The temperature dependence of a shallow-donor sample is commonly divided into three regimes:

1. **Freeze-out:** $N_D^+ < N_D$ and the carrier concentration is controlled by dopant ionization.
2. **Extrinsic or saturation regime:** the dopants are nearly fully ionized, intrinsic generation is still small, and $n \simeq N_D - N_A$ for net n-type material.
3. **Intrinsic regime:** thermally generated electron–hole pairs outnumber the net dopant contribution, so $n \simeq p \simeq n_i$ and μ approaches μ_{i}.

The boundaries are crossovers rather than phase transitions. Their locations depend on the dopant energy, concentration, compensation, host gap, effective densities of states, and the selected accuracy threshold.

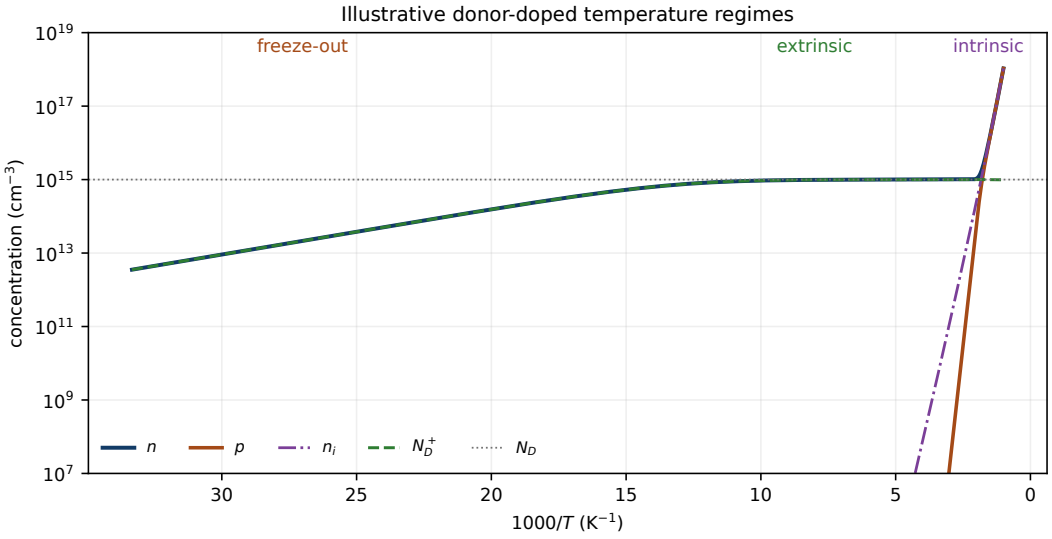


Figure 7.4: Illustrative donor-doped silicon calculation obtained by solving charge neutrality with partial donor ionization at every temperature. The model uses $N_D = 1 \times 10^{15}/\mathrm{cm}^3$, $E_{I,D} = 4.5 \times 10^1$ meV, $g_D = 2$, parabolic nondegenerate bands, and the same silicon gap and DOS convention used in chapter 6. The curves are model evidence, not a substitute for a material-specific dopant dataset.[4]

The plot is often presented against $1/T$ because activated regimes become approximately linear on a logarithmic vertical scale. Care is still required: $N_{\mathrm{C}} \propto T^{3/2}$, E_{g} varies with temperature, and μ is itself a solution of neutrality, so no entire curve is described by one activation energy.

7.11 Worked n-type silicon estimate at 3.00×10^2 K

Following the illustrative Sze–Lee calculation [7, Example 4 in Sec. 1.6.1], take

$$N_D = 1 \times 10^{16}/\mathrm{cm}^3, \qquad N_A = 0, \tag{7.51}$$

[4] *Student reproduction notebook:* figures/07_extrinsic-semiconductor/fig_07_04.ipynb.

with complete ionization and the convention

$$n_i = 9.65 \times 10^{9}/\mathrm{cm}^3, \qquad N_{\mathrm{C}} = 2.86 \times 10^{19}/\mathrm{cm}^3, \qquad k_{\mathrm{B}}T = 2.59 \times 10^{-2}\,\mathrm{eV}. \tag{7.52}$$

Because $N_D \gg n_i$,

$$n \simeq N_D = 1 \times 10^{16}/\mathrm{cm}^3. \tag{7.53}$$

The minority holes follow from mass action:

$$\begin{aligned} p &= \frac{n_i^2}{n} \\ &= \frac{(9.65 \times 10^{9})^2}{10^{16}} \\ &\simeq 9.3 \times 10^{3}/\mathrm{cm}^3. \end{aligned} \tag{7.54}$$

The distance from the conduction-band edge is

$$\begin{aligned} E_{\mathrm{C}} - \mu &= k_{\mathrm{B}}T \ln\left(\frac{N_{\mathrm{C}}}{N_D}\right) \\ &= 0.0259 \ln\left(\frac{2.86 \times 10^{19}}{10^{16}}\right) \\ &\simeq 2.05 \times 10^{-1}\,\mathrm{eV}. \end{aligned} \tag{7.55}$$

The chemical potential is therefore still inside the gap and more than several $k_{\mathrm{B}}T$ below E_{C}, so the nondegenerate approximation is self-consistent. Using μ_{i} as the reference gives

$$\mu - \mu_{\mathrm{i}} = k_{\mathrm{B}}T \ln\left(\frac{N_D}{n_i}\right) \simeq 3.58 \times 10^{-1}\,\mathrm{eV}. \tag{7.56}$$

7.12 Degenerate doping and impurity-band limits

When μ approaches or enters a band, the Boltzmann relations and the simple mass-action law must be abandoned. The full Fermi integrals in equations (7.19) and (7.20) are then required. High doping can also produce effects that are separate from degeneracy:

- dopant wave functions overlap and broaden discrete impurity levels into impurity bands;
- ionized-impurity and carrier–carrier interactions modify the band edges;
- the effective gap can narrow;
- screening changes dopant binding energies; and
- simple independent localized-state occupations cease to be adequate.

Sze and Lee emphasize degeneracy and band-gap narrowing at high doping [7, Sec. 1.6.2]; Rudan treats impurity-band overlap and the resulting effective band-edge shifts explicitly [6, Sec. 18.6].

If a model represents band-gap narrowing by shifts $\Delta E_{\mathrm{C}} > 0$ and $\Delta E_{\mathrm{V}} > 0$,

$$E'_{\mathrm{C}} = E_{\mathrm{C}} - \Delta E_{\mathrm{C}}, \qquad E'_{\mathrm{V}} = E_{\mathrm{V}} + \Delta E_{\mathrm{V}}, \tag{7.57}$$

then

$$E_{\mathrm{g}}' = E_{\mathrm{g}} - \Delta E_{\mathrm{g}}, \qquad \Delta E_{\mathrm{g}} = \Delta E_{\mathrm{C}} + \Delta E_{\mathrm{V}}. \tag{7.58}$$

Because intrinsic-like products depend exponentially on the gap, even a modest ΔE_{g} can strongly affect minority concentrations. The correction must be based on a declared empirical or many-body model rather than inserted as an unlabeled fitting constant.

7.13 Why local neutrality fails for nonuniform doping

Equation (7.18) is appropriate for a uniform neutral bulk region. If N_D or N_A varies with position, carriers diffuse. The immobile ionized dopants do not follow them, so a local space charge and an electric field develop. The general charge density is

$$\rho(\boldsymbol{r}) = q\left[p(\boldsymbol{r}) - n(\boldsymbol{r}) + N_D^{+}(\boldsymbol{r}) - N_A^{-}(\boldsymbol{r})\right], \tag{7.59}$$

and the electrostatic potential satisfies

$$-\nabla \cdot \left(\varepsilon_s \nabla \phi\right) = \rho(\phi, \boldsymbol{r}). \tag{7.60}$$

Carrier and dopant occupations depend on the shifted local energies, so equation (7.60) is nonlinear and must be solved self-consistently. Replacing it pointwise by $\rho = 0$ would erase depletion regions, built-in fields, and junction electrostatics.

7.14 Computational laboratory: solving neutrality

The following compact calculation uses nondegenerate bands but retains partial donor and acceptor ionization. It brackets the chemical potential and verifies the returned charge residual. Energy quantities are in electronvolts and concentrations are in $1/\mathrm{cm}^3$.

Listing 7.1: Bracketed equilibrium solve with incomplete ionization.

```
from math import exp

KB_EV = 8.617333262145e-5
T = 300.0
EV = 0.0
EC = 1.12
ED = EC - 0.045
EA = EV + 0.045
NC = 2.86e19
NV = 2.66e19
ND = 1.0e16
NA = 2.0e15
GD = 2.0
GA = 4.0
thermal = KB_EV * T


def populations(mu):
    n = NC * exp((mu - EC) / thermal)
```

```
    p = NV * exp((EV - mu) / thermal)
    donor_plus = ND / (1.0 + GD * exp((mu - ED) / thermal))
    acceptor_minus = NA / (1.0 + GA * exp((EA - mu) / thermal))
    return n, p, donor_plus, acceptor_minus


def residual(mu):
    n, p, donor_plus, acceptor_minus = populations(mu)
    return n + acceptor_minus - p - donor_plus


lower = EV - 0.5
upper = EC + 0.5
assert residual(lower) < 0.0 < residual(upper)

for _ in range(200):
    middle = 0.5 * (lower + upper)
    if residual(middle) > 0.0:
        upper = middle
    else:
        lower = middle

mu = 0.5 * (lower + upper)
n, p, donor_plus, acceptor_minus = populations(mu)
scale = max(n, p, donor_plus, acceptor_minus)
assert abs(residual(mu)) / scale < 1.0e-12
assert 0.0 <= donor_plus <= ND
assert 0.0 <= acceptor_minus <= NA
assert abs(n * p - NC * NV * exp(-(EC - EV) / thermal)) / (n * p) < 1.0e-12
```

The calculation returns

$$\begin{aligned} \mu - E_{\mathrm{V}} &= 9.084 \times 10^{-1}\,\mathrm{eV}, \\ n &= 7.968 \times 10^{15}/\mathrm{cm}^3, \\ p &= 1.461 \times 10^{4}/\mathrm{cm}^3, \\ N_D^+ &= 9.968 \times 10^{15}/\mathrm{cm}^3, \\ N_A^- &= 2.000 \times 10^{15}/\mathrm{cm}^3. \end{aligned} \tag{7.61}$$

The six-per-cubic-centimetre absolute residual remaining in double precision is less than 10^{-15} of the dominant concentration. The code does not assume $n = N_D - N_A$. If complete ionization and strong net n-type behavior are valid, that result should emerge from the solution and can then be used as a checked approximation.

7.15 Verification checklist

An extrinsic-semiconductor calculation should establish that

1. every dopant species has a declared chemical identity, lattice role, energy level, concentration, and degeneracy convention;
2. $0 \le N_D^+ \le N_D$ and $0 \le N_A^- \le N_A$;

3. the charge-neutrality residual is small compared with the dominant charge density;
4. the solution approaches the intrinsic result as $N_D, N_A \to 0$;
5. the complete-ionization quadratic is recovered when the ionized fractions approach unity;
6. the majority and minority limits satisfy $np = n_i^2$ only where the nondegenerate approximation is valid;
7. freeze-out, extrinsic, and intrinsic regimes are identified from the solved populations rather than imposed labels;
8. Fermi-integral and band-gap-narrowing corrections are activated before their simplified assumptions fail; and
9. a nonuniform doping profile is coupled to Poisson's equation instead of being forced into local neutrality.

7.16 Chapter summary

Doping adds localized charge states to the host semiconductor and shifts the equilibrium chemical potential. Donor and acceptor concentrations become mobile-carrier concentrations only through ionization statistics and charge neutrality. In the complete-ionization, nondegenerate regime, the familiar majority-carrier estimates $n \simeq N_D - N_A$ or $p \simeq N_A - N_D$ follow from a quadratic solution together with $np = n_i^2$. At low temperature, incomplete ionization produces freeze-out; at high temperature, intrinsic pair generation dominates; and at high doping, degeneracy, impurity bands, screening, and band-gap narrowing require richer models. The master principle is unchanged: solve the charged populations self-consistently, then justify every reduction used to simplify them.

Part II

Transport and Nonequilibrium

Chapter 8

Brownian Motion, Diffusion, and the Heat Equation

Transport describes the redistribution of particles, charge, energy, or another conserved quantity. A microscopic trajectory is irregular, while an ensemble of trajectories has reproducible moments and a deterministic density. This chapter derives the diffusion equation from a random walk, formulates the same limit as a stochastic differential equation, constructs Fick's law from particle crossings, and develops the matrix exponential used to propagate a discretized diffusion equation. The final sections apply these results to electron and hole currents in semiconductors.

The derivation follows the transport sequence of Sze and Lee [7, Chap. 2] and the distribution-function hierarchy developed by Rudan [6, Chaps. 6 and 19]. The stochastic formulation uses the Wiener and Itô framework [2, 9].

8.1 The random-walk limit

Consider a one-dimensional walker whose independent increments have duration Δt and values

$$\Delta X_s = \begin{cases} +\ell, & \text{with probability } 1/2, \\ -\ell, & \text{with probability } 1/2. \end{cases} \tag{8.1}$$

The first two moments are

$$\langle \Delta X_s \rangle = 0, \qquad \langle \Delta X_s \Delta X_{s'} \rangle = \ell^2 \delta_{ss'}. \tag{8.2}$$

After N steps,

$$X_N - X_0 = \sum_{s=1}^{N} \Delta X_s. \tag{8.3}$$

Independence removes the cross terms in the second moment, so

$$\langle X_N - X_0 \rangle = 0, \qquad \left\langle (X_N - X_0)^2 \right\rangle = N\ell^2. \tag{8.4}$$

Writing $t = N\Delta t$ and defining

$$D = \frac{\ell^2}{2\Delta t} \tag{8.5}$$

gives the one-dimensional mean-square-displacement law

$$\boxed{\left\langle (X(t) - X_0)^2 \right\rangle = 2Dt.} \tag{8.6}$$

For isotropic diffusion in d dimensions,

$$\boxed{\left\langle |\boldsymbol{R}(t) - \boldsymbol{R}_0|^2 \right\rangle = 2dDt.} \tag{8.7}$$

The linear dependence on time defines normal diffusion. Motion with a persistent velocity is ballistic and has a mean-square displacement proportional to t^2 at short times.

The diffusion limit requires observation lengths much larger than ℓ and observation times much longer than Δt. Correlated increments, broad waiting-time distributions, and long jumps produce transport laws that do not reduce to equation (8.7).

8.2 Brownian motion and drift–diffusion dynamics

A Wiener process $\boldsymbol{W}_t$ has independent Gaussian increments with

$$\mathbb{E}[\mathrm{d}\boldsymbol{W}_t] = \boldsymbol{0}, \qquad \mathbb{E}[\mathrm{d}\boldsymbol{W}_t\, \mathrm{d}\boldsymbol{W}_t^{\mathsf{T}}] = I\ \mathrm{d}t. \tag{8.8}$$

The Itô stochastic differential equation

$$\mathrm{d}\boldsymbol{R}_t = \boldsymbol{b}(\boldsymbol{R}_t, t)\ \mathrm{d}t + B(\boldsymbol{R}_t, t)\ \mathrm{d}\boldsymbol{W}_t \tag{8.9}$$

contains a deterministic drift velocity $\boldsymbol{b}$ and stochastic increments with diffusion tensor

$$\boldsymbol{D} = \frac{1}{2} BB^{\mathsf{T}}. \tag{8.10}$$

For constant isotropic diffusion, $B = \sqrt{2D}\, I$ and

$$\mathrm{d}\boldsymbol{R}_t = \boldsymbol{b}\ \mathrm{d}t + \sqrt{2D}\ \mathrm{d}\boldsymbol{W}_t. \tag{8.11}$$

Its ensemble moments are

$$\mathbb{E}[\boldsymbol{R}_t] = \boldsymbol{R}_0 + \boldsymbol{b}t, \qquad \mathrm{Cov}(\boldsymbol{R}_t) = 2Dt\, I. \tag{8.12}$$

Drift controls the mean position. Diffusion controls spreading about that mean. The uncentered second moment contains both contributions and therefore cannot be used as a diffusion estimate when $\boldsymbol{b} \neq \boldsymbol{0}$.

For a smooth scalar function $f(\boldsymbol{R}, t)$, Itô's formula gives

$$\mathrm{d}f = \left[\frac{\partial f}{\partial t} + \boldsymbol{b} \cdot \nabla f + \sum_{ij} D_{ij} \frac{\partial^2 f}{\partial x_i \partial x_j} \right] \mathrm{d}t + (\nabla f)^{\mathsf{T}} B\ \mathrm{d}\boldsymbol{W}_t. \tag{8.13}$$

The second-derivative term is the Itô correction. It converts stochastic path dynamics into diffusion of the ensemble density.

A direct numerical approximation is the Euler–Maruyama update

$$\boldsymbol{R}_{r+1} = \boldsymbol{R}_r + \boldsymbol{b}(\boldsymbol{R}_r, t_r)\Delta t + B(\boldsymbol{R}_r, t_r)\sqrt{\Delta t}\, \boldsymbol{\eta}_r, \tag{8.14}$$

where the components of $\boldsymbol{\eta}_r$ are independent standard normal random variables. Independent particles require independent random vectors. Convergence must be checked by reducing Δt and increasing the ensemble size [1, 4].

[1] *Student reproduction notebook:* `06: Euler-Maruyama drift-diffusion`.

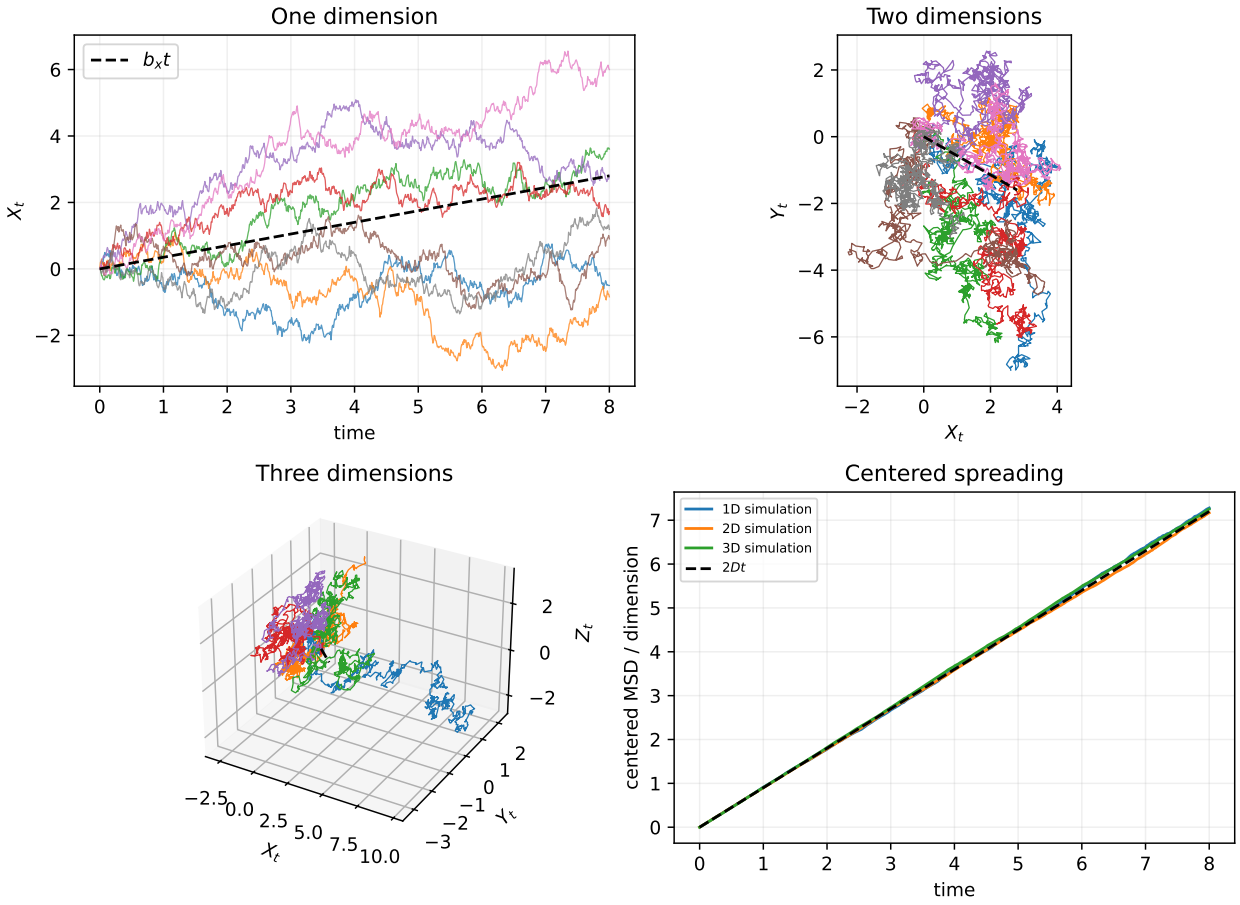


Figure 8.1: Euler–Maruyama simulations of drift–diffusion in one, two, and three dimensions. Individual paths fluctuate around the ensemble drift, while the centered covariance grows according to the prescribed diffusion tensor.[1]

8.3 From stochastic paths to an ensemble equation

Let $P(\boldsymbol{r},t)$ denote the probability density of equation (8.9). The adjoint of the Itô generator gives the Fokker–Planck equation

$$\frac{\partial P}{\partial t} = -\sum_i \frac{\partial}{\partial x_i}(b_i P) + \sum_{ij} \frac{\partial^2}{\partial x_i \partial x_j}(D_{ij} P). \tag{8.15}$$

For constant isotropic D,

$$\frac{\partial P}{\partial t} = -\nabla \cdot (\boldsymbol{b} P) + D\nabla^2 P. \tag{8.16}$$

Writing this equation as a conservation law identifies the probability current:

$$\frac{\partial P}{\partial t} + \nabla \cdot \boldsymbol{J}_P = 0, \qquad \boxed{\boldsymbol{J}_P = \boldsymbol{b}P - D\nabla P.} \tag{8.17}$$

The current contains advective transport with the drift and diffusive transport down the density gradient.

A point source containing N_0 particles in an infinite isotropic medium has the density

$$n(\boldsymbol{r},t) = \frac{N_0}{(4\pi D t)^{d/2}} \exp\!\left(-\frac{|\boldsymbol{r}|^2}{4Dt}\right). \tag{8.18}$$

Its spatial integral is N_0, its mean position is zero, and its mean-square radius is $2dDt$. These three relations test normalization, symmetry, and the diffusive width independently.

8.4 The master equation and Fick's law

The continuum equation also follows directly from a lattice walk. Let P_i^N be the probability at $x_i = i\ell$ after N steps. Nearest-neighbor motion gives

$$P_i^{N+1} = \frac{1}{2}P_{i-1}^N + \frac{1}{2}P_{i+1}^N. \tag{8.19}$$

Subtracting P_i^N, dividing by Δt, and using equation (8.5) yields

$$\frac{P_i^{N+1} - P_i^N}{\Delta t} = D\frac{P_{i+1}^N - 2P_i^N + P_{i-1}^N}{\ell^2}. \tag{8.20}$$

The continuum limit is

$$\boxed{\frac{\partial P}{\partial t} = D\frac{\partial^2 P}{\partial x^2}.} \tag{8.21}$$

The same crossing count produces the constitutive flux. If n_i is a one-dimensional number density, the expected signed particle current through the face between cells i and $i+1$ is

$$\Gamma_{i+1/2} = \frac{\ell}{2\Delta t}(n_i - n_{i+1}). \tag{8.22}$$

For a slowly varying density, $n_{i+1} - n_i = \ell\,\partial_x n + \mathcal{O}(\ell^2)$, so

$$\boxed{\boldsymbol{\Gamma} = -D\nabla n.} \tag{8.23}$$

This is Fick's first law. Its sign states that the net particle flux points down the density gradient.

Local conservation of particle number is

$$\frac{\partial n}{\partial t} + \nabla \cdot \mathbf{\Gamma} = S, \tag{8.24}$$

where S is the net volumetric source rate. For $S = 0$, substitution of equation (8.23) gives

$$\boxed{\frac{\partial n}{\partial t} = \nabla \cdot (D \nabla n).} \tag{8.25}$$

If D is uniform, this reduces to $\partial_t n = D\nabla^2 n$. A spatially varying D remains inside the divergence. Replacing $\nabla \cdot (D\nabla n)$ by $D\nabla^2 n$ would omit $(\nabla D) \cdot \nabla n$.

8.5 The heat equation

Let u be thermal energy per unit volume and $\boldsymbol{j}_Q$ the heat flux. Energy conservation without volumetric heating is

$$\frac{\partial u}{\partial t} + \nabla \cdot \boldsymbol{j}_Q = 0. \tag{8.26}$$

Fourier's law is

$$\boldsymbol{j}_Q = -\kappa \nabla T, \tag{8.27}$$

where κ is thermal conductivity. With approximately constant volumetric heat capacity C_V, $u = C_V T$, and therefore

$$C_V \frac{\partial T}{\partial t} = \nabla \cdot (\kappa \nabla T). \tag{8.28}$$

Uniform C_V and κ give

$$\boxed{\frac{\partial T}{\partial t} = \alpha \nabla^2 T, \qquad \alpha = \frac{\kappa}{C_V}.} \tag{8.29}$$

The heat equation and particle diffusion equation share a mathematical operator. Their fields, fluxes, constitutive coefficients, and conserved quantities remain distinct. Thermal conductivity κ, electrical conductivity σ, particle diffusivity D, and thermal diffusivity α are not interchangeable.

8.6 Matrix representation and exponential evolution

Consider M periodic cells of width h. Define the incidence matrix B so that row i computes $f_{i+1} - f_i$, with the final row wrapping to the first cell. The discrete gradient is

$$G = \frac{B}{h}. \tag{8.30}$$

For particle diffusion, the face flux and finite-volume balance are

$$\mathbf{\Gamma} = -DG\boldsymbol{n}, \qquad \frac{\mathrm{d}\boldsymbol{n}}{\mathrm{d}t} = \frac{1}{h} B^{\mathsf{T}} \mathbf{\Gamma}. \tag{8.31}$$

Combining them gives

$$\frac{\mathrm{d}\boldsymbol{n}}{\mathrm{d}t} = DL\boldsymbol{n}, \qquad \boxed{L = -\frac{B^{\mathsf{T}}B}{h^2}.} \tag{8.32}$$

The factorization shows that L is symmetric and negative semidefinite. Its constant eigenvector has eigenvalue zero because periodic diffusion conserves the spatial sum.

Let

$$L = V\Lambda V^{\mathsf{T}}, \qquad V^{\mathsf{T}}V = I. \tag{8.33}$$

The exact semidiscrete solution is

$$\boxed{\boldsymbol{n}(t) = e^{DLt}\boldsymbol{n}(0) = Ve^{D\Lambda t}V^{\mathsf{T}}\boldsymbol{n}(0).} \tag{8.34}$$

Every negative eigenvalue produces exponential decay. The zero eigenvalue preserves the mean. The propagator satisfies

$$e^{DL(t+s)} = e^{DLt}e^{DLs}. \tag{8.35}$$

The explicit-Euler update is

$$\boldsymbol{n}^{r+1} = (I + D\Delta tL)\boldsymbol{n}^{r}. \tag{8.36}$$

For the centered one-dimensional Laplacian, stability requires $D\Delta t/h^2 \le 1/2$. The exponential propagator has no corresponding step-size stability restriction. Dense diagonalization serves small teaching problems; large sparse systems require Krylov, implicit, or operator-splitting methods.

Boundary conditions alter both conservation and the spectrum. Periodic and zero-flux boundaries preserve the total amount in a closed domain. Fixed-value boundaries exchange particles or energy with a reservoir. Absorbing particle boundaries remove probability from the modeled domain.

The calculation in figure 8.2 constructs a 64×64 periodic Laplacian with 192 nonzero entries. Its eigenvalues lie in $[-163.84, 0]$ to numerical precision. The conserved-integral drift is below 3.2×10^{-14}, and the matrix-exponential semigroup error is approximately 1.1×10^{-15} in the infinity norm.

The same discrete Laplacian appears in a free-particle Hamiltonian,

$$H = -\frac{\hbar^2}{2m}L. \tag{8.37}$$

A Laplacian eigenvalue $\lambda_j \le 0$ produces the quantum multiplier $\exp[-\mathrm{i}E_jt/\hbar]$, whose magnitude is one, and the diffusive multiplier $\exp(D\lambda_jt)$, which decays for $\lambda_j < 0$. The spatial modes are identical; the evolution laws are not.

8.7 Range of validity

The diffusion equation is a continuum closure. It applies when the modeled length is large compared with the microscopic step length and the modeled time is long compared with the velocity-memory time. Constant scalar diffusivity assumes an isotropic, homogeneous medium. Ballistic motion, correlated steps, spatially varying materials, and anomalous waiting-time statistics require a kinetic, tensor, or non-Fickian description.

[2] *Student reproduction notebook:* `Lab 3: matrix diffusion and exponential evolution`.

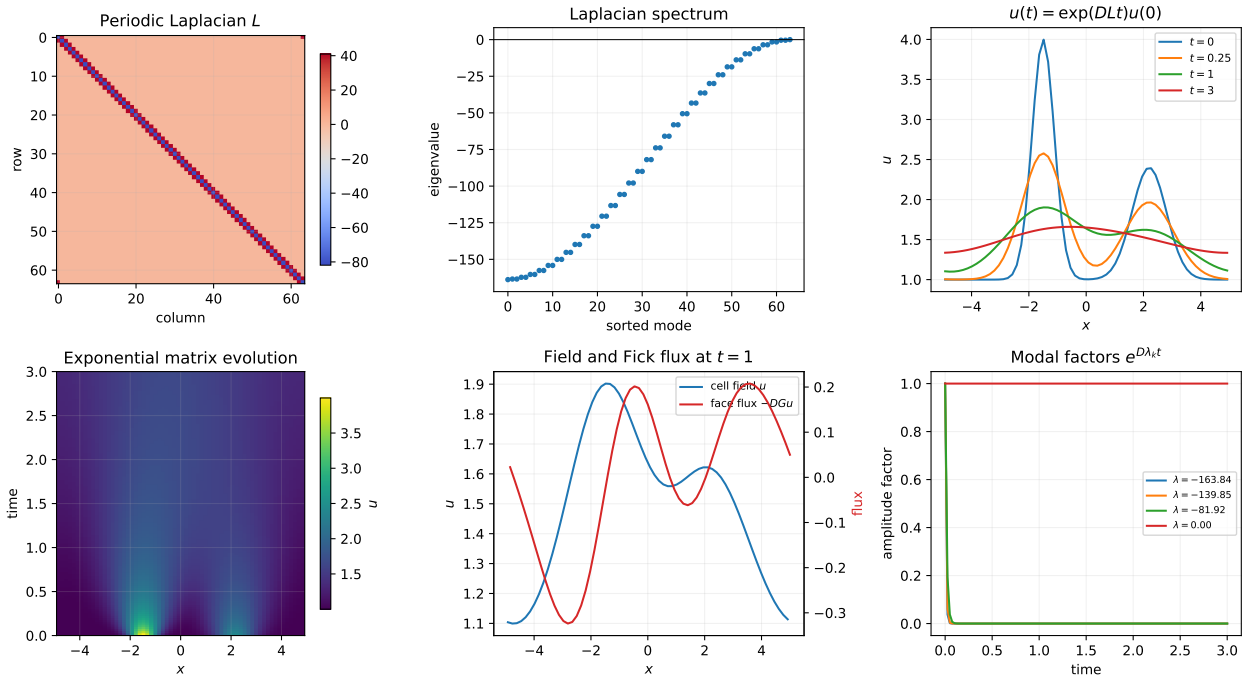


Figure 8.2: Matrix representation of periodic diffusion. The Laplacian spectrum is nonpositive, nonuniform modes decay exponentially, the constant mode is conserved, and the face flux follows from the discrete gradient.[2]

8.8 Chapter summary

Independent microscopic steps produce a mean-square displacement proportional to time. Their continuum limit is Brownian motion, whose ensemble density obeys the Fokker–Planck equation. Signed crossing counts produce Fick's law, and local conservation converts that constitutive law into the diffusion equation. Fourier's law gives the heat equation with the same Laplacian but a different physical field and coefficient. Spatial discretization yields $\dot{\boldsymbol{f}} = \chi L \boldsymbol{f}$, with exact semidiscrete evolution $\boldsymbol{f}(t) = e^{\chi L t}\boldsymbol{f}(0)$. The next chapter restores the velocity degree of freedom and derives the Maxwell–Boltzmann gas from underdamped Langevin dynamics.

Chapter 9

From Langevin Dynamics to the Maxwell–Boltzmann Gas

The diffusion equation resolves position after microscopic velocity memory has decayed. This chapter restores velocity through the underdamped Langevin equation. Thermal fluctuation and momentum relaxation produce the Maxwell–Boltzmann distribution. The same dynamics becomes ballistic at short time and Brownian at long time.

9.1 Underdamped Langevin dynamics

Let a particle of mass m and signed charge q_c have position $\boldsymbol{R}$ and velocity $\boldsymbol{V}$. A constant momentum-relaxation time τ gives

$$\mathrm{d}\boldsymbol{R} = \boldsymbol{V}\ \mathrm{d}t, \tag{9.1}$$

$$\mathrm{d}\boldsymbol{V} = -\frac{\boldsymbol{V}}{\tau}\ \mathrm{d}t + \frac{q_c}{m}\boldsymbol{E}\ \mathrm{d}t + \sqrt{\frac{2k_\mathrm{B}T}{m\tau}}\ \mathrm{d}\boldsymbol{W}_t. \tag{9.2}$$

The damping term removes directed momentum. The stochastic term restores the thermal fluctuations lost through damping. Their coefficients satisfy the fluctuation–dissipation relation.

With $\boldsymbol{E} = \boldsymbol{0}$, each velocity component is an Ornstein–Uhlenbeck process. Its stationary covariance is

$$\langle V_i V_j \rangle = \frac{k_\mathrm{B}T}{m}\delta_{ij}. \tag{9.3}$$

The associated three-dimensional velocity density is

$$f_\mathrm{MB}(\boldsymbol{v}) = \left(\frac{m}{2\pi k_\mathrm{B}T}\right)^{3/2} \exp\!\left(-\frac{m|\boldsymbol{v}|^2}{2k_\mathrm{B}T}\right). \tag{9.4}$$

Integration over velocity direction gives the Maxwell speed density

$$p(v) = 4\pi v^2 \left(\frac{m}{2\pi k_\mathrm{B}T}\right)^{3/2} \exp\!\left(-\frac{mv^2}{2k_\mathrm{B}T}\right), \qquad v \geq 0. \tag{9.5}$$

The second moment gives equipartition,

$$\frac{1}{2}m\left\langle |\boldsymbol{V}|^2 \right\rangle = \frac{3}{2}k_\mathrm{B}T. \tag{9.6}$$

The Maxwell–Boltzmann distribution therefore emerges from the same stochastic dynamics that produces Brownian diffusion.

9.2 Velocity memory and the Brownian limit

The equilibrium velocity autocorrelation in one Cartesian direction is

$$\langle V(t)V(0)\rangle = \frac{k_{\mathrm{B}}T}{m}e^{-t/\tau}. \tag{9.7}$$

Integrating the velocity twice gives

$$\left\langle [X(t)-X(0)]^2 \right\rangle = 2D\left[t-\tau\left(1-e^{-t/\tau}\right)\right], \tag{9.8}$$

with

$$\boxed{D=\frac{k_{\mathrm{B}}T\tau}{m}.} \tag{9.9}$$

For $t \ll \tau$, expansion of the exponential gives $\langle \Delta X^2\rangle = (k_{\mathrm{B}}T/m)t^2$. The particle retains its velocity and moves ballistically. For $t \gg \tau$, $\langle \Delta X^2\rangle = 2Dt$ apart from a constant offset. The overdamped Brownian model is the long-time limit of the thermal velocity process.

9.3 Chapter summary

The underdamped Langevin equation couples position, velocity, damping, and thermal noise. Its stationary velocity distribution is Maxwell–Boltzmann and satisfies equipartition. Exponential loss of velocity memory produces a ballistic mean-square displacement for $t \ll \tau$ and a diffusive mean-square displacement for $t \gg \tau$. The long-time diffusion coefficient is $D = k_{\mathrm{B}}T\tau/m$. The next chapter applies an electric field to the same velocity dynamics and obtains the Drude response.

Chapter 10

The Drude and Drude–Sommerfeld Electron Gases

An electric field shifts the mean velocity of a collision-dominated thermal gas. Classical Maxwell–Boltzmann occupation gives the Drude model. Fermi–Dirac occupation gives the Drude–Sommerfeld model and confines the linear response of a degenerate gas to the Fermi window.

10.1 The classical Drude response

Taking the ensemble average of equation (9.2) removes the zero-mean stochastic term:

$$m\frac{\mathrm{d}\langle \boldsymbol{V}\rangle}{\mathrm{d}t} = q_c\boldsymbol{E} - \frac{m}{\tau}\langle \boldsymbol{V}\rangle. \tag{10.1}$$

After a constant field is applied at $t = 0$,

$$\langle \boldsymbol{V}(t)\rangle = \frac{q_c\tau}{m}\boldsymbol{E}\left(1 - e^{-t/\tau}\right). \tag{10.2}$$

The steady drift velocity is

$$\boldsymbol{v}_\mathrm{d} = \frac{q_c\tau}{m}\boldsymbol{E}. \tag{10.3}$$

For carrier density n, the charge-current density is

$$\boldsymbol{J} = nq_c\boldsymbol{v}_\mathrm{d} = \frac{nq_c^2\tau}{m}\boldsymbol{E}. \tag{10.4}$$

The DC conductivity is therefore

$$\boxed{\sigma_0 = \frac{nq_c^2\tau}{m}.} \tag{10.5}$$

For electrons, $q_c = -q$. Electron drift is opposite the field, while conventional current is parallel to it.

For a harmonic field proportional to $e^{-\mathrm{i}\omega t}$, equation (10.1) gives

$$\boxed{\sigma(\omega) = \frac{\sigma_0}{1 - \mathrm{i}\omega\tau}.} \tag{10.6}$$

The response is nearly resistive for $\omega\tau \ll 1$. At $\omega\tau \gtrsim 1$, carrier momentum cannot relax within one field cycle and the current acquires a substantial phase lag.

The mechanical mobility magnitude is

$$\mu = \frac{|q_c|\tau}{m}. \tag{10.7}$$

Combining this result with equation (9.9) gives the Einstein relation

$$\boxed{D = \frac{\mu k_{\mathrm{B}} T}{|q_c|}}. \tag{10.8}$$

Diffusion and field-driven mobility contain the same momentum-relaxation time.

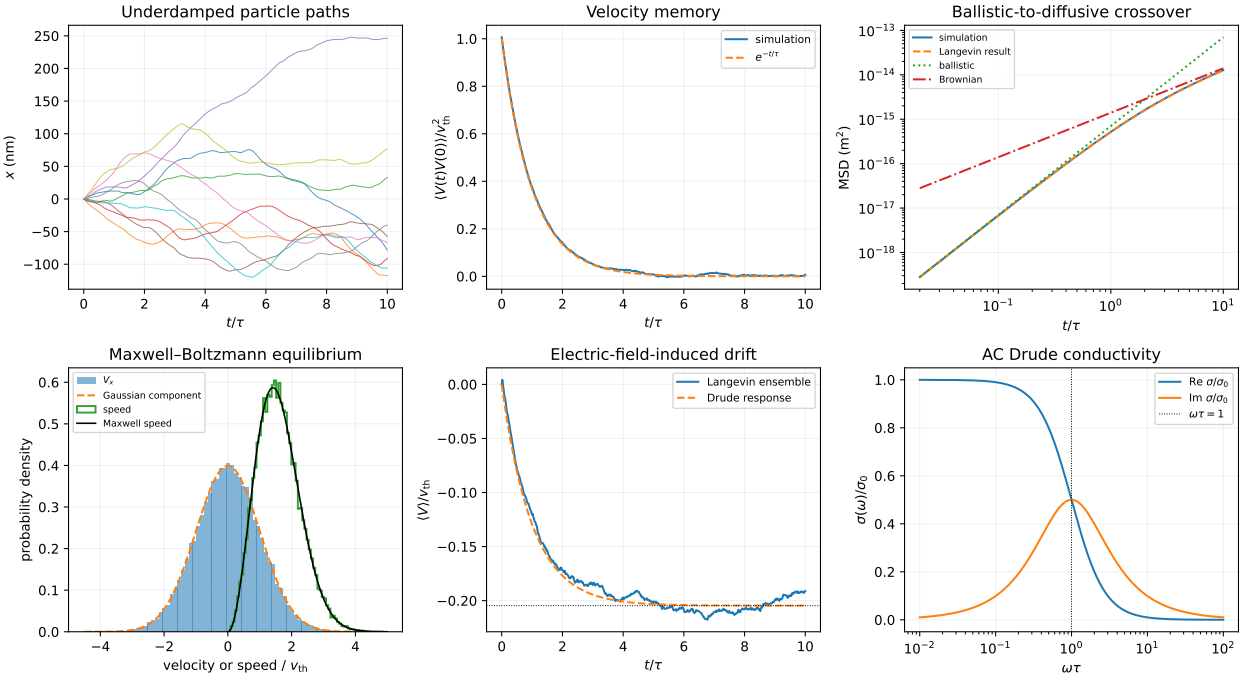


Figure 10.1: One Langevin model produces Maxwell–Boltzmann velocity statistics, the ballistic-to-diffusive crossover, field-induced Drude drift, and the frequency-dependent conductivity.[1]

10.2 The Sommerfeld correction

The classical Drude model assigns Maxwell–Boltzmann statistics to the electron gas. Electrons are fermions. Their equilibrium occupation is

$$f_0(E) = \frac{1}{e^{(E-\mu)/(k_{\mathrm{B}}T)} + 1}. \tag{10.9}$$

For a band $E_n(\boldsymbol{k})$, the semiclassical velocity is

$$\boldsymbol{v}_n(\boldsymbol{k}) = \frac{1}{\hbar}\nabla_{\boldsymbol{k}} E_n(\boldsymbol{k}). \tag{10.10}$$

[1] *Student reproduction notebook:* `Lab 4: Langevin, Maxwell-Boltzmann, and Drude transport`.

In the relaxation-time approximation, a weak uniform field changes the occupation by

$$\delta f_{n\boldsymbol{k}} = q_c \tau_{n\boldsymbol{k}} (\boldsymbol{E} \cdot \boldsymbol{v}_n(\boldsymbol{k})) \left(-\frac{\partial f_0}{\partial E} \right). \tag{10.11}$$

The resulting conductivity tensor is

$$\boxed{\sigma_{ij} = q_c^2 g_s \sum_n \int_{\mathrm{BZ}} \frac{\mathrm{d}^3 k}{(2\pi)^3} \tau_{n\boldsymbol{k}} v_{n,i} v_{n,j} \left(-\frac{\partial f_0}{\partial E} \right).} \tag{10.12}$$

The factor $-\partial f_0/\partial E$ is the Fermi window. In a degenerate gas, it restricts linear-response transport to states close to the chemical potential. The filled states far below the chemical potential do not produce a net current because opposite velocities cancel.

For an isotropic parabolic band and constant τ, equation (10.12) reduces to the Drude form $nq_c^2\tau/m$. The Drude–Sommerfeld model retains the relaxation-time equation of motion and replaces classical occupation by Fermi–Dirac occupation. This replacement corrects the electronic heat capacity and the population of current-carrying states without treating the microscopic origin of scattering.

10.3 Range of validity

The Drude model uses a parabolic free-electron dispersion, independent collisions, and a phenomenological momentum-relaxation time. The Sommerfeld correction replaces classical occupation by Fermi–Dirac occupation but retains the relaxation-time description. Energy-dependent and anisotropic scattering, strong fields, magnetic quantization, and coherent transport require a more complete collision or quantum-transport model.

10.4 Chapter summary

A constant electric field produces the steady drift velocity $\boldsymbol{v}_{\mathrm{d}} = q_c \tau \boldsymbol{E}/m$ and the Drude conductivity $\sigma_0 = nq_c^2\tau/m$. A harmonic field gives $\sigma(\omega) = \sigma_0/(1 - \mathrm{i}\omega\tau)$. Fermi–Dirac occupation replaces the classical gas by the Drude–Sommerfeld gas. Its conductivity is a Fermi-window integral of velocity products and relaxation times. The next chapter replaces the vacuum dispersion by crystal-band velocity and effective mass.

Chapter 11

Effective Mass and the Semiconductor Carrier Model

The Drude–Sommerfeld model describes carriers with a prescribed dispersion and relaxation time. A crystal supplies that dispersion through its Bloch bands. Band gradients determine velocity, band curvature determines effective mass, and equilibrium occupation determines the electron and hole populations. The resulting density moments form the semiconductor drift–diffusion model.

11.1 Crystal bands and effective mass

A carrier in a crystal responds to band curvature rather than to the vacuum mass alone. The semiclassical acceleration theorem is

$$\hbar\dot{\boldsymbol{k}} = q_c\boldsymbol{E}. \tag{11.1}$$

Differentiating equation (10.10) gives

$$\dot{\boldsymbol{v}} = q_c\boldsymbol{M}^{*-1}\boldsymbol{E}, \qquad \left(\boldsymbol{M}^{*-1}\right)_{ij} = \frac{1}{\hbar^2}\frac{\partial^2 E}{\partial k_i \partial k_j}. \tag{11.2}$$

This tensor is the effective mass already derived from band-edge curvature in equation (6.13).

For a parabolic band and scalar relaxation time, the tensor Drude relations are

$$\boxed{\boldsymbol{\mu} = |q_c|\tau\boldsymbol{M}^{*-1}, \qquad \boldsymbol{D} = k_{\mathrm{B}}T\tau\boldsymbol{M}^{*-1}, \qquad \boldsymbol{\sigma} = nq_c^2\tau\boldsymbol{M}^{*-1}.} \tag{11.3}$$

The Maxwell–Boltzmann velocity covariance becomes

$$\left\langle \boldsymbol{V}\boldsymbol{V}^{\mathsf{T}} \right\rangle = k_{\mathrm{B}}T\boldsymbol{M}^{*-1}. \tag{11.4}$$

The density-of-states mass controls the equilibrium population. The conductivity mass controls acceleration, mobility, and current. Anisotropic or multivalley bands require the full inverse-mass tensors and the population of each valley.

[1] *Student reproduction notebook:* `Lab 5: effective mass from bands to transport`.

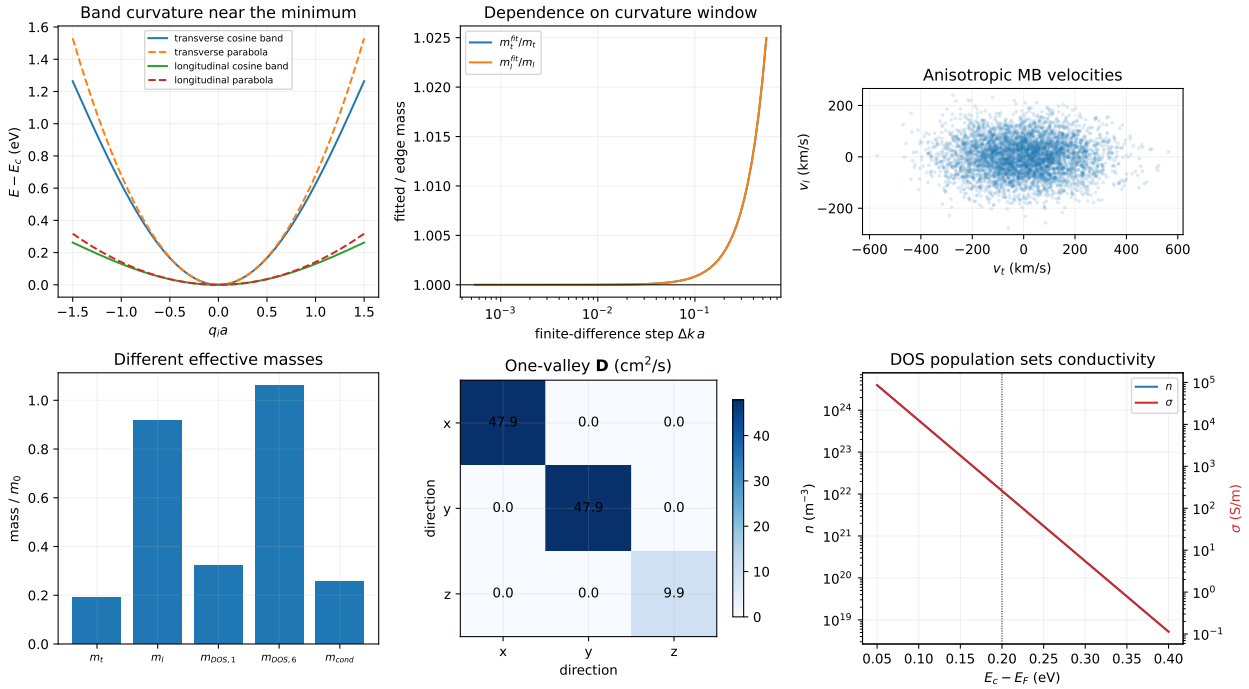


Figure 11.1: Band curvature determines the effective-mass tensor. The same tensor sets anisotropic thermal velocities, diffusion, mobility, and conductivity, while the density-of-states and conductivity masses preserve different properties of a multivalley band.[1]

11.2 The semiconductor carrier model

The equilibrium chapters determine n and p from the density of states, occupation law, and charge-neutrality condition. Transport promotes those scalar equilibrium populations to fields $n(\boldsymbol{r}, t)$ and $p(\boldsymbol{r}, t)$. At low field and in the diffusive regime, their conventional current densities are

$$\boxed{\boldsymbol{J}_n = qn\boldsymbol{\mu}_n\boldsymbol{E} + q\boldsymbol{D}_n\nabla n,} \tag{11.5}$$

$$\boxed{\boldsymbol{J}_p = qp\boldsymbol{\mu}_p\boldsymbol{E} - q\boldsymbol{D}_p\nabla p.} \tag{11.6}$$

For isotropic nondegenerate bands,

$$\frac{D_n}{\mu_n} = \frac{D_p}{\mu_p} = \frac{k_{\mathrm{B}}T}{q}. \tag{11.7}$$

For a general carrier gas, the thermodynamic form is

$$\frac{D}{\mu} = \frac{n}{q\,\partial n/\partial \mu_{\mathrm{ch}}}, \tag{11.8}$$

where μ_{ch} is the chemical potential measured as an energy. The Boltzmann result follows from $\partial n/\partial \mu_{\mathrm{ch}} = n/(k_{\mathrm{B}}T)$.

Generation and recombination close the carrier conservation equations:

$$\frac{\partial n}{\partial t} = \frac{1}{q}\nabla \cdot \boldsymbol{J}_n + G_n - R_n, \tag{11.9}$$

$$\frac{\partial p}{\partial t} = -\frac{1}{q}\nabla \cdot \boldsymbol{J}_p + G_p - R_p. \tag{11.10}$$

Poisson's equation supplies the electric field. Constitutive models for $\boldsymbol{\mu}$, $\boldsymbol{D}$, generation, recombination, and boundary fluxes complete the drift–diffusion system.

The model chain is therefore

$$\begin{aligned} \text{underdamped Langevin dynamics} &\longrightarrow \text{Maxwell–Boltzmann thermal gas} \\ &\longrightarrow \text{Brownian diffusion and classical Drude response} \\ &\longrightarrow \text{Fermi–Dirac Drude–Sommerfeld response} \\ &\longrightarrow \text{band velocity and effective mass} \\ &\longrightarrow \text{electron and hole drift–diffusion equations.} \end{aligned} \tag{11.11}$$

Each arrow changes the resolved variables or the occupation law. The classical gas resolves thermal velocities. The Sommerfeld model restores fermionic occupation. The effective-mass model replaces vacuum dynamics by local band dynamics. The carrier model takes density moments and closes them with mobility, diffusion, continuity, and electrostatics.

11.3 Range of validity

The Maxwell–Boltzmann gas describes a nondegenerate carrier population. The Drude–Sommerfeld gas describes degenerate occupation but retains a phenomenological relaxation time. The effective-mass reduction requires a locally parabolic band. The drift–diffusion closure requires lengths larger than the mean free path, times longer than the momentum-relaxation time, and fields within the linear-response regime. Full-band transport, inelastic and anisotropic collision operators, hot carriers, ballistic devices, and quantum coherence require a Boltzmann or quantum-transport description.

11.4 Chapter summary

The underdamped Langevin equation connects microscopic velocity relaxation to both equilibrium statistics and transport. With no applied field, it produces Maxwell–Boltzmann velocities and Brownian diffusion with $D = k_{\mathrm{B}}T\tau/m$. A field shifts the mean velocity and gives the Drude conductivity $nq_c^2\tau/m$. Fermi–Dirac occupation restricts the response of a degenerate gas to the Fermi window and produces the Drude–Sommerfeld model. Crystal-band velocity and curvature replace the vacuum dispersion by an effective-mass tensor. Density moments then yield the semiconductor electron and hole drift–diffusion currents and their continuity equations. The next chapter connects these constitutive coefficients to mobility, resistivity, and Hall measurements.

Chapter 12

Mobility, Resistivity, and the Hall Effect

The preceding chapters derived mobility from momentum relaxation and effective mass. A semiconductor measurement does not determine either quantity alone. It measures the combined response of band structure, lattice scattering, ionized impurities, and carrier populations. This chapter completes the low-field drift treatment of Sze and Lee, Chapter 2, by connecting mobility to resistivity and Hall data [7, Chap. 2].

12.1 Notation and units

Symbol	Meaning and SI unit unless stated otherwise
q, Q, s	Positive elementary-charge magnitude; carrier charge $Q = sq$; sign $s = -1$ for electrons and $s = +1$ for holes.
$\boldsymbol{E}$	Electric field, measured in volts per meter.
$\boldsymbol{B}$ and B_z	Magnetic flux density and its z component, measured in teslas.
n, p, N	Electron, hole, and generic one-carrier number densities, measured in inverse cubic meters.
m_n^*, m^*	Electron or generic conductivity effective mass, measured in kilograms.
τ_n, τ, τ_r	Electron, total, or mechanism-specific momentum-relaxation time, measured in seconds; r labels a scattering mechanism.
$\boldsymbol{v}_{n,\mathrm{d}}, v_{\mathrm{th}}, \ell$	Electron drift velocity and thermal speed in meters per second, and mean free path in meters.
$\mu, \mu_n, \mu_p, \mu_{\mathrm{L}}, \mu_{\mathrm{I}}, \mu_r$	Total, electron, hole, lattice-limited, impurity-limited, or mechanism-specific mobility, measured in square meters per volt-second.
T, k_{B}	Absolute temperature in kelvin and Boltzmann constant in joules per kelvin.
N_{I}	Total ionized-impurity density, measured in inverse cubic meters.
$\boldsymbol{J}, J_x, I_x$	Current density in amperes per square meter, its x component, and total x-directed current in amperes.
σ, ρ	Zero-field scalar conductivity in siemens per meter and resistivity in ohm-meters.
$\hat{\boldsymbol{\sigma}}, \hat{\boldsymbol{\rho}}, \sigma_0$	Magnetoconductivity tensor, magnetoresistivity tensor, and one-carrier zero-field conductivity.
ω_c, β	Cyclotron-frequency magnitude qB_z/m^* in inverse seconds and dimensionless magnetic-field parameter $\beta = \omega_c\tau = \mu B_z$.
$t, R_\square$	Sample thickness in meters and sheet resistance in ohms per square.
E_y, V_H, R_H	Hall field in volts per meter, Hall voltage in volts, and Hall coefficient in cubic meters per coulomb.
n_H, μ_H, r_H	Hall density in inverse cubic meters, Hall mobility in square meters per volt-second, and dimensionless Hall scattering factor.

12.2 Scattering-limited mobility

For an isotropic parabolic band, the low-field drift velocity and mobility are

$$\boldsymbol{v}_{n,\mathrm{d}} = -\mu_n \boldsymbol{E}, \qquad \mu_n = \frac{q\tau_n}{m_n^*}, \tag{12.1}$$

with the analogous positive drift relation for holes. The thermal speed and mean free path provide separate microscopic scales,

$$v_{\mathrm{th}} = \sqrt{\frac{3k_{\mathrm{B}}T}{m^*}}, \qquad \ell = v_{\mathrm{th}}\tau. \tag{12.2}$$

Thermal motion is much faster than the low-field drift motion. Mobility measures the small field-induced displacement of an otherwise randomized velocity distribution.

Lattice vibrations and ionized impurities produce distinct temperature trends. A reduced nondegenerate model gives

$$\mu_{\mathrm{L}} \propto T^{-3/2}, \qquad \mu_{\mathrm{I}} \propto \frac{T^{3/2}}{N_{\mathrm{I}}}. \tag{12.3}$$

Independent momentum-relaxation rates add. Therefore

$$\frac{1}{\tau} = \sum_r \frac{1}{\tau_r}, \qquad \boxed{\frac{1}{\mu} = \sum_r \frac{1}{\mu_r}.} \tag{12.4}$$

This Matthiessen construction explains why lightly doped material is lattice-limited at high temperature and heavily doped material is impurity-limited at low temperature. The power laws are asymptotic teaching models rather than universal mobility formulas.

12.3 Conductivity and resistivity

Electron and hole drift currents add as conventional current:

$$\boldsymbol{J}_{\text{drift}} = q(n\mu_n + p\mu_p)\boldsymbol{E}. \tag{12.5}$$

The conductivity and resistivity are

$$\boxed{\sigma = q(n\mu_n + p\mu_p),} \qquad \boxed{\rho = \frac{1}{\sigma}.} \tag{12.6}$$

In a majority-electron sample, $\rho \simeq (qn\mu_n)^{-1}$. Increasing donor density raises n but also strengthens ionized-impurity scattering and lowers μ_n. Resistivity therefore cannot be converted into dopant density without a mobility model and an ionization model.

A four-point probe separates current injection from voltage measurement. For a thin sheet with thickness t, the sheet resistance is $R_\square = \rho/t$. Geometric correction factors convert the measured voltage-to-current ratio into $R_\square$. The measurement determines resistivity; carrier type and carrier density require Hall information.

12.4 Hall response

12.4.1 Lorentz-force balance and sign convention

Let the conventional current flow along $+x$ and apply $\boldsymbol{B} = B_z\hat{\boldsymbol{z}}$ with $B_z > 0$. A carrier with charge $Q = sq$, where $s = -1$ for an electron and $s = +1$ for a hole, experiences the Lorentz force

$$\boxed{\boldsymbol{F}_{\text{L}} = Q\boldsymbol{v} \times \boldsymbol{B},} \qquad (F_{\text{L}})_y = -Qv_xB_z. \tag{12.7}$$

A hole carries current with $v_x > 0$, whereas an electron carries the same conventional current with $v_x < 0$ and $Q < 0$. Thus $Qv_x > 0$ in both cases and the magnetic force points toward the same physical edge, $-y$. Holes leave positive charge on that edge; electrons leave negative charge. The resulting Hall electric fields therefore point in opposite directions. This is the origin of the Hall-sign test.

Including momentum relaxation, the average drift velocity obeys

$$\boldsymbol{0} = Q\left(\boldsymbol{E} + \boldsymbol{v}_{\text{d}} \times \boldsymbol{B}\right) - \frac{m^*\boldsymbol{v}_{\text{d}}}{\tau}. \tag{12.8}$$

For a single carrier population in the open-circuit transverse direction, $v_y = 0$. Its transverse component reduces to

$$Q(E_y - v_x B_z) = 0, \qquad E_y = v_x B_z. \tag{12.9}$$

Since $J_x = NQv_x$, this immediately gives $E_y/J_x B_z = 1/(NQ) = s/(Nq)$. More generally, the measurable condition is zero *net* transverse current, $J_y = 0$; this distinction matters when multiple bands or carrier types conduct simultaneously.

Define the positive mobility $\mu = q\tau/m^*$, the zero-field conductivity $\sigma_0 = Nq\mu$, and the signed-field parameter

$$\beta = \omega_c \tau = \mu B_z, \qquad \omega_c = \frac{qB_z}{m^*}. \tag{12.10}$$

Solving equation (12.8) gives the one-carrier magnetoconductivity tensor

$$\begin{pmatrix} J_x \\ J_y \end{pmatrix} = \hat{\boldsymbol{\sigma}} \begin{pmatrix} E_x \\ E_y \end{pmatrix}, \qquad \hat{\boldsymbol{\sigma}} = \frac{\sigma_0}{1+\beta^2} \begin{pmatrix} 1 & s\beta \\ -s\beta & 1 \end{pmatrix}. \tag{12.11}$$

Its inverse is

$$\hat{\boldsymbol{\rho}} = \hat{\boldsymbol{\sigma}}^{-1} = \frac{1}{\sigma_0} \begin{pmatrix} 1 & -s\beta \\ s\beta & 1 \end{pmatrix}. \tag{12.12}$$

With $J_y = 0$, the second row yields

$$E_y = \frac{s\beta}{\sigma_0} J_x = \frac{s}{Nq} J_x B_z \equiv R_H J_x B_z. \tag{12.13}$$

Therefore

$$\boxed{R_H = -\frac{1}{qn} \quad \text{for electrons,} \qquad R_H = +\frac{1}{qp} \quad \text{for holes.}} \tag{12.14}$$

The sign follows from the stated directions of current, field, voltage leads, and magnetic field. Reversing any one of those conventions reverses the measured voltage. Notice also that $\rho_{xx} = 1/\sigma_0$ is independent of B_z in the ideal one-band, constant-τ Drude model. Longitudinal magnetoresistance therefore signals physics absent from that model, such as multiple carrier populations, anisotropic Fermi surfaces, or an energy-dependent scattering process.

12.4.2 Hall voltage and extracted quantities

For a rectangular bar of width w and thickness t, $J_x = I_x/(wt)$ and $E_y = V_H/w$. The width cancels, giving

$$V_H = \frac{R_H I_x B_z}{t}. \tag{12.15}$$

A small longitudinal voltage appears at the transverse contacts whenever they are not perfectly aligned. Because the Hall signal is odd in magnetic field while the zero-field longitudinal offset is even, the experimental Hall voltage is isolated by antisymmetrization:

$$V_H^{\mathrm{odd}}(B) = \frac{V_{xy}(+B) - V_{xy}(-B)}{2}, \qquad R_H = \frac{t}{I_x} \frac{\mathrm{d}V_H^{\mathrm{odd}}}{\mathrm{d}B}. \tag{12.16}$$

Linearity of V_H^{odd} with both I_x and B is an internal check of the low-field, ohmic measurement.

Under the one-carrier approximation, the measured Hall density and Hall mobility are

$$n_H = \frac{1}{q|R_H|}, \qquad \boxed{\mu_H = |R_H|\sigma = \frac{|R_H|}{\rho}.} \tag{12.17}$$

For example, an electron sample with $n = 2.0 \times 10^{22}\ \mathrm{m^{-3}}$ and $\mu_n = 0.12\ \mathrm{m^2\,V^{-1}\,s^{-1}}$ has $R_H = -3.12 \times 10^{-4}\ \mathrm{m^3\,C^{-1}}$. At $I_x = 1.0$ mA, $B_z = 0.50$ T, and $t = 0.50$ mm, equation (12.15) gives $V_H = -0.312$ mV. The conductivity is $3.85 \times 10^2\ \mathrm{S\,m^{-1}}$, so the extracted Hall mobility is $0.120\ \mathrm{m^2\,V^{-1}\,s^{-1}}$ in the constant-τ limit.

12.4.3 Hall scattering factor

Conductivity averages one power of the relaxation time over the carriers that respond to the field, whereas the weak-field Hall response averages two. For an isotropic parabolic band this distinction is summarized by

$$R_H = \frac{s r_H}{Nq}, \qquad r_H = \frac{\langle \tau^2 \rangle_{\mathrm{tr}}}{\langle \tau \rangle_{\mathrm{tr}}^2}, \qquad \mu_H = r_H \mu. \tag{12.18}$$

Consequently $n_H = N/r_H$ rather than N. For a nondegenerate three-dimensional parabolic band with $\tau(E) \propto E^r$,

$$r_H = \frac{\Gamma(5/2 + 2r)\Gamma(5/2)}{\Gamma(5/2 + r)^2}. \tag{12.19}$$

Constant relaxation time gives $r_H = 1$; the reduced acoustic-phonon law $r = -1/2$ gives $r_H = 3\pi/8 \simeq 1.18$; and the ionized-impurity law $r = 3/2$ gives $r_H \simeq 1.93$. Band warping, degeneracy, and anisotropic scattering require a tensor Boltzmann calculation rather than this scalar factor.

12.4.4 Electron–hole conduction

When electrons and holes both contribute, their zero-field conductivities add, but their transverse responses have opposite signs. The finite-field tensor components are

$$\sigma_{xx}(B) = q\left[\frac{n\mu_n}{1 + (\mu_n B)^2} + \frac{p\mu_p}{1 + (\mu_p B)^2}\right], \tag{12.20}$$

$$\sigma_{xy}(B) = qB\left[-\frac{n\mu_n^2}{1 + (\mu_n B)^2} + \frac{p\mu_p^2}{1 + (\mu_p B)^2}\right]. \tag{12.21}$$

Here $\hat{\boldsymbol{\sigma}} = \left(\begin{smallmatrix} \sigma_{xx} & \sigma_{xy} \\ -\sigma_{xy} & \sigma_{xx} \end{smallmatrix}\right)$, so $\rho_{yx} = \sigma_{xy}/(\sigma_{xx}^2 + \sigma_{xy}^2)$. The weak-field Hall coefficient becomes

$$\boxed{R_H = \frac{p\mu_p^2 - n\mu_n^2}{q\,(p\mu_p + n\mu_n)^2}.} \tag{12.22}$$

The mobility-squared weighting has three direct consequences. The Hall sign can be controlled by a less numerous but more mobile carrier population; R_H vanishes when $p\mu_p^2 = n\mu_n^2$ even though both populations are finite; and $1/(q|R_H|)$ is not the total density. Curvature of $V_H(B)$ and magnetoresistance provide evidence that a one-carrier extraction is insufficient. In that regime, longitudinal and transverse data must be fitted together over magnetic field rather than reduced to one Hall density.

[1] *Student reproduction notebook:* figures/12_mobility-resistivity-hall/fig_12_01.ipynb.

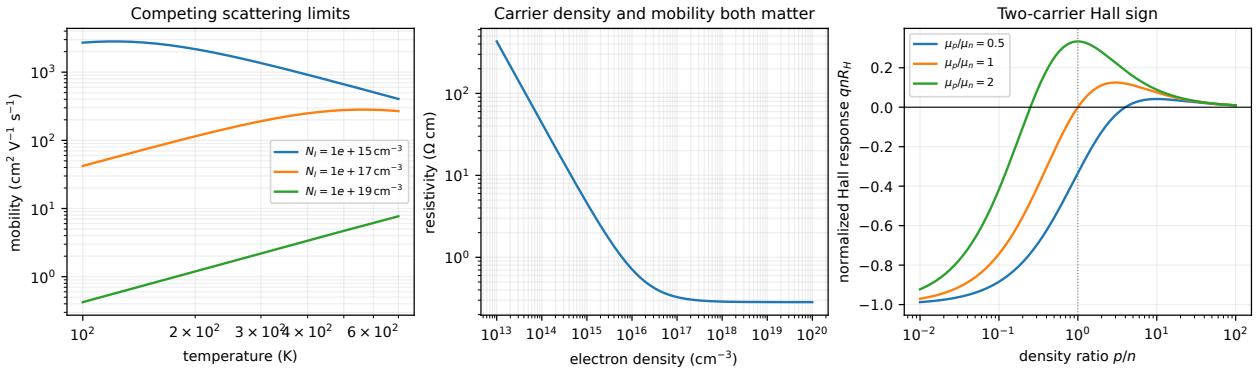


Figure 12.1: Competing lattice and impurity mobility limits, the resulting resistivity trend, and the mobility-squared sign competition in a two-carrier Hall response.[1]

12.5 Range of validity

The inverse-density extraction in equation (12.14) assumes one isotropic carrier population and low electric field. The finite-field Drude tensors remain useful beyond $\mu B \ll 1$, but real tensor bands require a band-resolved conductivity calculation. Degenerate statistics require Fermi-window averages of the energy-dependent scattering time. A linear Hall slope alone cannot uniquely determine the parameters of several carrier populations; reliable multicarrier fitting needs both $\rho_{xx}(B)$ and $\rho_{yx}(B)$ over a field range. The empirical power laws in equation (12.3) identify trends but do not replace measured or microscopic scattering models.

12.6 Chapter summary

Effective mass converts momentum relaxation into mobility. Lattice and ionized-impurity scattering contribute additive inverse mobilities. The Lorentz force produces the transverse Hall field, and inversion of the magnetoconductivity tensor fixes its sign and magnitude. A one-carrier Hall measurement yields a Hall density and Hall mobility only after accounting for the Hall scattering factor. With electrons and holes together, mobility- squared weighting controls the Hall sign and invalidates the inverse-density interpretation. These measurements provide the transport parameters needed by the drift–diffusion equations.

Chapter 13

Generation, Recombination, and Carrier Continuity

Equilibrium fixes the product $np = n_i^2$. Illumination, carrier injection, and impact ionization drive the semiconductor away from that constraint. Generation and recombination then determine how the carrier populations return toward equilibrium. The continuity equations combine these local processes with drift and diffusion [7, Chap. 2].

13.1 Notation and units

Symbol	Meaning and SI unit unless stated otherwise
n, p; n_n, p_n	Generic electron and hole densities; Sze–Lee electron and hole densities in an n-type sample, measured in inverse cubic meters.
n_0, p_0; n_{n0}, p_{n0}	Generic equilibrium densities; Sze–Lee equilibrium densities in an n-type sample, measured in inverse cubic meters.
$\Delta n, \Delta p$	Departures from equilibrium, defined by $\Delta n = n - n_0$ and $\Delta p = p - p_0$, measured in inverse cubic meters.
n_i	Intrinsic equilibrium carrier density, measured in inverse cubic meters.
g_c, g_v, f_c, f_v	Conduction- and valence-band densities of states in states per joule per cubic meter, and dimensionless nonequilibrium electron occupation functions.
$G, R, G_{\mathrm{th}}, G_L, U$	Total generation, total recombination, thermal generation, optical generation, and net recombination rates, measured in inverse cubic meter-seconds.
G_n, G_p, R_n, R_p	Electron and hole generation and recombination rates in the continuity equations.
B	Band-to-band recombination coefficient, measured in cubic meters per second.
C_n, C_p	Electron- and hole-initiated Auger coefficients, measured in sixth-power meters per second.
τ_n, τ_p, τ_g	Electron lifetime, hole lifetime, and depletion-region generation lifetime, measured in seconds.
E_t, E_i	Recombination-center energy and intrinsic reference energy, measured in joules or electron-volts under one consistent convention.
N_t, F_t, d_t	Bulk recombination-center density in inverse cubic meters, center electron-occupancy probability, and dimensionless center degeneracy.
$\sigma_n, \sigma_p, v_{\mathrm{th}}$	Electron and hole capture cross sections in square meters and carrier thermal speed in meters per second.
α_n, α_p	Capture coefficients $v_{\mathrm{th}}\sigma_n$ and $v_{\mathrm{th}}\sigma_p$, measured in cubic meters per second.
e_n, e_p	Electron and hole emission probabilities per occupied or unoccupied center, measured in inverse seconds.
n_1, p_1, n_B, p_B	Unit-degeneracy and degeneracy-corrected SRH auxiliary densities, measured in inverse cubic meters.
S, N_{st}, p_s	Surface recombination velocity in meters per second, surface-center density in inverse square meters, and surface hole density in inverse cubic meters.
E_{Fn}, E_{Fp}	Electron and hole quasi-Fermi energies.
$\boldsymbol{J}_n, \boldsymbol{J}_p$	Conventional electron and hole current densities, measured in amperes per square meter.
$\Delta\sigma, \mu_n, \mu_p$	Photoconductivity change in siemens per meter and electron and hole mobilities in square meters per volt-second.
D_p, L_p	Hole diffusivity in square meters per second and hole diffusion length in meters.
v_d, N	Packet drift velocity in meters per second and injected carrier number per unit area in inverse square meters.
x, t	Position in meters and time in seconds.
q, k_{B}, T	Positive elementary charge, Boltzmann constant, and absolute temperature.

13.2 Excess carriers and lifetime

Definition 13.1: Nonequilibrium carrier densities

Sze and Lee use n_n and p_n for the electron and hole densities in an n-type semiconductor. The first subscript identifies the semiconductor type; the final subscript 0 identifies thermal

equilibrium. Their excess-carrier definitions are [7, Sec. 2.3.1, Eqs. (36) and (38)]

$$\boxed{n_n = n_{n0} + \Delta n, \qquad p_n = p_{n0} + \Delta p,} \qquad \Delta n = n_n - n_{n0}, \quad \Delta p = p_n - p_{n0}. \tag{13.1}$$

This chapter subsequently uses n, p for the actual densities and n_0, p_0 for their equilibrium values. Thus $n \equiv n_n$, $p \equiv p_n$, $n_0 \equiv n_{n0}$, and $p_0 \equiv p_{n0}$ in the n-type example. The excess densities vanish at equilibrium, are positive for carrier excess, and are negative for carrier depletion.

Rudan connects these macroscopic densities to the nonequilibrium state occupations [6, Sec. 20.2, Eqs. (20.3)–(20.4)]:

$$n(\boldsymbol{r}, t) = \int_{E_C}^{\infty} g_c(E) f_c(\boldsymbol{r}, E, t)\, \mathrm{d}E, \qquad p(\boldsymbol{r}, t) = \int_{-\infty}^{E_V} g_v(E)[1 - f_v(\boldsymbol{r}, E, t)]\, \mathrm{d}E.$$

Thus Δn and Δp represent changes in occupied conduction-band states and empty valence-band states, not new kinds of particles.

Optical absorption, thermal generation, and impact ionization create electron–hole pairs and therefore equal electron and hole source rates. In a quasineutral bulk region with no trapping imbalance or boundary flux, charge neutrality enforces $\Delta n = \Delta p$. For selective contacts, trapping, boundary extraction, or unipolar electrical injection, solve Δn and Δp independently.

Definition 13.2: Net recombination rate

Sze and Lee define U as the recombination rate R in excess of the thermal generation rate G_{th} [7, Sec. 2.3.1, Eqs. (39)–(44)]. Under steady optical generation, this net rate equals the optical pair-generation rate G_L. Under low-level injection in n-type material, $\Delta p, p_{n0} \ll n_{n0}$, and U is linear in the excess minority-hole density:

$$\boxed{U \equiv R - G_{\mathrm{th}} = G_L = \frac{p_n - p_{n0}}{\tau_p} = \frac{\Delta p}{\tau_p}.} \tag{13.2}$$

Positive U removes electron–hole pairs, and $U = 0$ at thermal equilibrium. The equality $U = G_L$ applies to the steady illuminated state; the definition $U \equiv R - G_{\mathrm{th}}$ applies independently of the external source.

For a uniform optical generation rate G_L, the rate equation is

$$\frac{\mathrm{d}\Delta p}{\mathrm{d}t} = G_L - \frac{\Delta p}{\tau_p}. \tag{13.3}$$

Constant illumination gives $\Delta p_{\mathrm{ss}} = G_L \tau_p$. After the source is removed at $t = 0$,

$$\boxed{\Delta p(t) = \Delta p(0) e^{-t/\tau_p}.} \tag{13.4}$$

Photoconductivity decay measures this time constant because $\Delta\sigma = q(\mu_n \Delta n + \mu_p \Delta p)$.

13.3 Direct and recombination-center-assisted processes

13.3.1 Band-to-band recombination and detailed balance

A band-to-band event annihilates one conduction-band electron and one valence-band hole. The number of available electron–hole pairs is proportional to np, so the recombination rate is

$$R_{\mathrm{dir}} = Bnp, \tag{13.5}$$

where B contains the transition matrix element, band-structure factors, and state counting. In thermal equilibrium, recombination does not stop. It is balanced event by event, on average, by the reverse thermal generation process. At fixed lattice temperature,

$$R_{\mathrm{dir},0} = Bn_0p_0 = Bn_i^2 = G_{\mathrm{th,dir}}. \tag{13.6}$$

Subtracting the equilibrium reverse rate from the nonequilibrium forward rate gives

$$\boxed{U_{\mathrm{dir}} \equiv R_{\mathrm{dir}} - G_{\mathrm{th,dir}} = B(np - n_i^2).} \tag{13.7}$$

This subtraction is detailed balance: $U_{\mathrm{dir}} = 0$ at equilibrium, $U_{\mathrm{dir}} > 0$ for net recombination, and $U_{\mathrm{dir}} < 0$ for net generation.

A direct-gap material places the conduction-band minimum and valence-band maximum at the same crystal momentum. A photon carries the transition energy while contributing negligible crystal momentum, so radiative band-to-band recombination is efficient. An indirect-gap transition also requires a phonon to supply crystal momentum. Sze and Ng use representative room-temperature coefficients of about 10^{-10} $\mathrm{cm^3\,s^{-1}}$ for a direct-gap material and 10^{-15} $\mathrm{cm^3\,s^{-1}}$ for an indirect-gap material [8, Sec. 1.5.4, Eqs. (88)–(91)]. Under low-level injection in n-type material, $n \simeq n_0$ and $p = p_0 + \Delta p$, so

$$U_{\mathrm{dir}} \simeq Bn_0\Delta p, \qquad \tau_{p,\mathrm{dir}} = \frac{1}{Bn_0}. \tag{13.8}$$

The lifetime therefore belongs to a specified mechanism and injection regime; it is not a density-independent material constant.

13.3.2 The two-step recombination-center mechanism

Sze and Lee introduce a localized state at energy E_t inside the gap and call it an *intermediate-level state* or *recombination center* [7, Sec. 2.3.3, Fig. 13 and Eq. (50)]. Rudan traces such states to lattice defects, dislocations, residual impurities, and impurities introduced during processing [6, Secs. 20.2.2–20.2.3].

The center provides a two-step path between bands. First, an empty center captures a conduction electron and becomes occupied. Second, the occupied center captures a valence-band hole; equivalently, the trapped electron drops into the empty valence state. The center returns to its initial empty state, while one conduction electron and one valence hole have disappeared. The reverse sequence generates an electron–hole pair: a valence electron fills the center and the center subsequently emits the electron into the conduction band. The center is a catalyst for the interband transfer and is not consumed by the cycle.

Each step exchanges a phonon with the lattice. A center near midgap divides an interband transition into two energy steps of order $E_g/2$. The equilibrium phonon population falls rapidly with phonon

energy, so the two-step path is more effective than one phonon-assisted step across E_g in an indirect-gap semiconductor.

Let N_t be the center density and F_t the probability that a center is occupied by an electron. Sze–Lee Appendix I derives four elementary rates [7, Appendix I, Eqs. (2)–(14)]:

$$R_a = v_{\mathrm{th}}\sigma_n n N_t(1-F_t) \qquad \text{electron capture,} \tag{13.9}$$

$$R_b = e_n N_t F_t \qquad \text{electron emission,} \tag{13.10}$$

$$R_c = v_{\mathrm{th}}\sigma_p p N_t F_t \qquad \text{hole capture,} \tag{13.11}$$

$$R_d = e_p N_t(1-F_t) \qquad \text{hole emission.} \tag{13.12}$$

The capture coefficients

$$\alpha_n = v_{\mathrm{th}}\sigma_n, \qquad \alpha_p = v_{\mathrm{th}}\sigma_p \tag{13.13}$$

have units of volume per time. In R_a, the factor n counts available conduction electrons and $N_t(1-F_t)$ counts empty centers. In R_c, the factor p counts available holes and N_tF_t counts occupied centers. Electron emission requires an occupied center, so $R_b \propto N_tF_t$; hole emission requires an empty center, so $R_d \propto N_t(1-F_t)$. The emission probabilities e_n and e_p have units of inverse time.

13.3.3 Detailed balance and center degeneracy

Definition 13.3: Unit degeneracy and center degeneracy

Let g_0 be the number of microscopic configurations of an empty center and g_1 the number for the occupied center. The center degeneracy is $d_t = g_1/g_0$. Its equilibrium occupation is

$$F_{t0} = \left[1 + \frac{1}{d_t} e^{(E_t - E_F)/(k_{\mathrm{B}}T)}\right]^{-1}.$$

Unit degeneracy means $d_t = 1$: the empty and occupied charge states have equal statistical multiplicity. Center degeneracy means $d_t \neq 1$: several microscopic states represent one charge state. For example, one empty configuration and two allowed one-electron spin configurations give $d_t = 2$ when double occupation is excluded. Degeneracy does not mean $N_t = 1$, and it does not mean that several center energies are present. N_t counts centers; d_t counts configurations of one center level.

At thermal equilibrium, every elementary process is balanced by its reverse:

$$R_{a0} = R_{b0}, \qquad R_{c0} = R_{d0}. \tag{13.14}$$

Using the four rates and cancelling N_t gives

$$\frac{e_n}{\alpha_n} = n_0 \frac{1-F_{t0}}{F_{t0}}, \tag{13.15}$$

$$\frac{e_p}{\alpha_p} = p_0 \frac{F_{t0}}{1-F_{t0}}. \tag{13.16}$$

For nondegenerate equilibrium carriers, $n_0 = n_i e^{(E_F - E_i)/(k_{\mathrm{B}}T)}$ and $p_0 = n_i e^{(E_i - E_F)/(k_{\mathrm{B}}T)}$. Substitution of the center occupation then yields

$$e_n = \alpha_n n_B, \qquad e_p = \alpha_p p_B, \qquad n_B = \frac{n_i}{d_t} e^{(E_t - E_i)/(k_{\mathrm{B}}T)}, \qquad p_B = d_t n_i e^{(E_i - E_t)/(k_{\mathrm{B}}T)}, \tag{13.17}$$

with $n_B p_B = n_i^2$. For $d_t = 1$, these auxiliary densities become

$$n_1 = n_i e^{(E_t - E_i)/(k_\mathrm{B}T)}, \qquad p_1 = n_i e^{(E_i - E_t)/(k_\mathrm{B}T)}. \tag{13.18}$$

Unit degeneracy places half occupation at $E_F = E_t$. General center degeneracy places half occupation at $E_F = E_t - k_\mathrm{B}T \ln d_t$. The degeneracy therefore shifts the statistical occupancy of the level even though its physical energy E_t is unchanged.

13.3.4 Nonequilibrium occupancy and the SRH rate

Away from equilibrium, the net electron and hole transfer rates through the center are

$$U_n = R_a - R_b = N_t \alpha_n \left[n(1 - F_t) - n_B F_t \right], \tag{13.19}$$

$$U_p = R_c - R_d = N_t \alpha_p \left[p F_t - p_B (1 - F_t) \right]. \tag{13.20}$$

Electron capture fills the center, whereas hole capture empties it. Center occupation therefore obeys

$$N_t \frac{\partial F_t}{\partial t} = U_n - U_p. \tag{13.21}$$

The steady-occupancy condition is $U_n = U_p$. Solving this equality gives

$$F_t = \frac{\alpha_n n + \alpha_p p_B}{\alpha_n (n + n_B) + \alpha_p (p + p_B)}. \tag{13.22}$$

Substitution into either net transfer rate eliminates the unmeasured occupancy. Defining

$$\tau_n = \frac{1}{N_t \alpha_n} = \frac{1}{v_\mathrm{th} \sigma_n N_t}, \qquad \tau_p = \frac{1}{N_t \alpha_p} = \frac{1}{v_\mathrm{th} \sigma_p N_t}, \tag{13.23}$$

gives the degeneracy-corrected SRH rate

$$\boxed{U_\mathrm{SRH} = \frac{np - n_i^2}{\tau_p (n + n_B) + \tau_n (p + p_B)}.} \tag{13.24}$$

The numerator follows directly from detailed balance because $n_B p_B = n_i^2$. It fixes the direction: $np > n_i^2$ gives net recombination, $np < n_i^2$ gives net generation, and $np = n_i^2$ gives zero net rate. For unit degeneracy, $n_B = n_1$ and $p_B = p_1$, so

$$U_\mathrm{SRH} = \frac{np - n_i^2}{\tau_p (n + n_1) + \tau_n (p + p_1)}. \tag{13.25}$$

The lifetimes vary inversely with center density and capture cross section. Sze and Ng document this scaling for impurity diffusion and radiation-induced defects in silicon [8, Sec. 1.5.4, Eqs. (95a)–(97)].

The algebraic SRH rate assumes one center energy and steady center occupancy. Rudan applies it dynamically when N_t is small enough that stored center charge and its time derivative are negligible relative to band and dopant charge. Otherwise, equation (13.21) must be solved with the carrier continuity equations. Mobile charge in localized states also invalidates the approximation; Rudan identifies that case with gap conduction in polycrystalline material [6, Sec. 20.2.3].

Sze–Lee Eq. (51) sets $\sigma_n = \sigma_p$ and rewrites the energy dependence with a hyperbolic cosine. Rudan locates the most effective center at

$$E_{t,\max} \simeq E_i + \frac{k_\mathrm{B}T}{2} \ln\left(d_t^2 \frac{\sigma_p}{\sigma_n}\right). \tag{13.26}$$

Equal capture cross sections and $d_t = 1$ reduce this result to $E_{t,\max} = E_i$. Midgap is therefore the symmetric limit, not an independent assumption [6, Sec. 20.2.3, Eqs. (20.25)–(20.30)].

The surface expression comes from a separate two-dimensional population of recombination centers in Sze–Lee Sec. 2.3.4. Their Eqs. (53)–(54) give, for low-level injection in n-type material,

$$U_s = v_\mathrm{th}\sigma_p N_{st}(p_s - p_{n0}) \equiv S(p_s - p_{n0}), \qquad S = v_\mathrm{th}\sigma_p N_{st}. \tag{13.27}$$

The units of S are velocity. Surface passivation lowers N_{st} or the capture effectiveness and therefore lowers S; it does not alter the bulk center density N_t.

13.3.5 Depletion and high-injection limits

The same SRH expression describes net generation. In a depleted region with $n, p \ll n_1, p_1$, a unit-degeneracy midgap center gives

$$U_\mathrm{SRH} \simeq -\frac{n_i}{\tau_p + \tau_n} \equiv -\frac{n_i}{\tau_g}. \tag{13.28}$$

The negative sign denotes electron–hole generation. For $\tau_n = \tau_p = \tau$, the generation lifetime is $\tau_g = 2\tau$. This limit supplies the depletion-region generation current in a reverse-biased junction [8, Sec. 1.5.4]; the weak-injection limit instead gives $U_\mathrm{SRH} = \Delta p/\tau_p$ in n-type material.

At high carrier density, Auger recombination transfers the released energy to a third carrier rather than to a photon or one localized center. Rudan's detailed-balance reduction gives

$$U_\mathrm{A} = (C_n n + C_p p)\,(np - n_i^2), \tag{13.29}$$

where C_n and C_p describe electron- and hole-initiated events [6, Sec. 20.3, Eqs. (20.41)–(20.43)]. The factors n^2p and np^2 make Auger recombination dominant at high injection or heavy doping. Impact ionization is the inverse three-carrier process and dominates instead in a strong electric field. Far from equilibrium, use the separate Auger and impact-ionization rates rather than the detailed-balance reduction in equation (13.29).

A local bulk model combines independent channels as

$$U_\mathrm{bulk} = U_\mathrm{dir} + U_\mathrm{SRH} + U_\mathrm{A}. \tag{13.30}$$

External optical or electrical generation remains a separate source in the continuity equations. This separation prevents an imposed source from being counted again inside a net thermal recombination law.

13.4 Quasi-Fermi levels

One equilibrium Fermi level cannot represent two independently displaced carrier populations. Nondegenerate electron and hole densities define separate quasi-Fermi levels:

$$n = n_i e^{(E_{Fn} - E_i)/(k_\mathrm{B}T)}, \qquad p = n_i e^{(E_i - E_{Fp})/(k_\mathrm{B}T)}. \tag{13.31}$$

Their separation is

$$\boxed{E_{Fn} - E_{Fp} = k_{\mathrm{B}} T \ln\left(\frac{np}{n_i^2}\right).} \tag{13.32}$$

The levels coincide in equilibrium. Optical generation or electrical injection produces a positive splitting when $np > n_i^2$. Spatial gradients of the quasi-Fermi levels drive electron and hole currents.

13.5 Continuity and diffusion length

Local particle conservation gives

$$\frac{\partial n}{\partial t} = \frac{1}{q}\nabla \cdot \boldsymbol{J}_n + G_n - R_n, \qquad \frac{\partial p}{\partial t} = -\frac{1}{q}\nabla \cdot \boldsymbol{J}_p + G_p - R_p. \tag{13.33}$$

The opposite divergence signs follow from the opposite carrier charges. These equations combine with the drift–diffusion currents from equations (11.5) and (11.6) and Poisson's equation.

For minority holes in a neutral n-type region with zero field and no volume generation, steady state gives

$$D_p \frac{\mathrm{d}^2 \Delta p}{\mathrm{d}x^2} - \frac{\Delta p}{\tau_p} = 0. \tag{13.34}$$

The diffusion length

$$\boxed{L_p = \sqrt{D_p \tau_p}} \tag{13.35}$$

sets the spatial relaxation scale. For injection at $x = 0$ into a semi-infinite sample,

$$\Delta p(x) = \Delta p(0) e^{-x/L_p}. \tag{13.36}$$

A surface boundary replaces a prescribed concentration by a flux condition,

$$D_p \left.\frac{\partial p}{\partial x}\right|_s = S[p_s - p_0], \tag{13.37}$$

with the sign chosen for the outward surface normal.

13.6 Drift, diffusion, and recombination of a packet

A localized minority-carrier pulse in a uniform field provides a direct transport measurement. In one dimension,

$$\frac{\partial \Delta p}{\partial t} = D_p \frac{\partial^2 \Delta p}{\partial x^2} - v_d \frac{\partial \Delta p}{\partial x} - \frac{\Delta p}{\tau_p}. \tag{13.38}$$

An impulse containing N carriers per unit area evolves as

$$\Delta p(x,t) = \frac{N}{\sqrt{4\pi D_p t}} \exp\left[-\frac{(x - v_d t)^2}{4 D_p t} - \frac{t}{\tau_p}\right]. \tag{13.39}$$

The packet center gives $v_d = \mu_p E$, its variance gives $2 D_p t$, and its integrated amplitude gives e^{-t/τ_p}. One experiment therefore separates mobility, diffusivity, and lifetime.

[1] *Student reproduction notebook:* figures/13_generation-recombination-continuity/fig_13_01.ipynb.

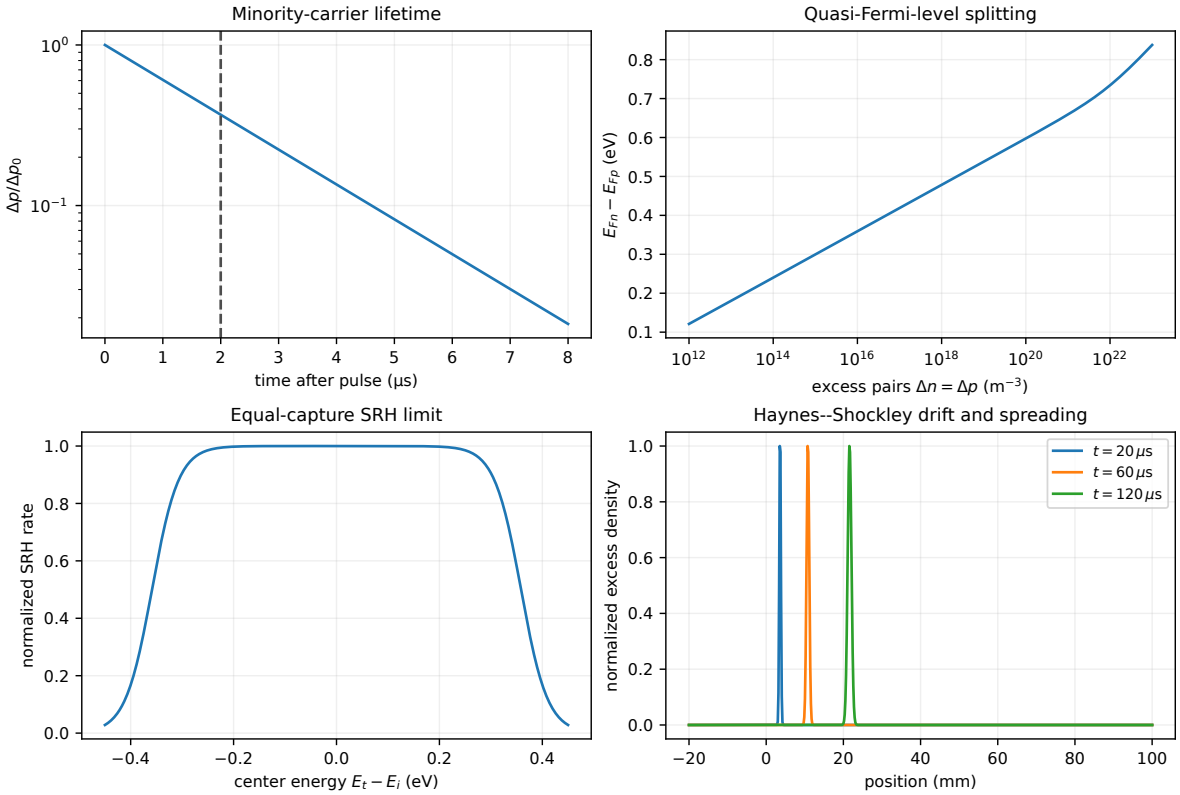


Figure 13.1: Temporal lifetime decay, quasi-Fermi-level splitting, the energy selectivity of trap-assisted recombination, and the simultaneous drift, diffusion, and recombination of a minority-carrier packet.[1]

13.7 Finite-difference continuity and dynamic center kinetics

The analytic packet in equation (13.39) supplies a numerical oracle, but arbitrary generation profiles, nonlinear SRH rates, and surface recombination require a numerical solution. Let $u_i^m = \Delta p(x_i, t_m)$ on a uniform grid with spacing Δx. The centered diffusion stencil is

$$\left.\frac{\partial^2 u}{\partial x^2}\right|_i \simeq \frac{u_{i-1} - 2u_i + u_{i+1}}{\Delta x^2}. \tag{13.40}$$

For positive drift velocity, the second-order upwind derivative is

$$\left.\frac{\partial u}{\partial x}\right|_i \simeq \frac{3u_i - 4u_{i-1} + u_{i-2}}{2\Delta x}; \tag{13.41}$$

the stencil reverses for negative velocity. Upwinding follows the direction from which carriers arrive and prevents the instability of a centered advection stencil at large cell Péclet number.

For the linear lifetime model, the spatial discretization produces

$$\frac{\mathrm{d}\boldsymbol{u}}{\mathrm{d}t} = \boldsymbol{A}\boldsymbol{u} + \boldsymbol{G}. \tag{13.42}$$

The theta method advances this system through

$$(\boldsymbol{I} - \theta\Delta t\boldsymbol{A})\,\boldsymbol{u}^{m+1} = [\boldsymbol{I} + (1-\theta)\Delta t\boldsymbol{A}]\,\boldsymbol{u}^m + \Delta t\boldsymbol{G}. \tag{13.43}$$

Crank–Nicolson uses $\theta = 1/2$ and is second order in time; backward Euler uses $\theta = 1$ and damps unresolved transients more strongly. The packet validation uses periodic boundaries so only discretization error remains. Its discrete carrier number must satisfy

$$\frac{\Delta x \sum_i u_i(t)}{\Delta x \sum_i u_i(0)} = e^{-t/\tau_p} \tag{13.44}$$

when generation is zero.

13.7.1 Nonlinear steady SRH boundary-value problem

For quasineutral pair injection, set $n = n_0 + u$ and $p = p_0 + u$. A spatially varying source produces the nonlinear steady equation

$$D_p\frac{\mathrm{d}^2 u}{\mathrm{d}x^2} + G(x) - U_{\mathrm{SRH}}(n_0 + u, p_0 + u) = 0. \tag{13.45}$$

At an exposed left surface and an ideal collecting right contact, use

$$D_p u'(0) = Su(0), \qquad u(L) = 0. \tag{13.46}$$

The interior finite-difference residual is

$$F_i = u_{i-1} - 2u_i + u_{i+1} + \frac{\Delta x^2}{D_p}\left[G_i - U_{\mathrm{SRH}}(n_0 + u_i, p_0 + u_i)\right]. \tag{13.47}$$

A bounded nonlinear least-squares solve enforces $u_i \geq 0$. The first boundary row discretizes the Robin condition in equation (13.46); the last row fixes the collecting-contact density. This calculation retains the full injection dependence of SRH recombination instead of replacing it by one constant lifetime.

13.7.2 Time-dependent center occupation

The algebraic SRH formula imposes $U_n = U_p$ at every time. To test that reduction, integrate the carrier and center equations directly:

$$\frac{\mathrm{d}n}{\mathrm{d}t} = G - U_n, \qquad \frac{\mathrm{d}p}{\mathrm{d}t} = G - U_p, \qquad N_t \frac{\mathrm{d}F_t}{\mathrm{d}t} = U_n - U_p. \tag{13.48}$$

Capture and emission can be much faster than carrier decay, so the system is stiff. A backward-differentiation-formula solver resolves both time scales. For the acceptor-like charge convention, the exact invariant

$$p - n - N_t F_t = \text{constant} \tag{13.49}$$

provides a stronger diagnostic than solver success alone.

Numerical check	Result	Acceptance condition
Packet relative L_2 error	4.999×10^{-3}	below 10^{-2}
Integrated packet ratio at 40 μs	0.6065297	agrees with $e^{-1/2}$
Nonlinear SRH residual	8.41×10^{-10}	below 2×10^{-6}
Charge-invariant drift	2.32×10^{-12}	below 10^{-9}

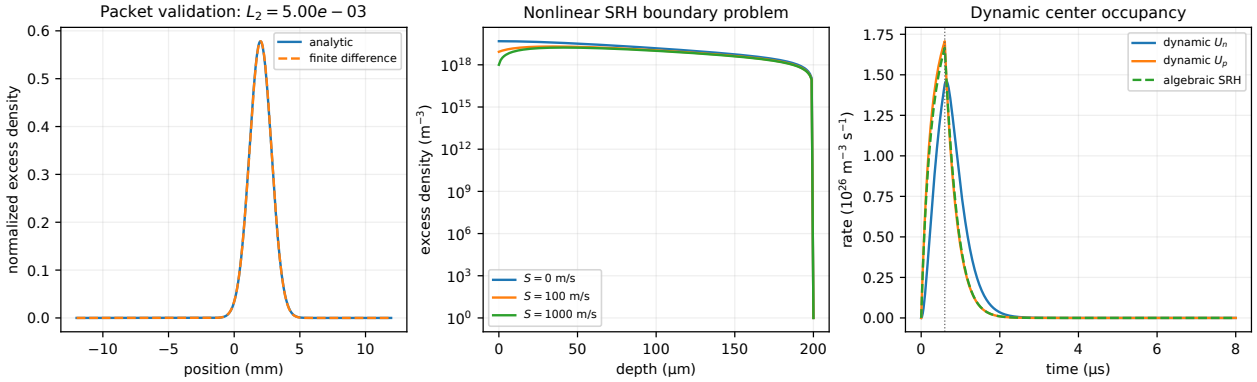


Figure 13.2: Finite-difference validation against a drifting, diffusing, and recombining analytic packet; a nonlinear SRH profile with optical generation and surface recombination; and dynamic center rates compared with the steady-occupancy SRH reduction.[2]

13.8 Range of validity

A constant lifetime represents the low-injection linearization of one dominant channel. The full SRH rate requires the center energy, degeneracy, density, capture cross sections, and charge state. Its algebraic form assumes steady center occupancy and negligible stored center charge; otherwise the occupancy requires an additional rate equation. Several center levels contribute separate rates. Auger recombination replaces the lifetime law at high carrier density, while depleted regions exhibit net SRH generation. Quasi-Fermi levels require each carrier population to remain internally thermalized. Diffusion-length solutions assume uniform coefficients and neutral regions; depletion fields and spatially varying material parameters require the coupled drift–diffusion–Poisson system.

[2] *Student reproduction notebook:* `figures/13_generation-recombination-continuity/fig_13_01.ipynb`.

13.9 Chapter summary

Generation displaces carrier populations from $np = n_i^2$, and recombination restores equilibrium. Low-level excess carriers decay with a lifetime and over a diffusion length. Direct, recombination-center-assisted, and Auger channels produce distinct density dependences; the SRH channel reverses sign to generate carriers in depleted material. Quasi-Fermi levels quantify nonequilibrium carrier populations. Continuity combines these local rates with carrier current, while a moving minority-carrier packet exposes mobility, diffusivity, and lifetime separately.

Chapter 14

Barrier Transport and Space-Charge-Limited Current

Drift and diffusion describe carriers that remain inside a band. Interfaces and strong injection introduce different transport controls. Thermionic emission selects carriers above a barrier, tunneling transmits wave amplitude through a barrier, and space charge couples injected density back to the electric field [7, Chap. 2].

14.1 Notation and units

Symbol	Meaning and SI unit unless stated otherwise
q, k_{B}, T	Positive elementary charge, Boltzmann constant, and absolute temperature.
E_C, E_F	Conduction-band edge and Fermi energy, measured in joules or electron-volts under one consistent convention.
χ, V_n, Φ_B	Electron affinity, conduction-band-to-Fermi-level separation, and emission barrier, each expressed in volts so multiplication by q gives energy.
N_C, n_{th}	Conduction-band effective density of states and thermionically eligible electron density, measured in inverse cubic meters.
A^*	Effective Richardson constant, measured in amperes per square meter-kelvin squared.
J_{th}	Thermionic current density, measured in amperes per square meter.
U_0, E	Rectangular-barrier energy and incident carrier energy, measured in joules or electron-volts under one consistent convention.
$d, \kappa, m^*, \hbar$	Barrier width in meters, evanescent decay constant in inverse meters, effective mass in kilograms, and reduced Planck constant in joule-seconds.
$\mathcal{T}(E)$	Dimensionless transmission probability at energy E.
$x_i, \Delta x, N$	Finite-difference grid point, grid spacing in meters, and number of grid points.
$t_0, \boldsymbol{H}, \boldsymbol{G}^r$	Finite-difference hopping energy, Hamiltonian matrix, and retarded Green matrix.
$\Sigma_{L,R}, \Gamma_{L,R}$	Lead self-energies and broadenings, measured in joules.
ρ, n, p, N_D^+, N_A^-	Charge density in coulombs per cubic meter and mobile or ionized carrier densities in inverse cubic meters.
$\phi, \mathcal{E}$	Electrostatic potential in volts and one-dimensional electric-field magnitude in volts per meter.
J, μ, ϵ_s	Current density, mobility, and semiconductor permittivity in amperes per square meter, square meters per volt-second, and farads per meter.
V, L, v_s	Applied voltage in volts, layer length in meters, and saturation velocity in meters per second.

14.2 Thermionic emission

The electron affinity χ is the energy separation, expressed in volts, between the conduction-band edge and the vacuum level. If $E_C - E_F = qV_n$, an electron must acquire energy $q(\chi + V_n)$ to leave an isolated n-type semiconductor. In the nondegenerate limit, the population above that barrier scales as

$$n_{\mathrm{th}} \propto N_C \exp\left[-\frac{q(\chi + V_n)}{k_{\mathrm{B}}T}\right]. \tag{14.1}$$

The exponential factor dominates the emission probability. A modest change in barrier energy therefore changes the emitted population by orders of magnitude.

The directed flux contains an additional velocity moment. For a parabolic band and an ideal absorbing interface, the current takes the Richardson form

$$J_{\mathrm{th}} = A^* T^2 \exp\left(-\frac{q\Phi_B}{k_{\mathrm{B}}T}\right), \tag{14.2}$$

where A^* contains the transport effective mass and Φ_B is the relevant barrier in volts. Interface reflection, image-force lowering, tunneling, and nonparabolic bands modify this ideal law.

14.3 Quantum tunneling through a barrier

Consider a rectangular barrier with energy height U_0 and width d. For carrier energy $0 < E < U_0$, the wavefunction inside the barrier contains evanescent terms with decay constant

$$\kappa = \frac{\sqrt{2m^*(U_0 - E)}}{\hbar}. \tag{14.3}$$

Continuity of the wavefunction and its derivative gives

$$\boxed{\mathcal{T}(E) = \left[1 + \frac{U_0^2 \sinh^2(\kappa d)}{4E(U_0 - E)}\right]^{-1}}. \tag{14.4}$$

When $\kappa d \gg 1$,

$$\mathcal{T}(E) \propto e^{-2\kappa d}. \tag{14.5}$$

The transmission increases for a thinner barrier, a lower barrier, a higher incident energy, or a smaller effective mass. Tunneling and thermionic emission therefore sample different parts of the same energy landscape: thermionic carriers pass over a barrier, while tunneling carriers retain finite transmission below it.

14.4 Space charge from injected carriers

The semiconductor charge density is

$$\rho = q(p - n + N_D^+ - N_A^-). \tag{14.6}$$

Charge neutrality makes this quantity nearly zero in a uniform bulk region. Strong injection breaks that cancellation. For electron injection with $n \gg N_D^+$ and negligible holes, the charge magnitude is qn and Poisson's equation links the carrier density to the field gradient.

Assume a one-dimensional layer of length L, constant mobility, negligible diffusion, and an injecting contact at which the field vanishes. Current and Poisson equations give

$$J = qn\mu\mathcal{E}, \qquad \frac{\mathrm{d}\mathcal{E}}{\mathrm{d}x} = \frac{qn}{\epsilon_s}. \tag{14.7}$$

Eliminating n yields

$$\mathcal{E}\frac{\mathrm{d}\mathcal{E}}{\mathrm{d}x} = \frac{J}{\epsilon_s\mu}. \tag{14.8}$$

The ohmic injecting-contact condition is $\mathcal{E}(0) = 0$. Integration gives

$$\mathcal{E}(x) = \sqrt{\frac{2Jx}{\epsilon_s\mu}}. \tag{14.9}$$

The applied voltage is the field integral,

$$V = \int_0^L \mathcal{E}(x)\,\mathrm{d}x = \frac{2}{3}\sqrt{\frac{2J}{\epsilon_s\mu}}L^{3/2}. \tag{14.10}$$

Solving for J produces the trap-free Mott–Gurney law

$$\boxed{J = \frac{9}{8}\epsilon_s\mu\frac{V^2}{L^3}.} \tag{14.11}$$

The quadratic voltage dependence distinguishes space-charge-limited current from an ohmic current proportional to V.

When carrier speed has reached a field-independent saturation value v_s, $J = qnv_s$. Poisson's equation then gives a linear field:

$$\mathcal{E}(x) = \frac{Jx}{\epsilon_s v_s}, \qquad V = \frac{JL^2}{2\epsilon_s v_s}. \tag{14.12}$$

Therefore

$$\boxed{J = 2\epsilon_s v_s\frac{V}{L^2}.} \tag{14.13}$$

Velocity saturation changes both the voltage exponent and the length scaling. Traps introduce additional stored charge and produce other exponents or thresholds.

14.5 Finite-difference barrier and space-charge calculations

The closed-form barrier and space-charge laws provide validation targets for numerical methods that also apply to nonrectangular potentials and nonuniform charge. Three finite-difference problems are treated separately: the open quantum barrier, Poisson electrostatics for prescribed charge, and the nonlinear space-charge field with unknown current.

[1] *Student reproduction notebook:* figures/14_barrier-space-charge-transport/fig_14_01.ipynb.

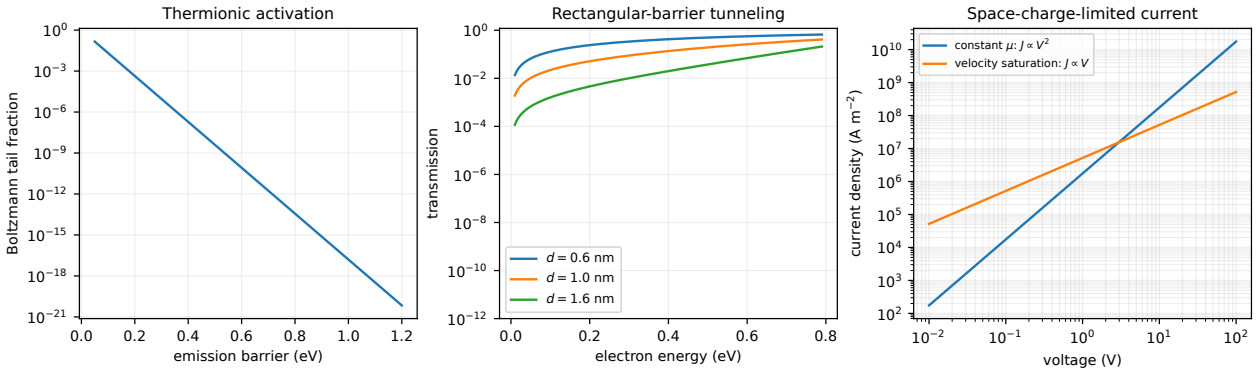


Figure 14.1: Thermionic activation by the Boltzmann tail, tunneling through finite-width barriers, and the distinct voltage scalings of constant-mobility and velocity-saturated space-charge-limited current.[1]

14.5.1 Open-boundary Schrödinger equation

On a cell-centered grid inside the barrier, the second derivative is

$$\left.\frac{\mathrm{d}^2\psi}{\mathrm{d}x^2}\right|_i \simeq \frac{\psi_{i-1} - 2\psi_i + \psi_{i+1}}{\Delta x^2}. \tag{14.14}$$

Define

$$t_0 = \frac{\hbar^2}{2m^*\Delta x^2}. \tag{14.15}$$

The finite-difference Hamiltonian has diagonal entries $2t_0 + U_i$ and nearest- neighbor entries $-t_0$. Setting the wavefunction to zero at the ends would turn the barrier into a closed box and eliminate transmitted flux. Transport requires open boundaries connected to semi-infinite leads.

A zero-potential discrete lead has dispersion

$$E = 2t_0[1 - \cos(k\Delta x)]. \tag{14.16}$$

Its retarded boundary self-energy and broadening are

$$\Sigma = -t_0 e^{ik\Delta x}, \qquad \Gamma = -2\,\mathrm{Im}\,\Sigma. \tag{14.17}$$

With one lead attached to each end, the open retarded Green matrix is

$$\boldsymbol{G}^r(E) = \left[E\boldsymbol{I} - \boldsymbol{H} - \Sigma_L - \Sigma_R\right]^{-1}. \tag{14.18}$$

Only the end-to-end element is required in one dimension:

$$\boxed{\mathcal{T}_{mFD}(E) = \Gamma_L\Gamma_R\left|G^r_{1N}(E)\right|^2.} \tag{14.19}$$

This formulation preserves incoming and outgoing flux and converges to equation (14.4) as $\Delta x \to 0$.

14.5.2 Poisson equation for prescribed charge

For electrostatic potential ϕ, Poisson's equation is

$$\frac{\mathrm{d}^2\phi}{\mathrm{d}x^2} = -\frac{\rho(x)}{\epsilon_s}. \tag{14.20}$$

At an interior node,

$$\phi_{i-1} - 2\phi_i + \phi_{i+1} = -\frac{\Delta x^2}{\epsilon_s}\rho_i. \tag{14.21}$$

Dirichlet contact potentials replace the first and last matrix rows. The result is a tridiagonal linear system. A uniform charge with $\phi(0) = \phi(L) = 0$ has the analytic parabola

$$\phi(x) = \frac{\rho}{2\epsilon_s}x(L - x), \tag{14.22}$$

which tests both the sign convention and second-order stencil.

14.5.3 Current-constrained nonlinear space charge

For unipolar injection, eliminate the density from $J = qnv(\mathcal{E})$ and Poisson's equation:

$$\frac{\mathrm{d}\mathcal{E}}{\mathrm{d}x} = \frac{J}{\epsilon_s v(\mathcal{E})}. \tag{14.23}$$

The current J is not supplied; it is fixed by the applied voltage. At each cell midpoint, the finite-difference residual is

$$F_i = \mathcal{E}_{i+1} - \mathcal{E}_i - \frac{\Delta x\, J}{\epsilon_s v[(\mathcal{E}_i + \mathcal{E}_{i+1})/2]}. \tag{14.24}$$

The unknown vector contains all nodal fields and J. It is closed by

$$\mathcal{E}_0 = 0, \qquad \sum_i \frac{\mathcal{E}_i + \mathcal{E}_{i+1}}{2}\Delta x = V. \tag{14.25}$$

A bounded nonlinear least-squares solver enforces nonnegative field and current. Using $v = \mu\mathcal{E}$ recovers the square-root field profile and Mott–Gurney current. Using $v = v_s$ recovers the linear field profile and the saturated current law.

Numerical check	Result	Acceptance condition
Barrier transmission relative L_2 error, $N = 100$	5.88×10^{-5}	below 10^{-4}
Uniform-charge Poisson error	7.61×10^{-15}	below 10^{-10}
Mott–Gurney current relative error, $N = 81$	8.53×10^{-4}	below 10^{-3}
Constant-velocity current relative error	6.67×10^{-11}	below 10^{-8}

[2] *Student reproduction notebook:* figures/14_barrier-space-charge-transport/fig_14_01.ipynb.

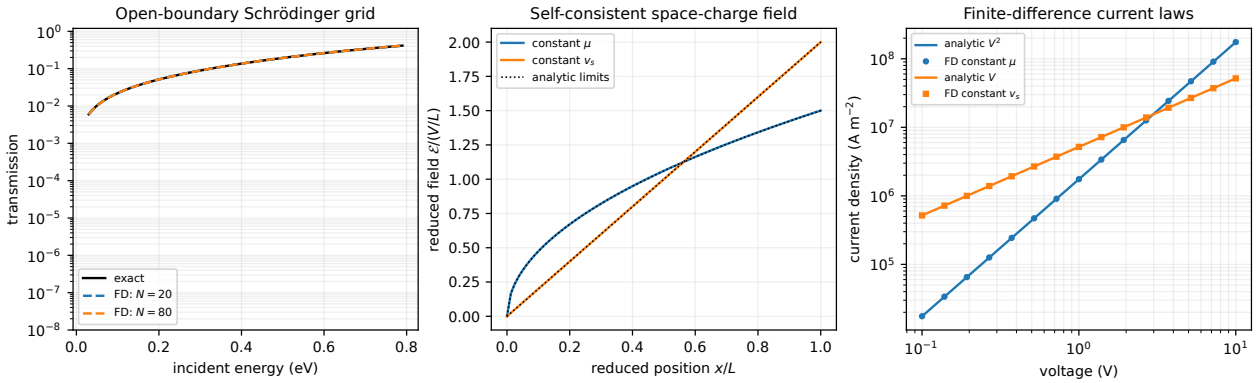


Figure 14.2: Convergence of open-boundary finite-difference tunneling to the exact rectangular-barrier result, numerical space-charge field profiles in the constant-mobility and constant-velocity limits, and recovery of their quadratic and linear current–voltage laws.[2]

14.6 Range of validity

The thermionic law assumes a thermalized nondegenerate population and an ideal interface. The rectangular-barrier result and its finite-difference model are one-dimensional, coherent, elastic, and constant-mass. The quantum grid must resolve both the lead wavelength and the evanescent decay length; its discrete lead band must contain every incident energy. The Mott–Gurney calculation assumes trap-free injection, one carrier type, negligible diffusion, and an ohmic contact with $\mathcal{E}(0) = 0$. A nonohmic contact replaces that boundary condition and changes the current. Real interfaces combine thermionic emission, field lowering, tunneling, trapping, and contact resistance. Their relative importance is determined by barrier shape, temperature, field, and scattering length.

14.7 Chapter summary

Thermionic emission depends exponentially on barrier energy relative to thermal energy. Quantum tunneling depends exponentially on barrier width and evanescent decay constant. Open-boundary finite differences reproduce the exact rectangular-barrier transmission and extend to arbitrary barrier shapes. Centered finite differences solve Poisson's equation for nonuniform charge. Under strong injection, simultaneous solution of the field profile and voltage constraint produces space-charge-limited current. Constant mobility yields $J \propto V^2/L^3$, while velocity saturation yields $J \propto V/L^2$.

Chapter 15

High-Field Carrier Transport

Constant mobility is a linear-response approximation. At high field, carriers gain enough energy between collisions to alter their scattering rates, occupy other valleys, or create additional electron–hole pairs. These processes set the current scale of short devices and prepare the breakdown analysis of a p–n junction [7, Chap. 2].

15.1 Notation and units

Symbol	Meaning and SI unit unless stated otherwise
E	Electric-field magnitude, measured in volts per meter.
v_d, μ_0, v_s, γ	Drift speed, low-field mobility, saturation speed, and dimensionless interpolation exponent.
$P, \boldsymbol{J}, \boldsymbol{E}$	Joule power density in watts per cubic meter, current-density vector, and electric-field vector.
n_1, n_2, n	Lower-valley, upper-valley, and total electron densities, measured in inverse cubic meters.
$\mu_1, \mu_2, \bar{\mu}$	Lower-valley, upper-valley, and population-averaged mobilities, measured in square meters per volt-second.
σ	Electrical conductivity, measured in siemens per meter.
E_g	Semiconductor band-gap energy, measured in joules or electron-volts.
α_n, α_p	Electron and hole impact-ionization coefficients, measured in inverse meters.
A_n, A_p	Ionization-rate prefactors, measured in inverse meters.
B_n, B_p	Characteristic ionization fields, measured in volts per meter.
G_A	Avalanche electron–hole pair-generation rate, measured per cubic meter per second.
J_n, J_p, q	Electron and hole conventional current densities in amperes per square meter and positive elementary-charge magnitude in coulombs.

15.2 Velocity saturation

Low-field drift obeys $v_d = \mu_0 E$. This form fails when the drift speed approaches the characteristic thermal or phonon-emission speed. A smooth empirical interpolation is

$$\boxed{v_d(E) = \frac{\mu_0 E}{[1 + (\mu_0 E / v_s)^{\gamma}]^{1/\gamma}}.} \tag{15.1}$$

It has the limits

$$v_d \simeq \mu_0 E \quad (\mu_0 E \ll v_s), \qquad v_d \longrightarrow v_s \quad (\mu_0 E \gg v_s). \tag{15.2}$$

The differential mobility $\mathrm{d}v_d/\,\mathrm{d}E$ decreases with field even while the velocity remains positive and monotone. In a short device, a moderate voltage can produce a field large enough that current is controlled by v_s rather than by μ_0.

The local Joule power density is

$$P = \boldsymbol{J} \cdot \boldsymbol{E}. \tag{15.3}$$

High-field transport therefore requires an energy-balance model when the carrier temperature departs from the lattice temperature. A field-dependent velocity law alone does not determine heat removal.

15.3 Transferred-electron effect

A multivalley conduction band can produce a stronger departure from linear response. Let a low valley have population n_1 and mobility μ_1, and a higher satellite valley have population n_2 and mobility μ_2. The total conductivity is

$$\sigma = q(n_1\mu_1 + n_2\mu_2), \qquad n = n_1 + n_2, \tag{15.4}$$

and the population-weighted mobility is

$$\bar{\mu} = \frac{n_1\mu_1 + n_2\mu_2}{n_1 + n_2}. \tag{15.5}$$

If high field transfers electrons into a valley with $\mu_2 \ll \mu_1$, $\bar{\mu}$ can fall faster than E rises. The drift velocity then satisfies

$$\frac{\mathrm{d}v_d}{\mathrm{d}E} < 0 \tag{15.6}$$

over a finite field interval. This negative differential mobility is a band population effect, not a reversal of carrier velocity. It underlies the transferred-electron response of n-type GaAs.

15.4 Impact ionization

A sufficiently energetic carrier can create an electron–hole pair in a collision. Conservation of energy and momentum makes the threshold kinetic energy exceed the band gap. The microscopic probability is summarized by the ionization coefficients $\alpha_n(E)$ and $\alpha_p(E)$, the mean number of pairs produced per unit distance by electrons and holes. A common empirical form is

$$\alpha_{n,p}(E) = A_{n,p} \exp\left(-\frac{B_{n,p}}{|E|}\right). \tag{15.7}$$

The exponential field dependence produces an abrupt transition from negligible generation to strong multiplication.

Electron and hole currents both contribute to the avalanche pair-generation rate. In one dimension,

$$\boxed{G_A = \frac{\alpha_n|J_n| + \alpha_p|J_p|}{q}.} \tag{15.8}$$

This term enters both carrier continuity equations because each event creates one electron and one hole. Feedback occurs when the generated carriers are accelerated and generate further pairs. A junction depletion region supplies the extended high-field region needed for avalanche multiplication.

[1] *Student reproduction notebook:* `figures/15_high-field-transport/fig_15_01.ipynb`.

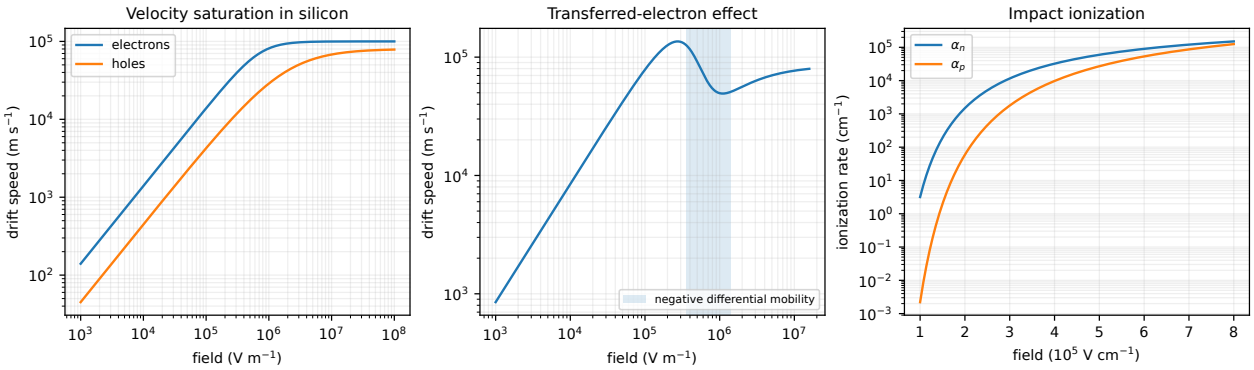


Figure 15.1: Velocity saturation in silicon, negative differential mobility from intervalley transfer, and the rapid field activation of electron and hole impact-ionization coefficients.[1]

15.5 Transport-regime hierarchy

The field scale must be compared with material and device scales. Linear response requires $v_d \ll v_s$. Velocity saturation requires a field-dependent constitutive law. Transferred-electron transport requires multiple valleys and an energy-dependent population model. Impact ionization requires carrier energy gain large enough to create pairs and must be included in continuity. Strong carrier heating requires an energy-balance or Boltzmann description. Quantum-coherent short structures require a transmission-based model rather than a local velocity law.

15.6 Chapter summary

High fields invalidate constant mobility. Silicon approaches a saturation velocity as inelastic scattering removes field-supplied energy. Multivalley materials can show negative differential mobility when carriers transfer to a low-mobility satellite valley. Still larger fields activate impact ionization, which adds a nonlinear generation term to carrier continuity. These mechanisms complete the Chapter 2 transport sequence and provide the physical inputs for junction current, depletion-region generation, and breakdown.

References

[1] Desmond J. Higham. "An Algorithmic Introduction to Numerical Simulation of Stochastic Differential Equations". In: *SIAM Review* 43.3 (2001), pp. 525–546. DOI: 10.1137/S0036144500378302.

[2] Kiyosi Itô. "Stochastic Integral". In: *Proceedings of the Imperial Academy* 20.8 (1944), pp. 519–524. DOI: 10.3792/pia/1195572786.

[3] Charles Kittel. *Introduction to Solid State Physics*. 8th ed. Hoboken, New Jersey: John Wiley & Sons, 2005.

[4] Peter E. Kloeden and Eckhard Platen. *Numerical Solution of Stochastic Differential Equations*. Springer, 1992. DOI: 10.1007/978-3-662-12616-5.

[5] Donald A. Neamen. *Semiconductor Physics and Devices: Basic Principles*. 4th ed. New York: McGraw-Hill, 2012.

[6] Massimo Rudan. *Physics of Semiconductor Devices*. New York: Springer, 2015. DOI: 10.1007/978-1-4939-1151-6.

[7] Simon M. Sze and Ming-Kwei Lee. *Semiconductor Devices: Physics and Technology*. 3rd ed. Hoboken, New Jersey: John Wiley & Sons, 2012. ISBN: 9780470537947.

[8] Simon M. Sze and Kwok K. Ng. *Physics of Semiconductor Devices*. 3rd ed. Hoboken, New Jersey: John Wiley & Sons, 2007. DOI: 10.1002/0470068329.

[9] Norbert Wiener. "Differential-Space". In: *Journal of Mathematics and Physics* 2 (1923), pp. 131–174.

[10] Peter Y. Yu and Manuel Cardona. *Fundamentals of Semiconductors: Physics and Materials Properties*. 4th ed. Berlin: Springer, 2010. DOI: 10.1007/978-3-642-00710-1.